# CCNA Security

## Chapter Eight

## Implementing Virtual Private Networks

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# Lesson Planning

- This lesson should take 3-4 hours to present
- The lesson should include lecture, demonstrations, discussions and assessments
- The lesson can be taught in person or using remote instruction

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# Major Concepts

- Describe the purpose and operation of VPN types
- Describe the purpose and operation of GRE VPNs
- Describe the components and operations of IPsec VPNs
- Configure and verify a site-to-site IPsec VPN with pre-shared key authentication using CLI
- Configure and verify a site-to-site IPsec VPN with pre-shared key authentication using SDM
- Configure and verify a Remote Access VPN

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# Lesson Objectives

Upon completion of this lesson, the successful participant will be able to:

1. Describe the purpose and operation of VPNs
2. Differentiate between the various types of VPNs
3. Identify the Cisco VPN product line and the security features of these products
4. Configure a site-to-site VPN GRE tunnel
5. Describe the IPSec protocol and its basic functions
6. Differentiate between AH and ESP
7. Describe the IKE protocol and modes
8. Describe the five steps of IPSec operation

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# Lesson Objectives

9. Describe how to prepare IPSec by ensuring that ACLs are compatible with IPSec
10. Configure IKE policies using the CLI
11. Configure the IPSec transform sets using the CLI
12. Configure the crypto ACLs using the CLI
13. Configure and apply a crypto map using the CLI
14. Describe how to verify and troubleshoot the IPSec configuration
15. Describe how to configure IPSec using SDM
16. Configure a site-to-site VPN using the Quick Setup VPN Wizard in SDM
17. Configure a site-to-site VPN using the step-by-step VPN Wizard in SDM

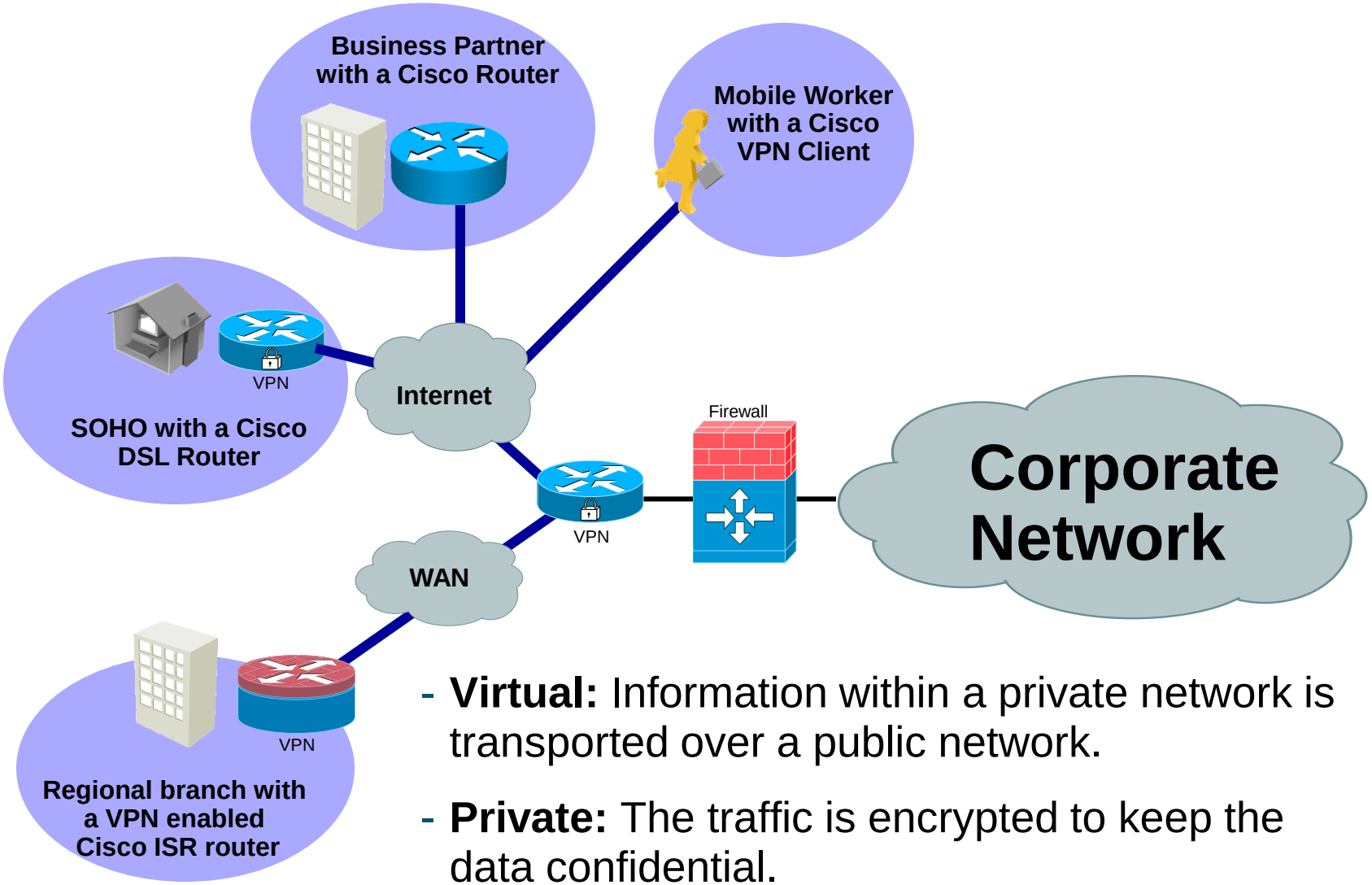
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# Lesson Objectives

18. Verify, monitor and troubleshoot VPNs using SDM
19. Describe how an increasing number of organizations are offering telecommuting options to their employees
20. Differentiate between Remote Access IPSec VPN solutions and SSL VPNs
21. Describe how SSL is used to establish a secure VPN connection
22. Describe the Cisco Easy VPN feature
23. Configure a VPN Server using SDM
24. Connect a VPN client using the Cisco VPN Client software

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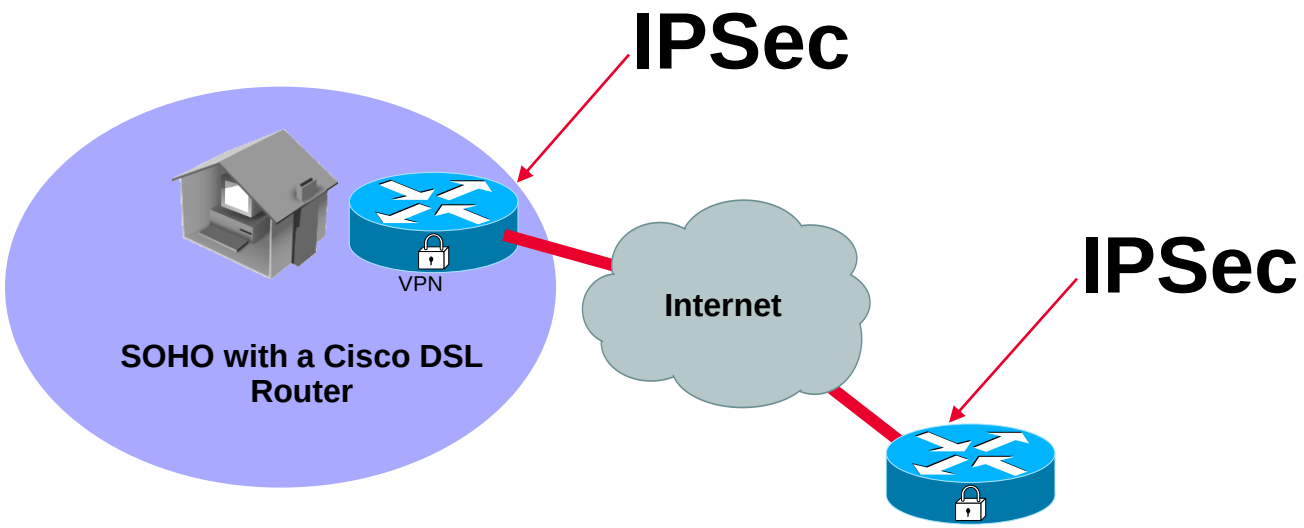
# What is a VPN?




- **Virtual:** Information within a private network is transported over a public network.
- **Private:** The traffic is encrypted to keep the data confidential.

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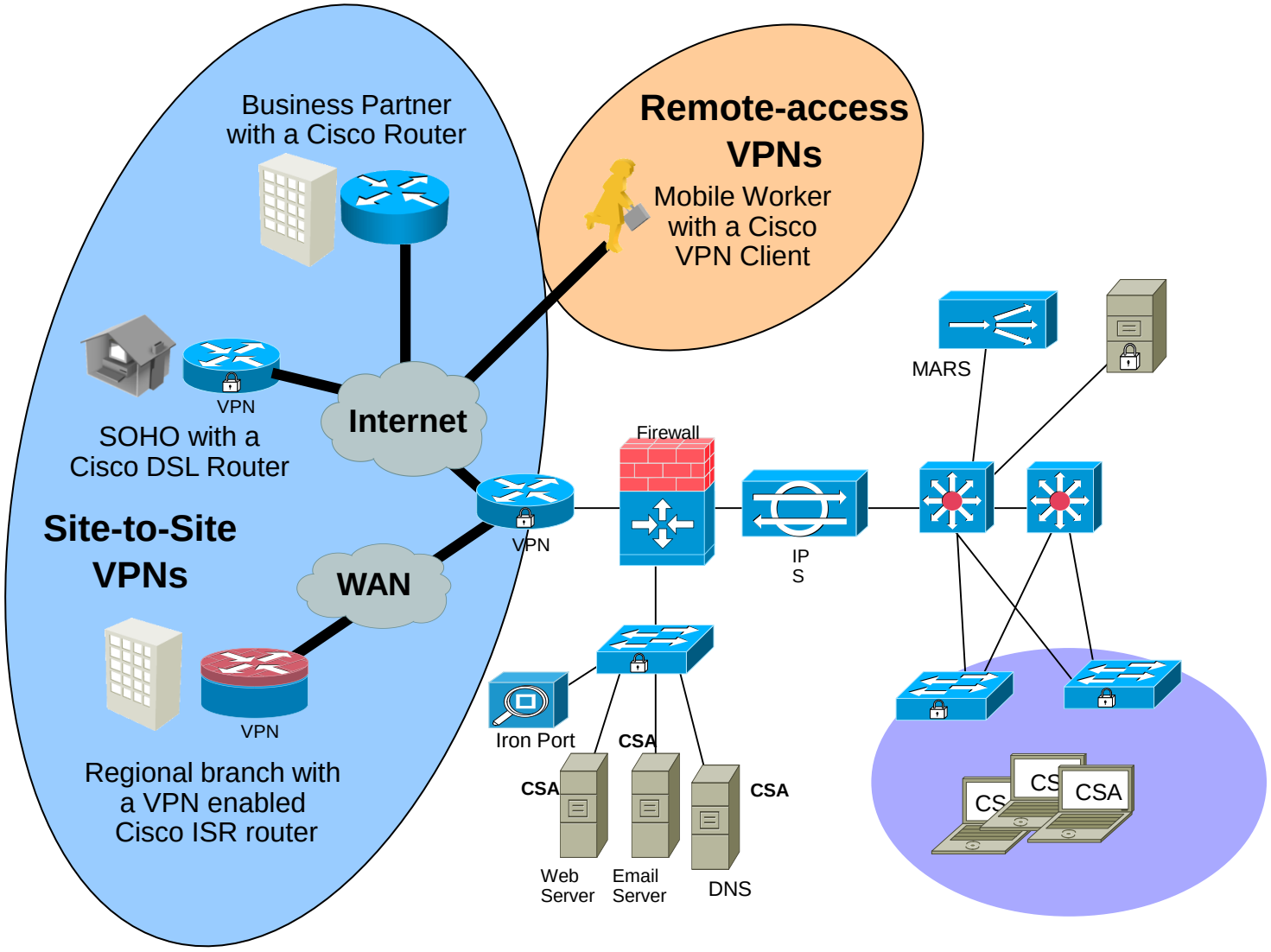
# Layer 3 VPN




- Generic routing encapsulation (GRE)
- Multiprotocol Label Switching (MPLS)
- IPSec

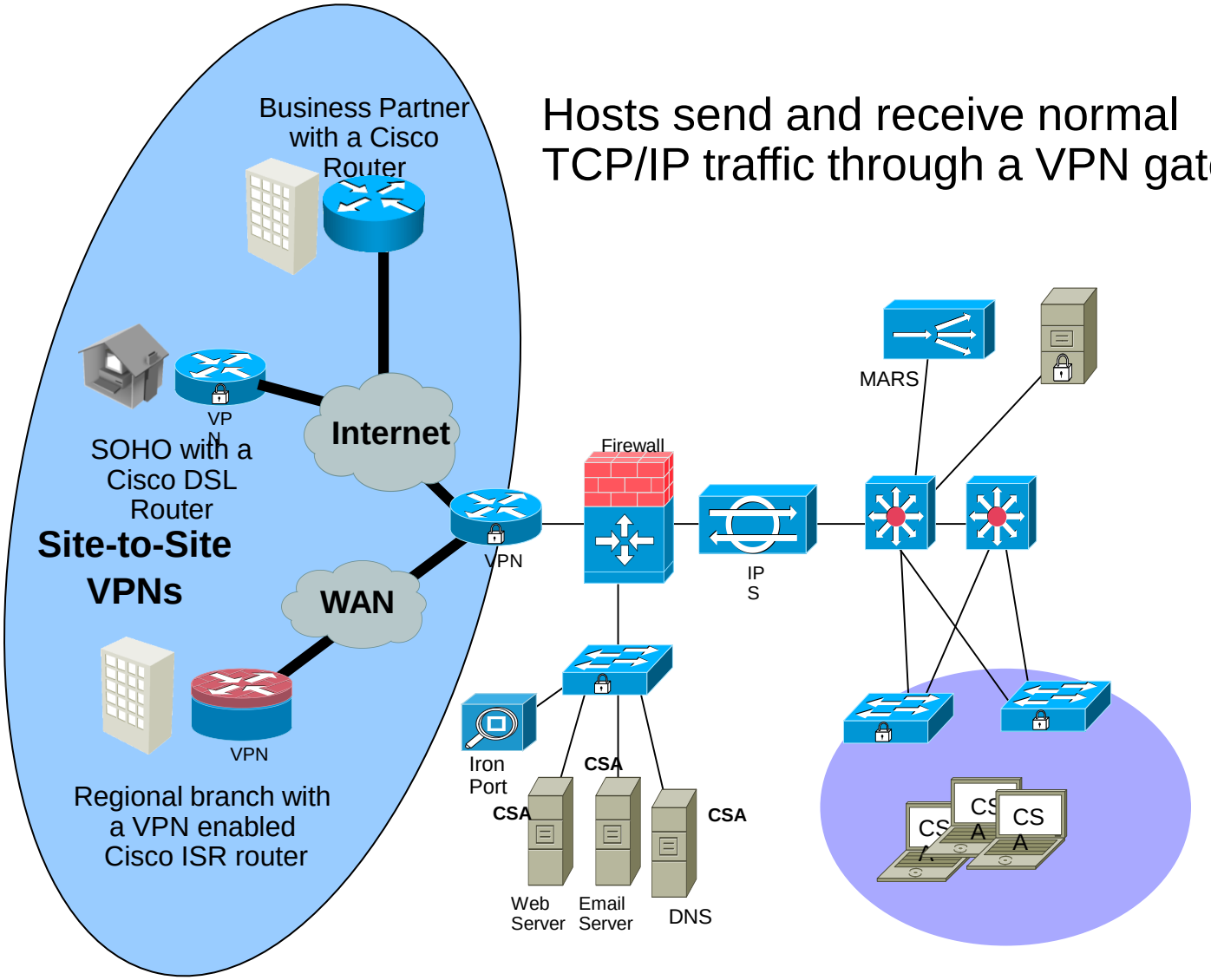
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# Types of VPN Networks




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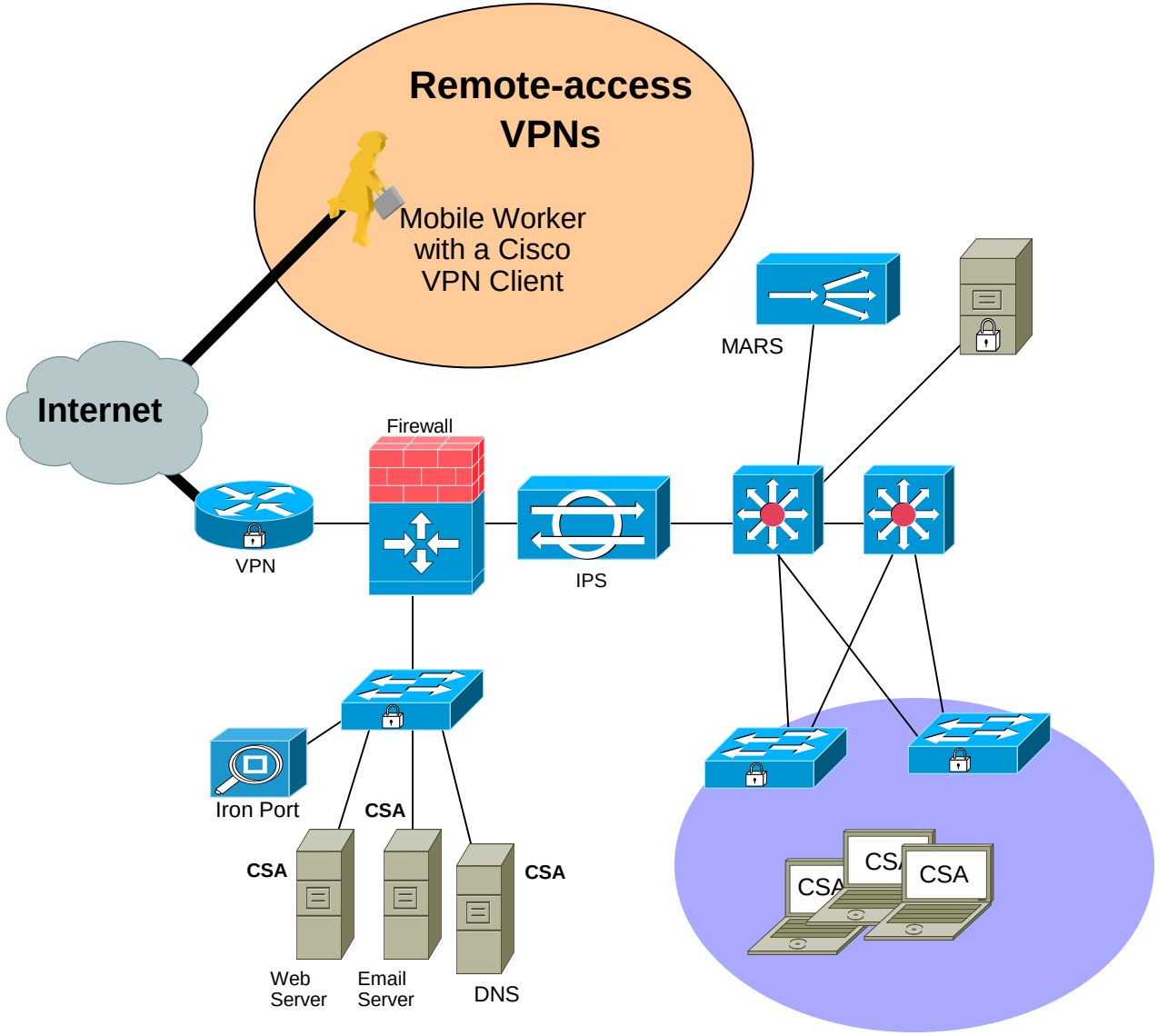
# Site-to-Site VPN




Hosts send and receive normal TCP/IP traffic through a VPN gateway

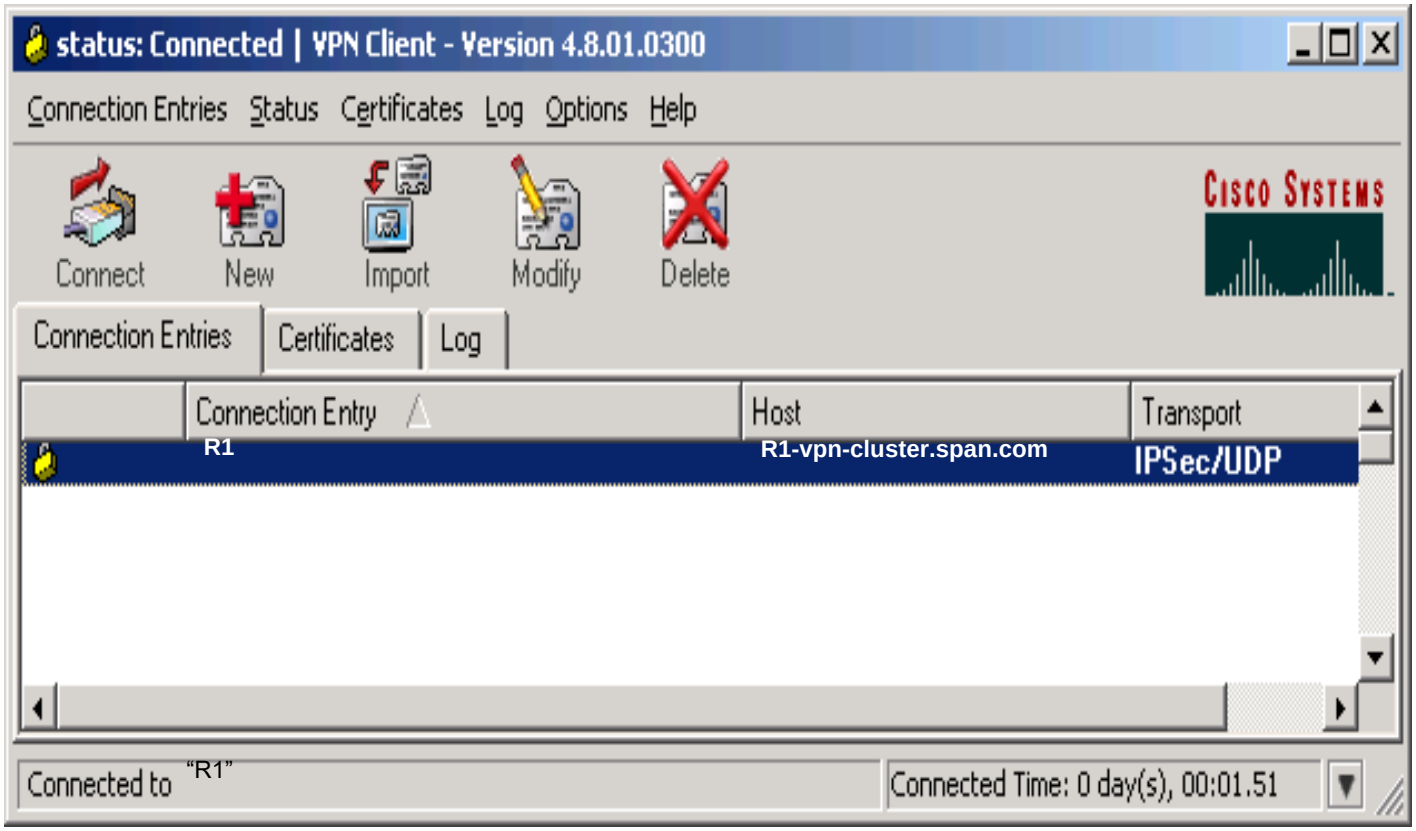
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# Remote-Access VPNs




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# VPN Client Software

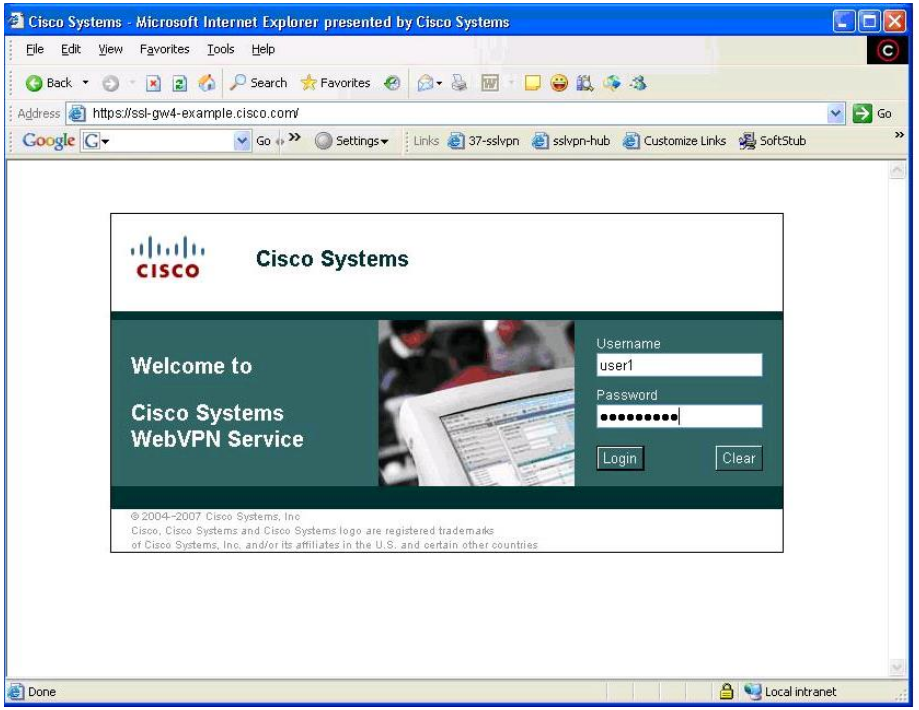



In a remote-access VPN, each host typically has Cisco VPN Client software

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# Cisco IOS SSL VPN

- Provides remote-access connectivity from any Internet-enabled host
- Uses a web browser and SSL encryption
- Delivers two modes of access:
  - Clientless
  - Thin client



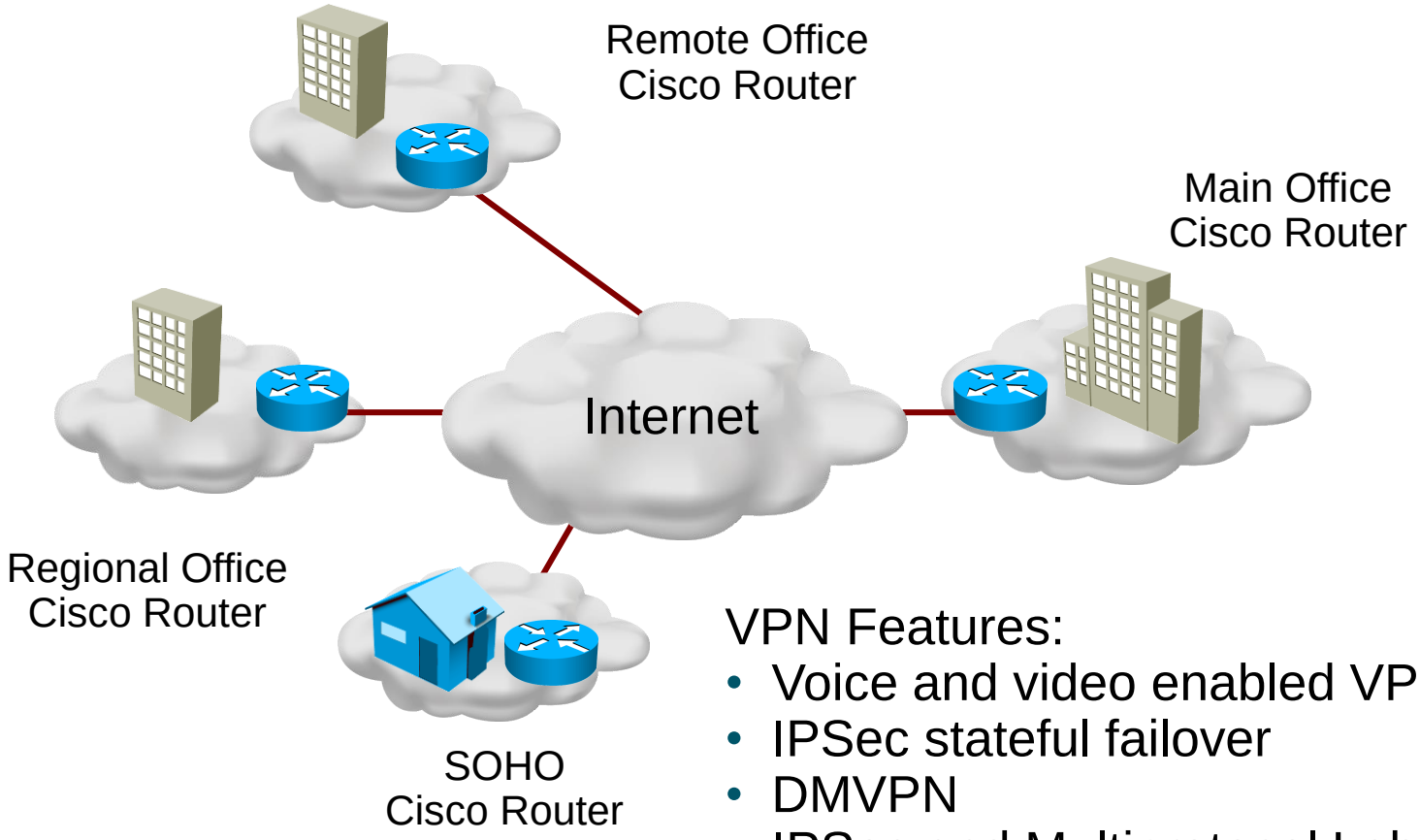

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# Cisco VPN Product Family

| Product Choice | Remote-Access VPN | Site-to-Site VPN |
|---|---|---|
| Cisco VPN-Enabled Router | Secondary role | Primary role |
| Cisco PIX 500 Series Security Appliances | Secondary role | Primary role |
| Cisco ASA 5500 Series Adaptive Security Appliances | Primary role | Secondary role |
| Cisco VPN 3000 Series Concentrators | Primary role | Secondary role |
| Home Routers | Primary role | |

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# Cisco VPN-Optimized Routers

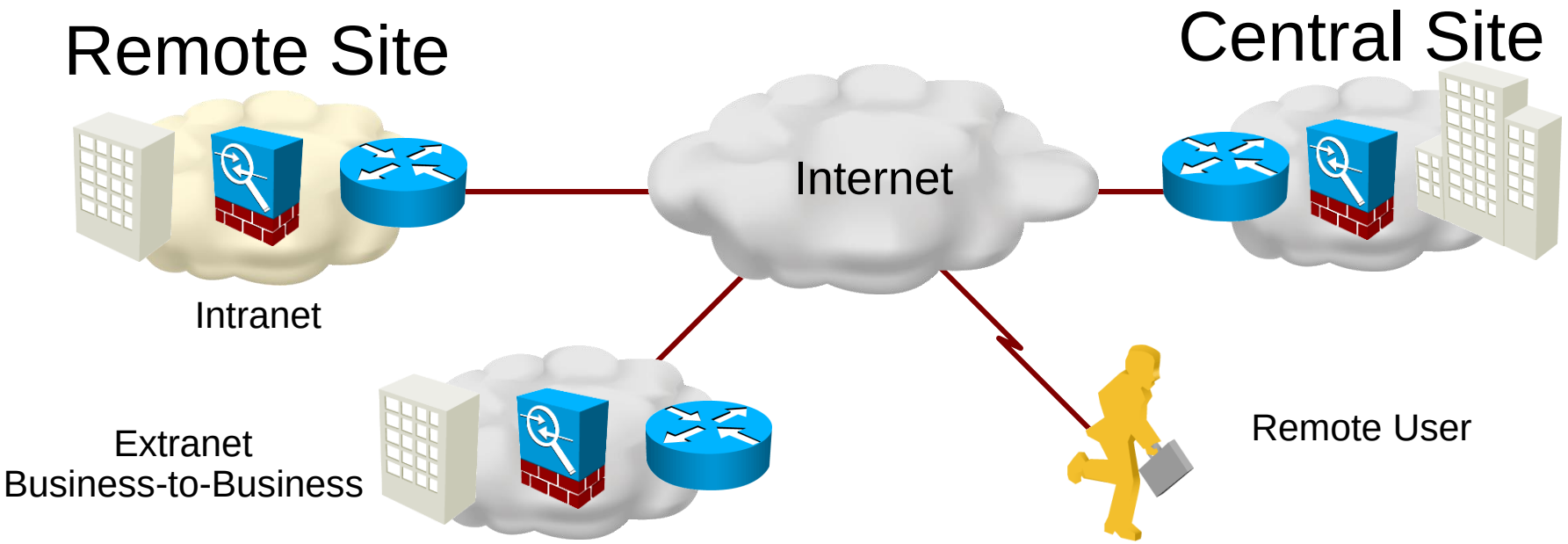



VPN Features:

- Voice and video enabled VPN (V3PN)
- IPSec stateful failover
- DMVPN
- IPSec and Multiprotocol Label Switching (MPLS) integration
- Cisco Easy VPN

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# Cisco ASA 5500 Series Adaptive Security Appliances

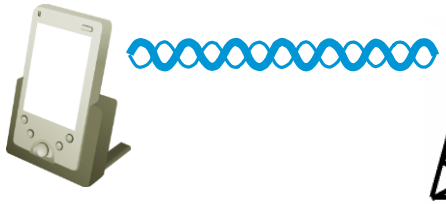



- Flexible platform
- Resilient clustering
- Cisco Easy VPN
- Automatic Cisco VPN

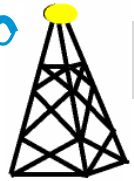
- Cisco IOS SSL VPN
- VPN infrastructure for contemporary applications
- Integrated web-based management

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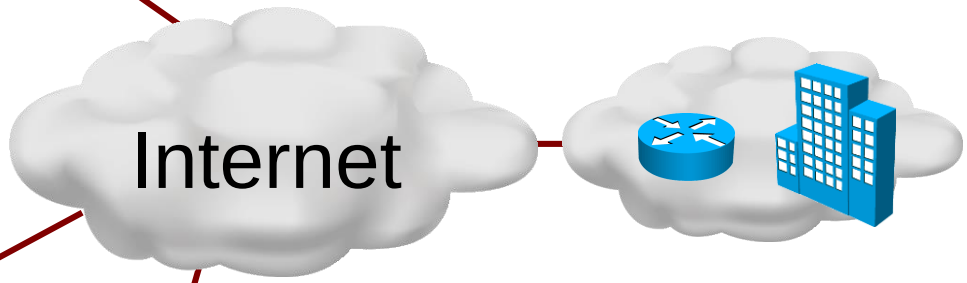
# IPSec Clients

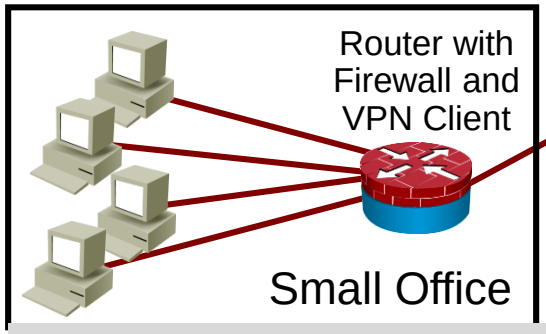


Certicom PDA IPsec VPN Client



A wireless client that is loaded on a pda





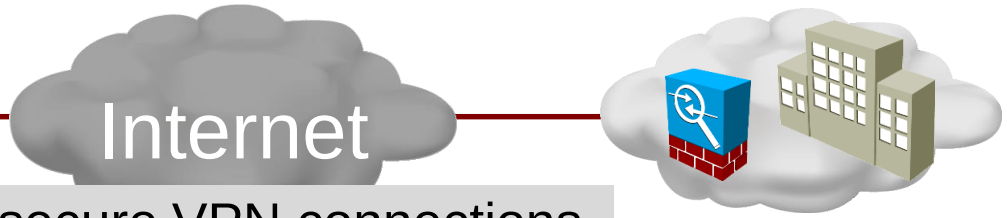

Cisco VPN Software Client

Software loaded on a PC

A network appliance that connects SOHO LANs to the VPN



Cisco AnyConnect VPN Client




Provides remote users with secure VPN connections

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# Hardware Acceleration Modules

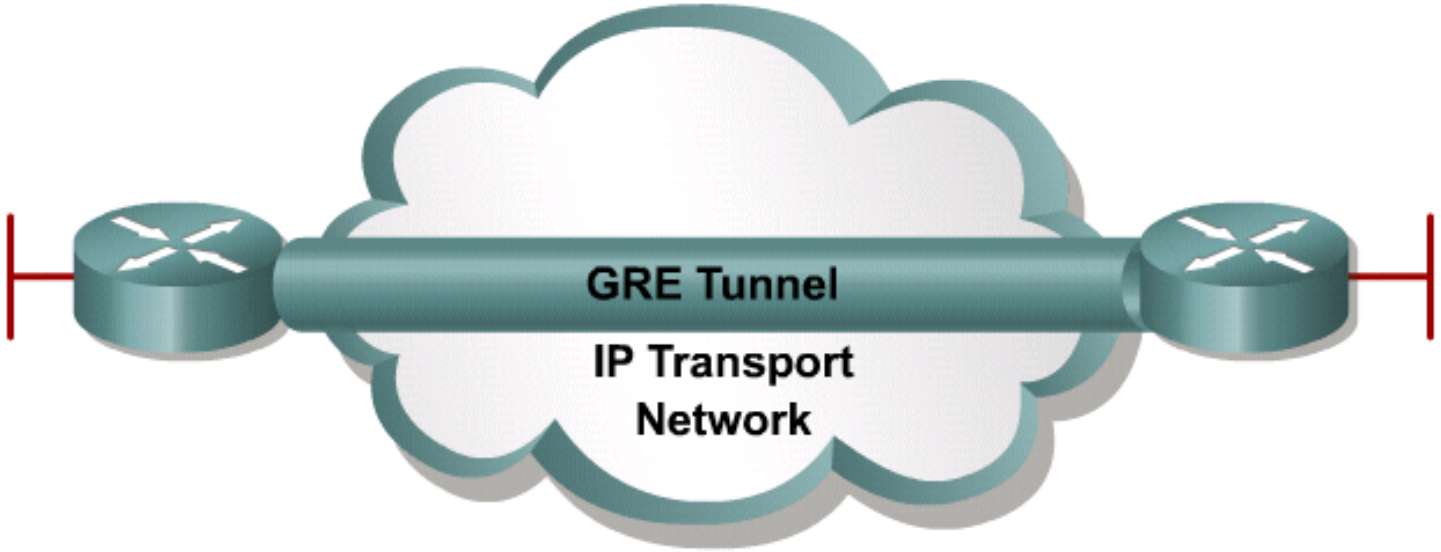
- AIM
- Cisco IPSec VPN Shared Port Adapter (SPA)
- Cisco PIX VPN Accelerator Card+ (VAC+)
- Enhanced Scalable Encryption Processing (SEP-E)



Cisco IPsec VPN SPA

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# GRE VPN Overview



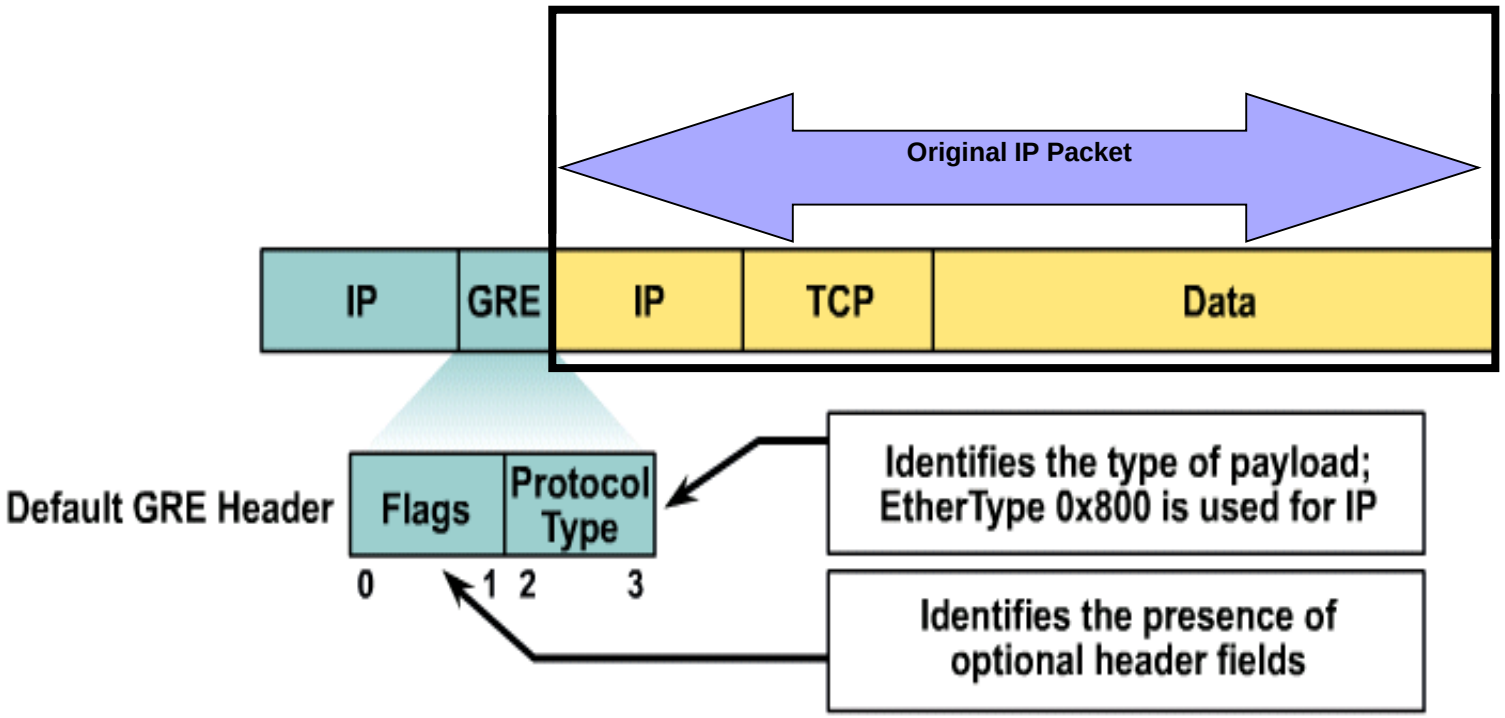

**OSI Layer 3 tunneling protocol:**

- Encapsulates a wide variety of protocol packet types inside IP tunnels
- Creates a virtual point-to-point link to Cisco routers at remote points over an IP internetwork
- Uses IP for transport
- Uses an additional header to support any other OSI Layer 3 protocol as payload (for example, IP, IPX, AppleTalk)

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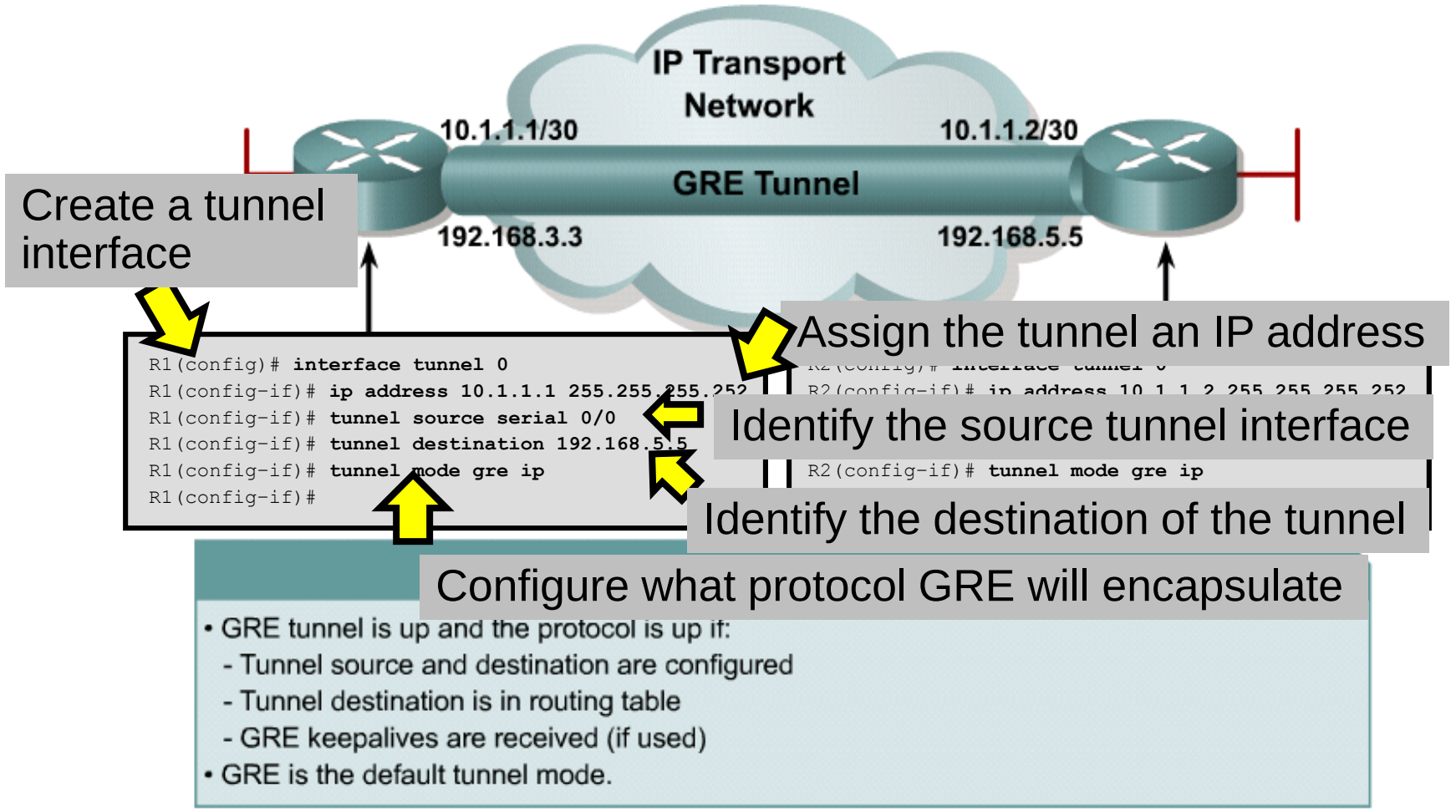
# Encapsulation

## Encapsulated with GRE



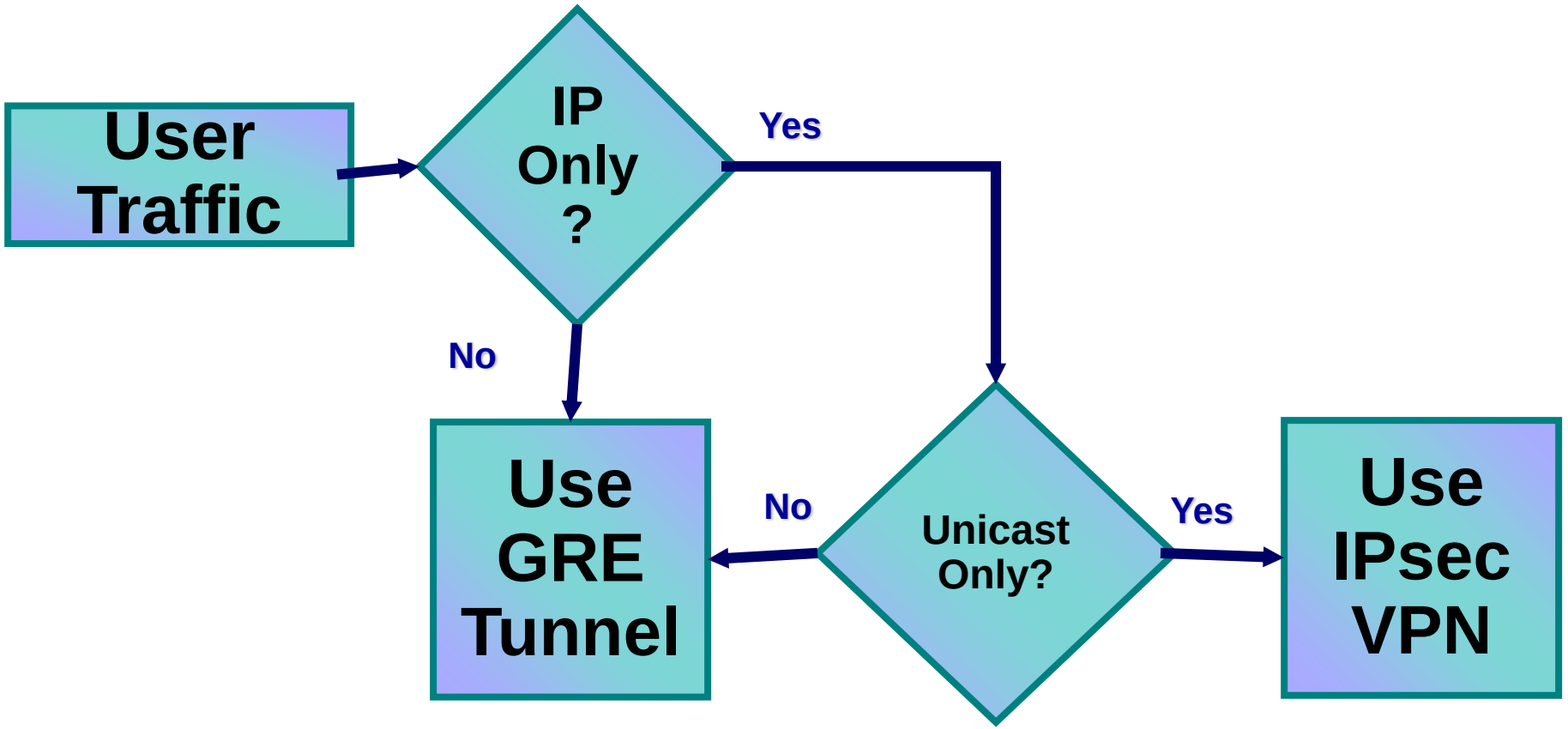

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# Configuring a GRE Tunnel




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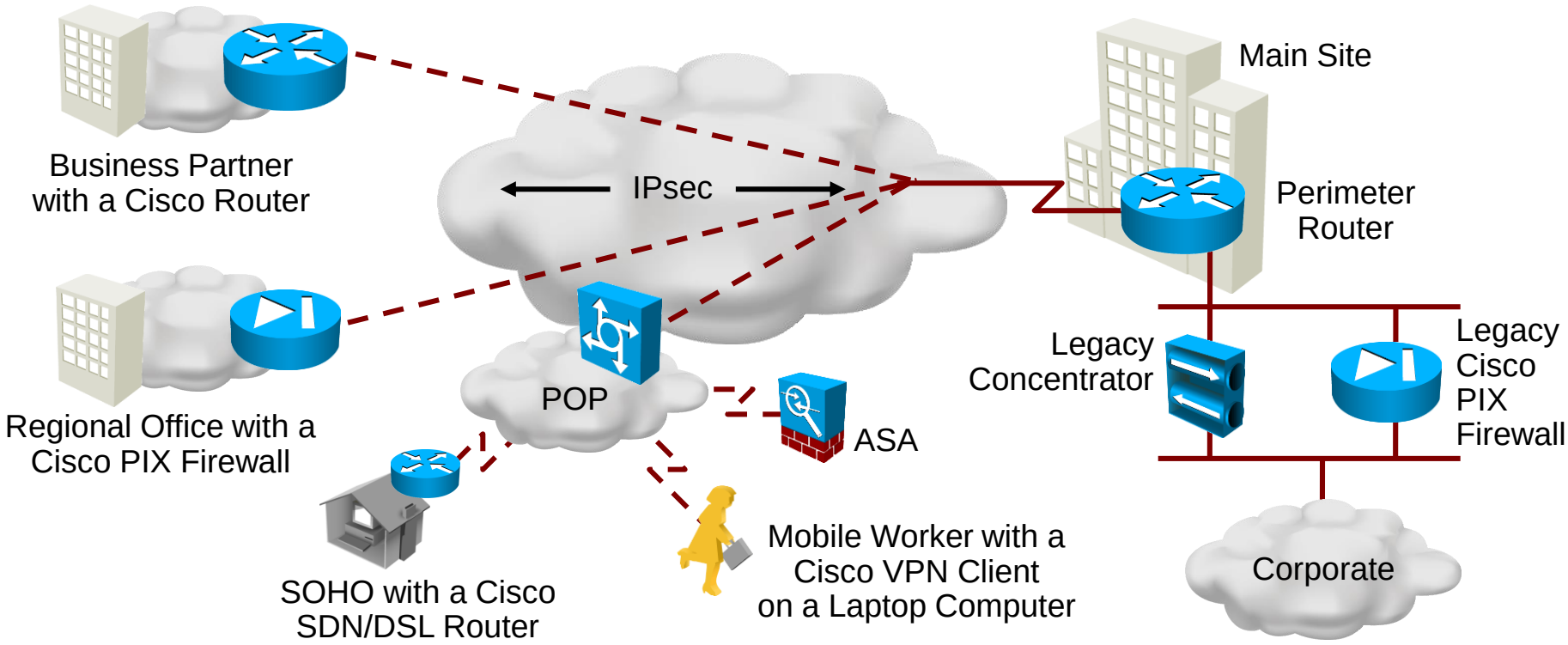
# Using GRE




GRE does not provide encryption

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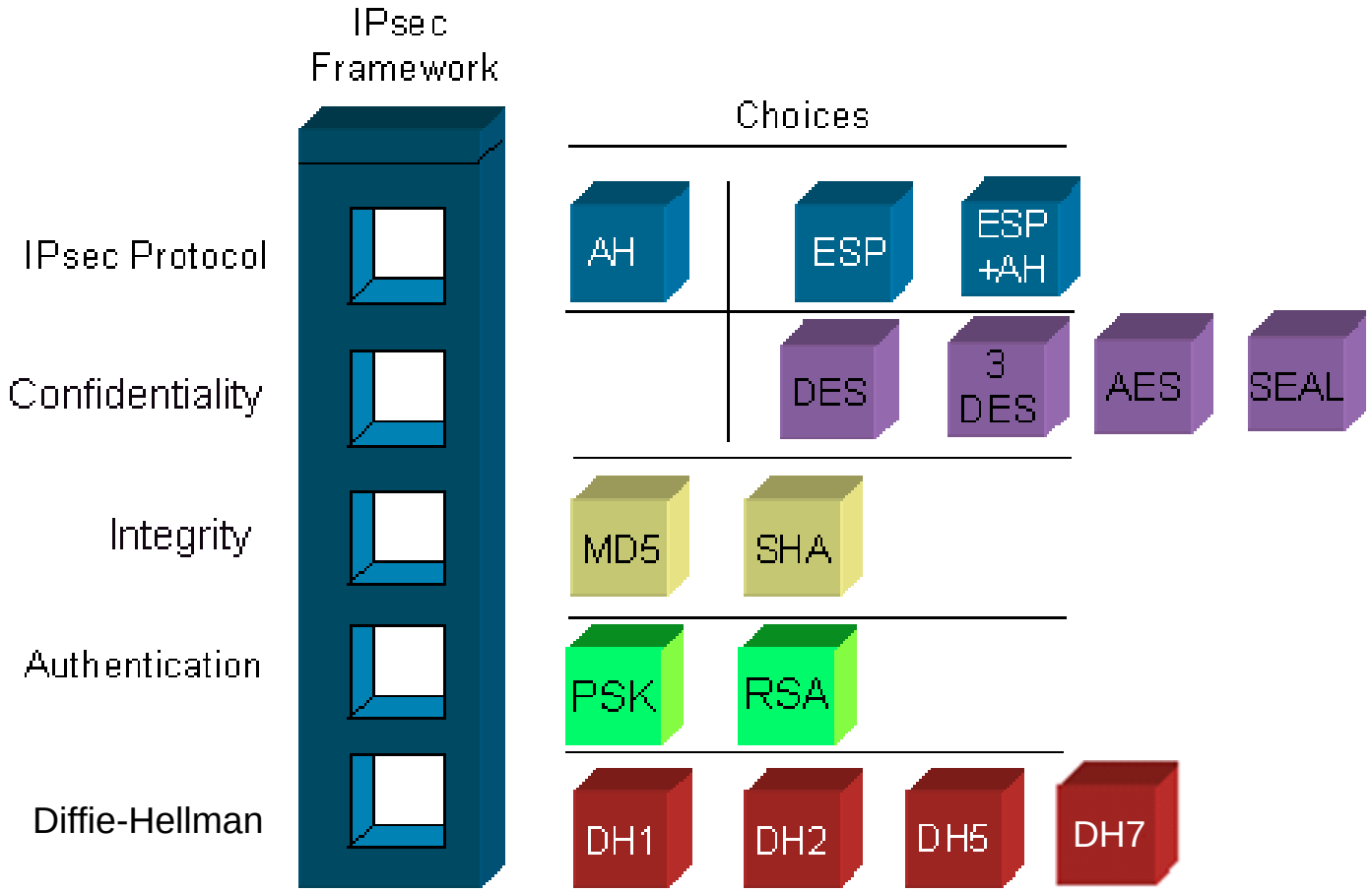
# IPSec Topology




- Works at the network layer, protecting and authenticating IP packets.
  - It is a framework of open standards which is algorithm-independent.
  - It provides data confidentiality, data integrity, and origin authentication.

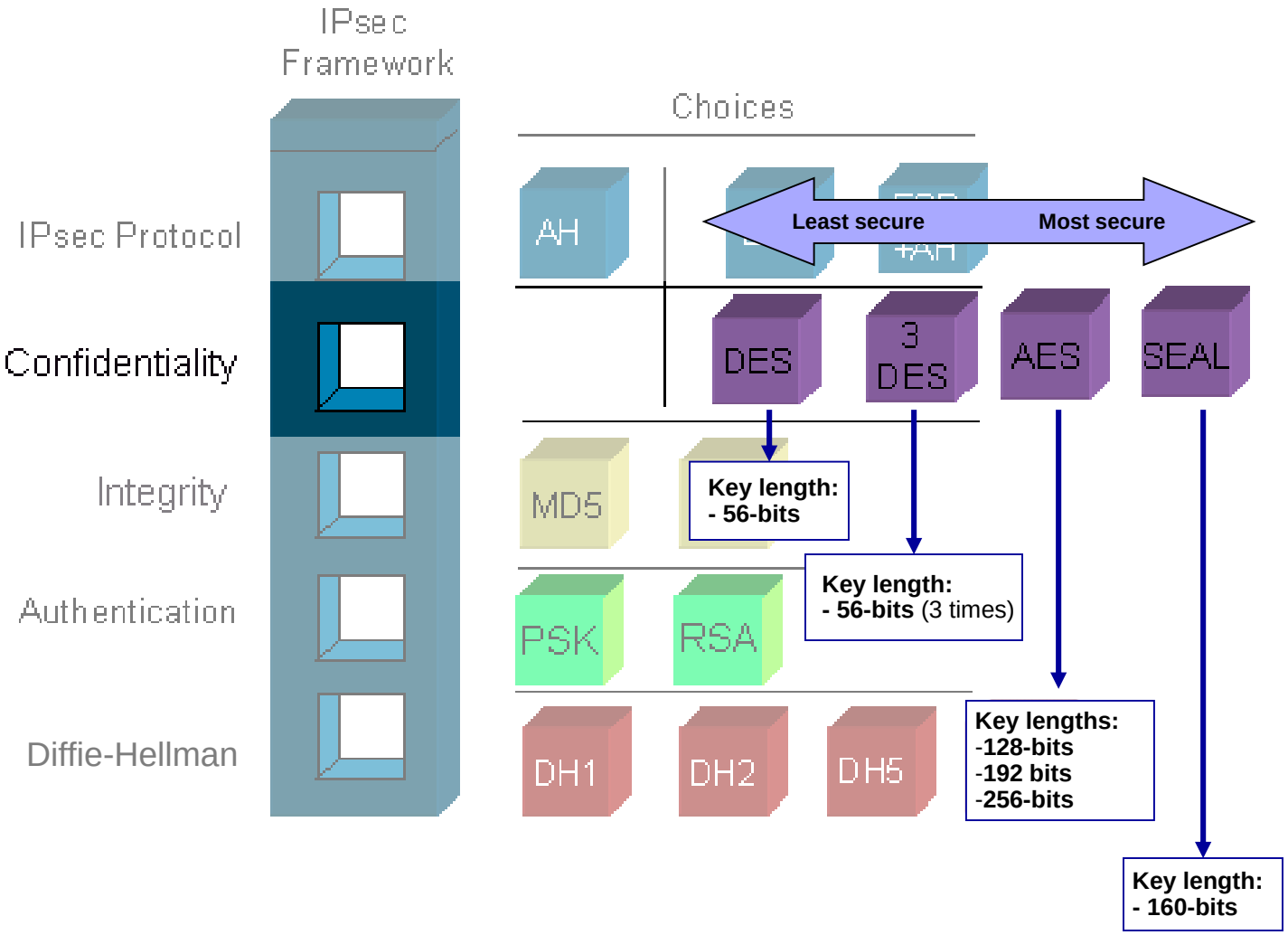
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# IPSec Framework



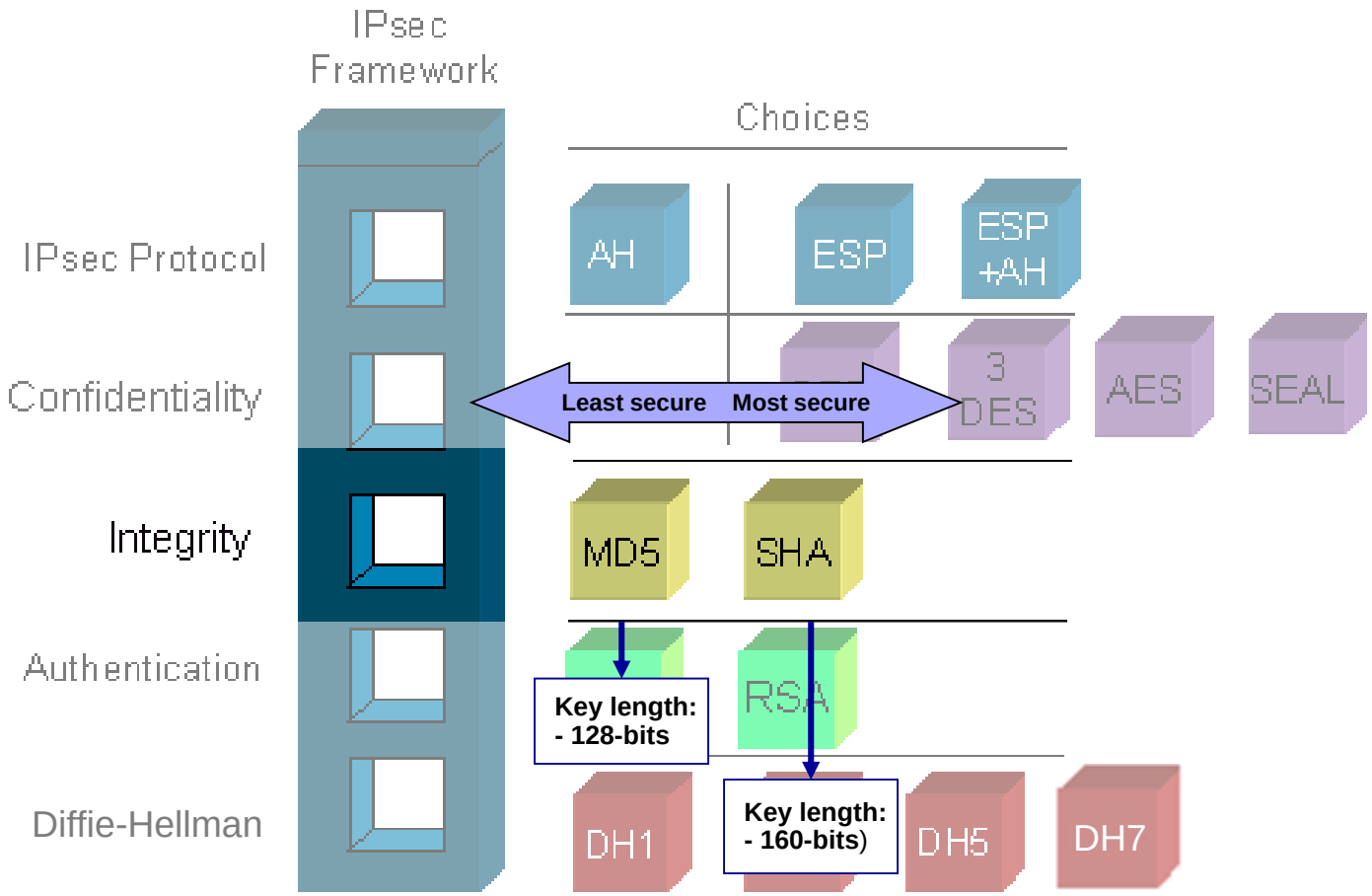

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# Confidentiality



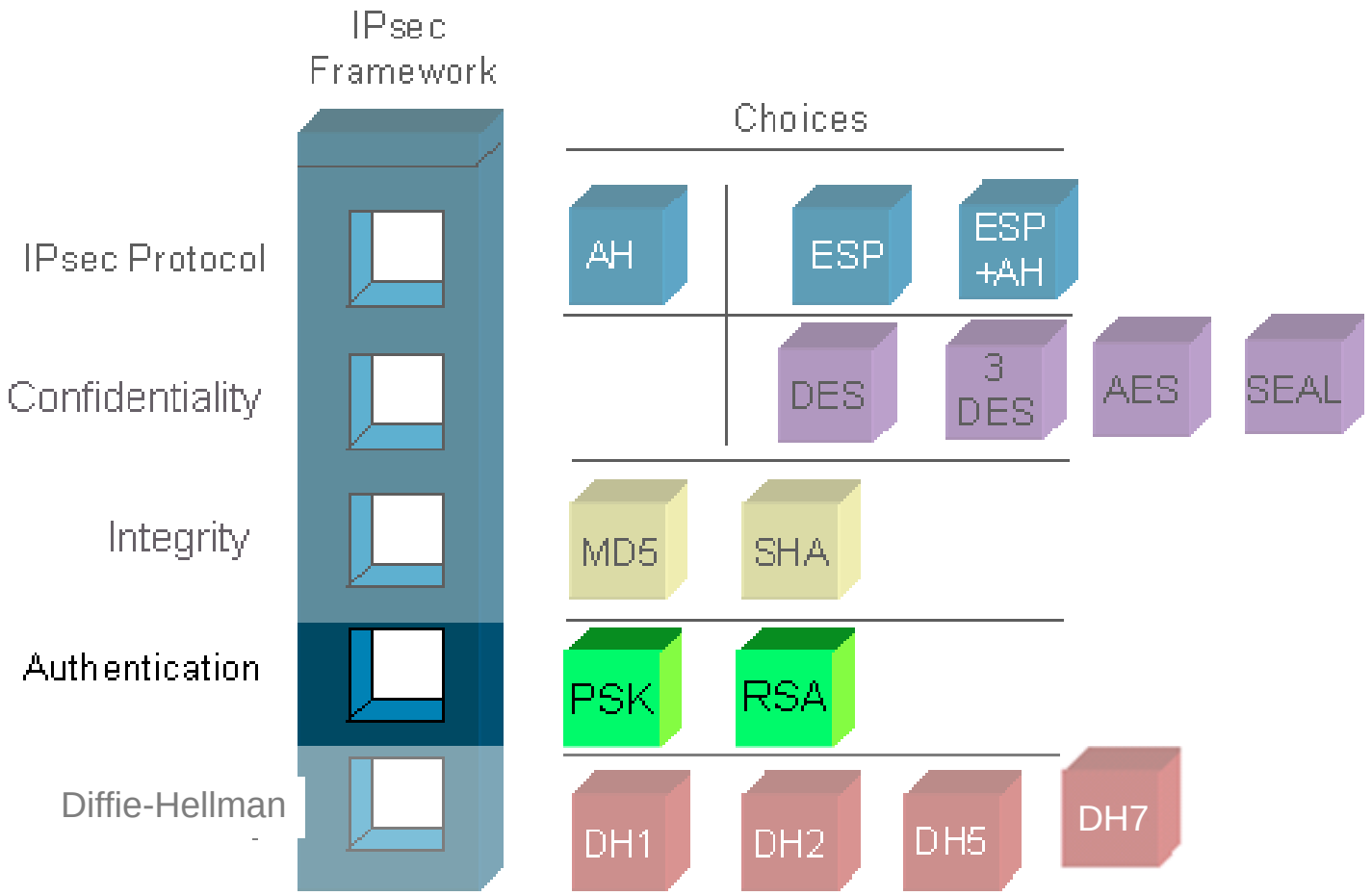

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# Integrity



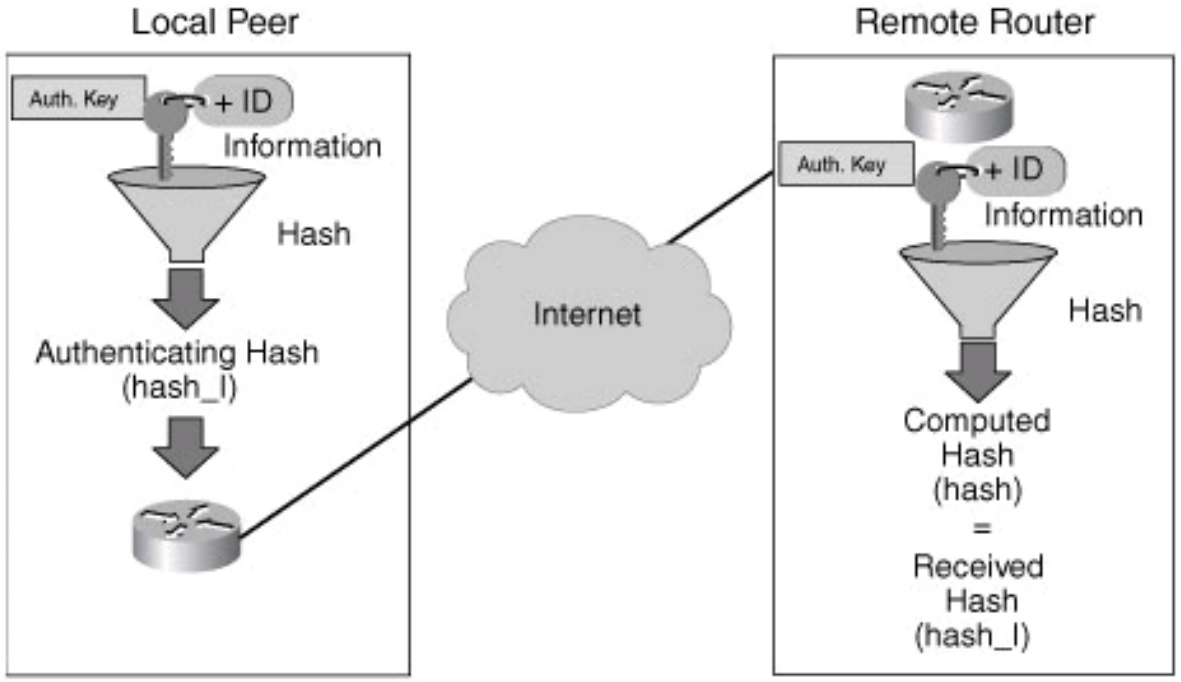

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# Authentication




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# Pre-shared Key (PSK)

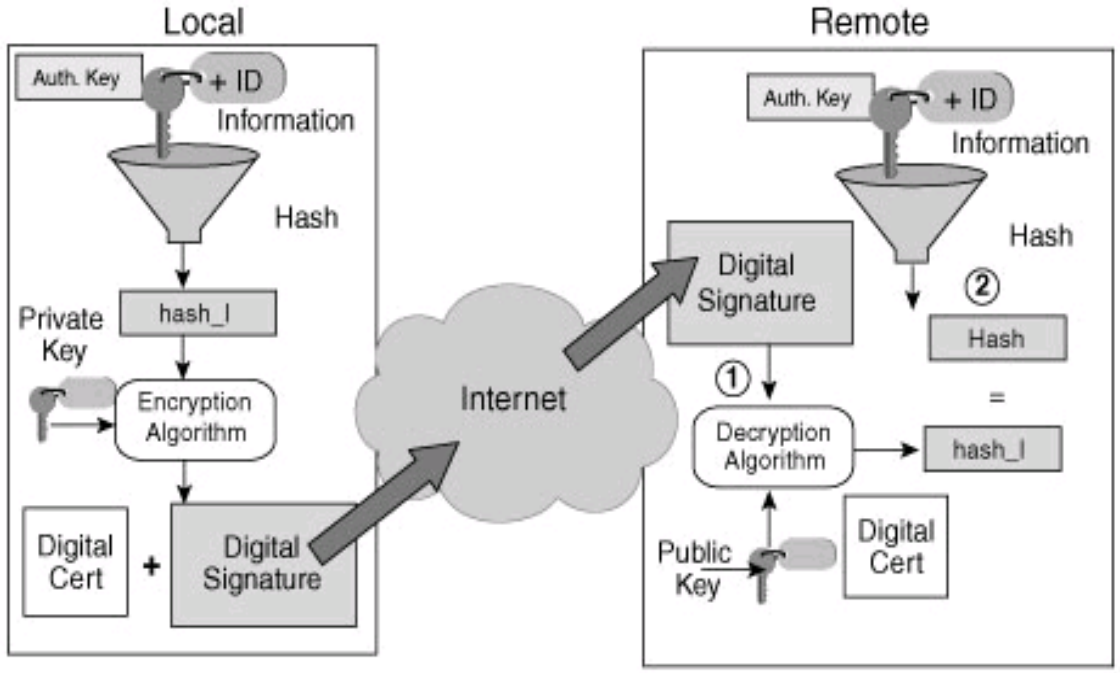



•At the local device, the authentication key and the identity information (device-specific information) are sent through a hash algorithm to form hash_I. One-way authentication is established by sending hash_I to the remote device. If the remote device can independently create the same hash, the local device is authenticated.

• The authentication process continues in the opposite direction. The remote device combines its identity information with the preshared-based authentication key and sends it through the hash algorithm to form hash_R. hash_R is sent to the local device. If the local device can independently create the same hash, the remote device is authenticated.

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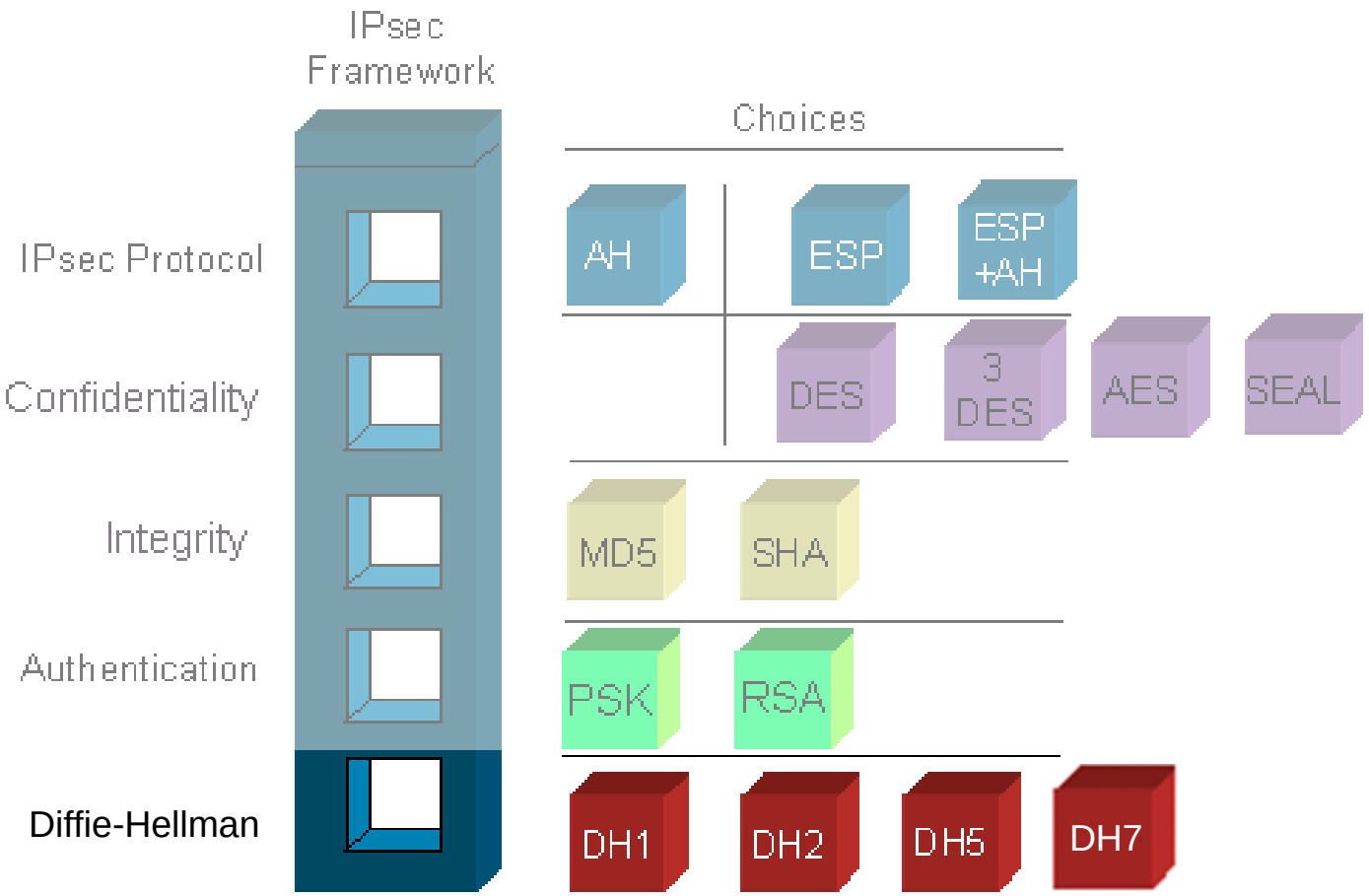
# RSA Signatures




- At the local device, the authentication key and identity information (device-specific information) are sent through the hash algorithm forming hash_I. hash_I is encrypted using the local device's private encryption key creating a digital signature. The digital signature and a digital certificate are forwarded to the remote device. The public encryption key for decrypting the signature is included in the digital certificate. The remote device verifies the digital signature by decrypting it using the public encryption key. The result is hash_I.
- Next, the remote device independently creates hash_I from stored information. If the calculated hash_I equals the decrypted hash_I, the local device is authenticated. After the remote device authenticates the local device, the authentication process begins in the opposite direction and all steps are repeated from the remote device to the local device.

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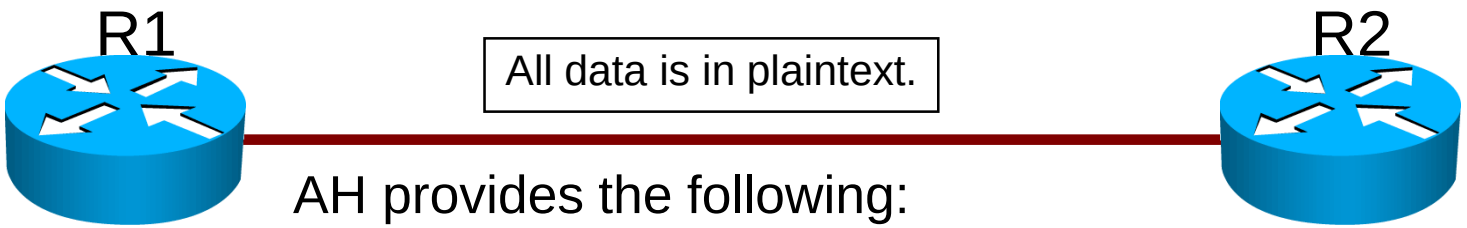
# Secure Key Exchange




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# IPSec Framework Protocols

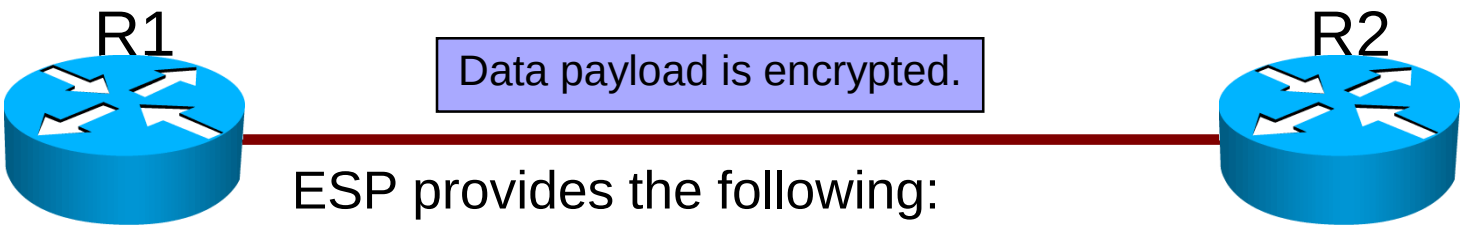
## Authentication Header




AH provides the following:

- Authentication
- Integrity

## Encapsulating Security Payload




ESP provides the following:

- **Encryption**
- Authentication
- Integrity

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# Authentication Header

1. The IP Header and data payload are hashed

IP Header + Data + Key

Hash

Authentication Data
(**00ABCDEF**)

IP HDR | **AH** | Data

2. The hash builds a new AH header which is prepended to the original packet

3. The new packet is transmitted to the IPSec peer router

R2

IP HDR | **AH** | Data

IP Header + Data + Key

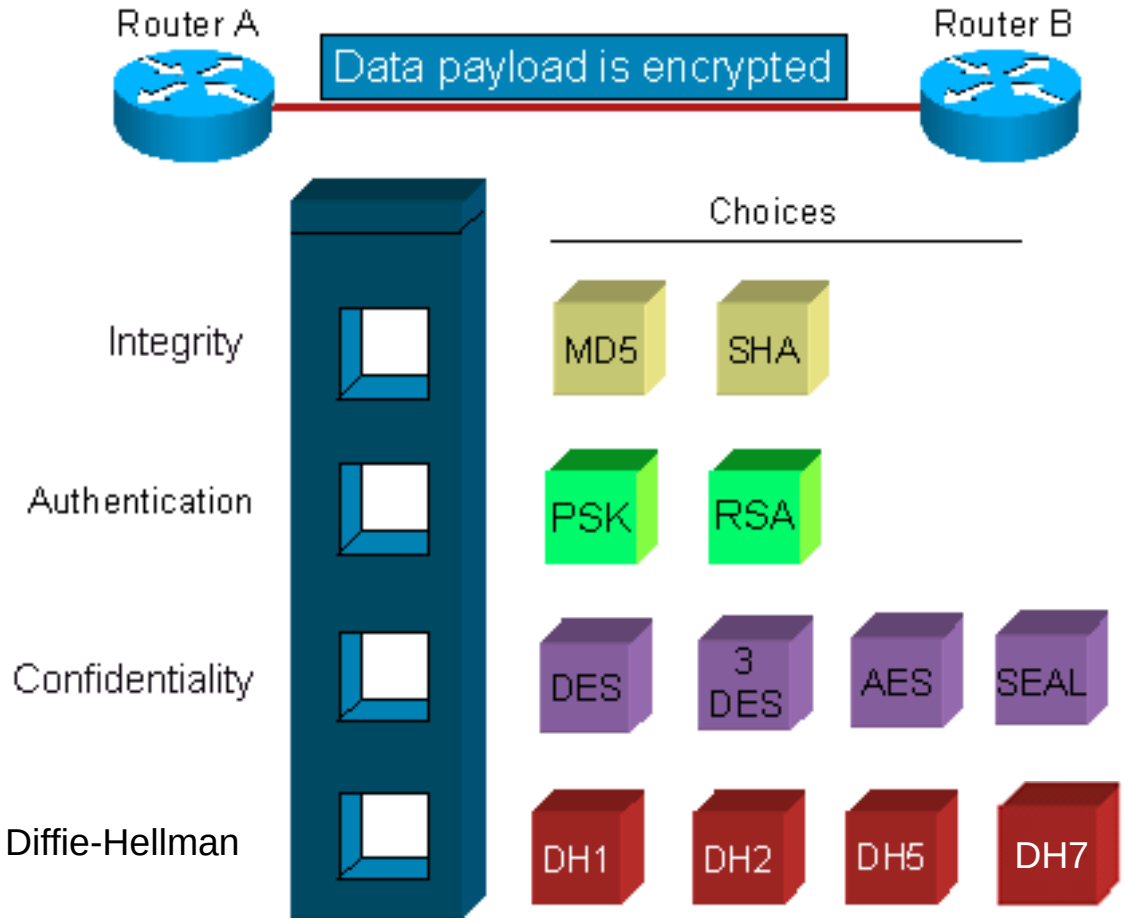
Hash

Recomputed Hash (**00ABCDEF**) = Received Hash (**00ABCDEF**)

4. The peer router hashes the IP header and data payload, extracts the transmitted hash and compares

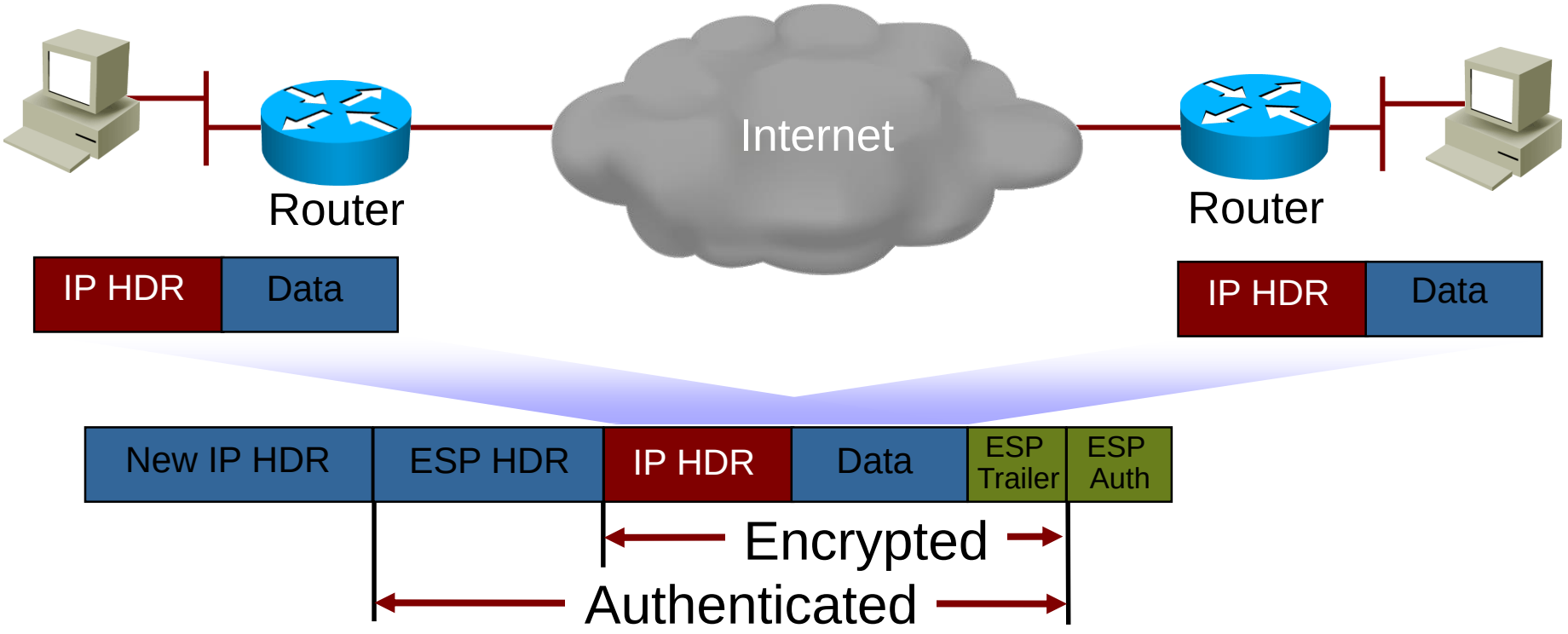
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# ESP




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# Function of ESP




- Provides confidentiality with encryption
- Provides integrity with authentication

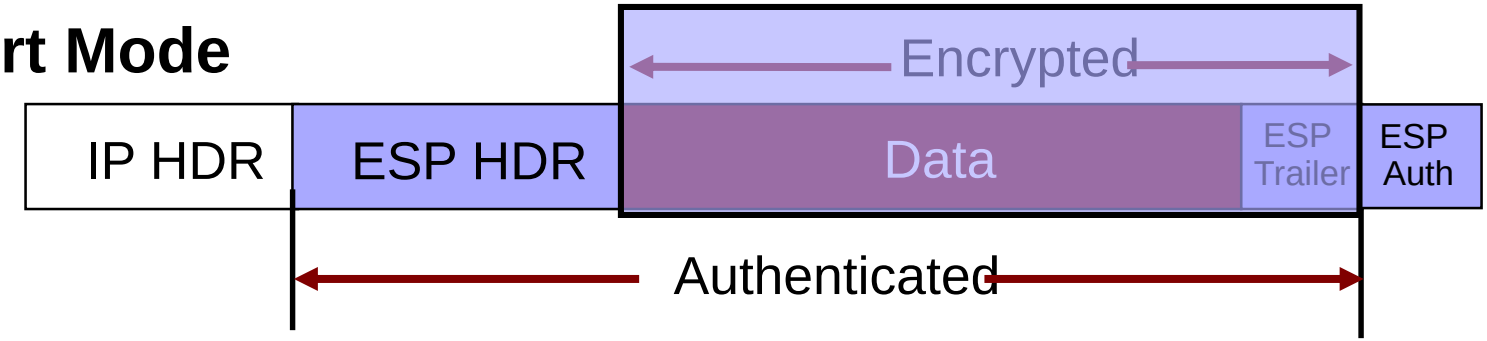
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# Mode Types

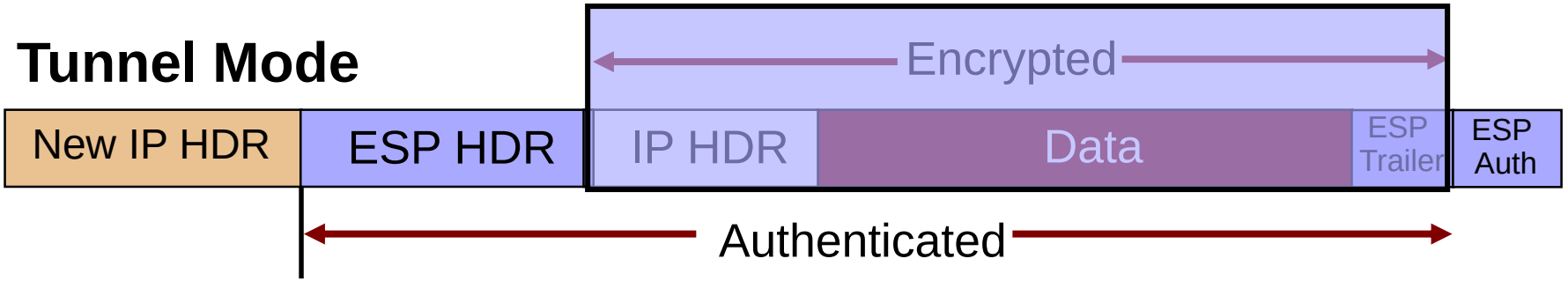



Original data prior to selection of IPSec protocol mode

**Transport Mode**

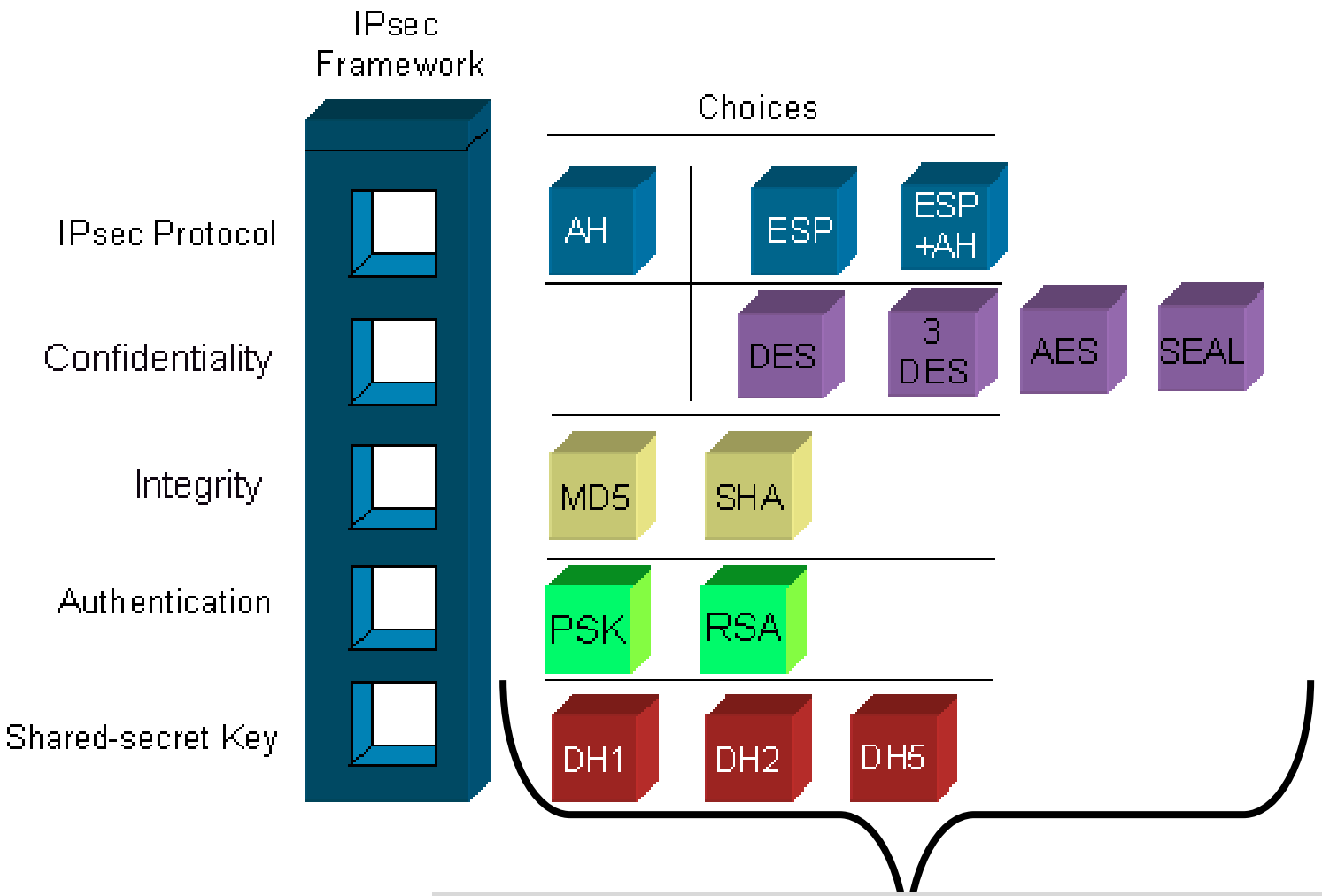



**Tunnel Mode**




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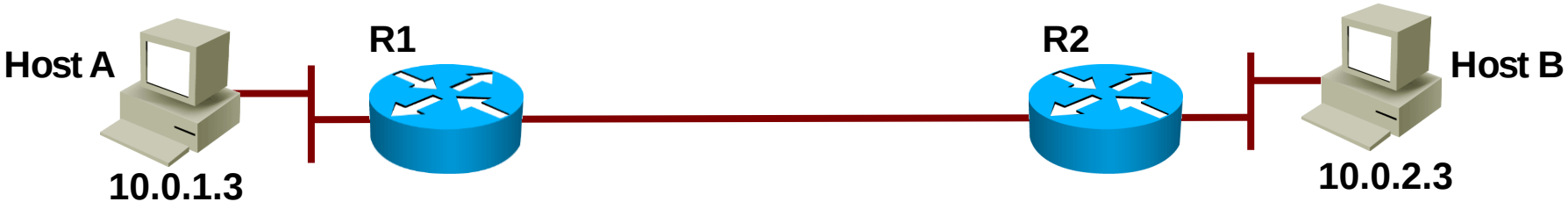
# Security Associations



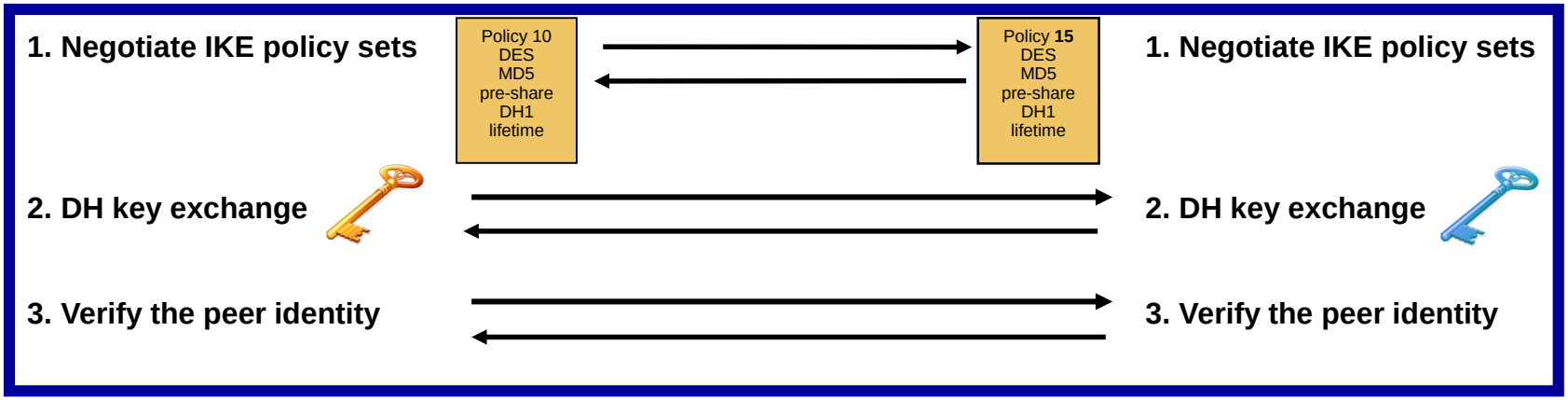

IPSec parameters are configured using IKE

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# IKE Phases




## IKE Phase 1 Exchange

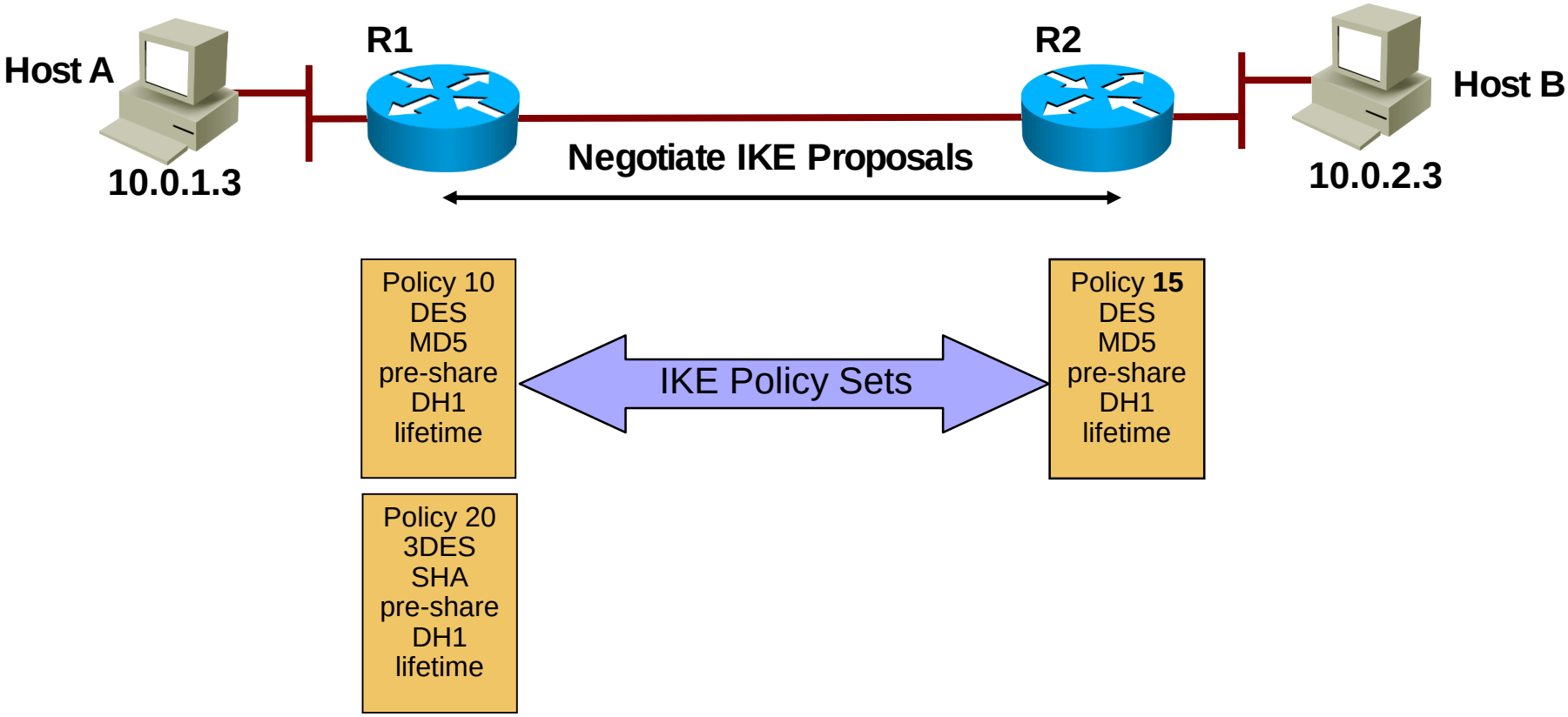



## IKE Phase 2 Exchange




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# IKE Phase 1 – First Exchange

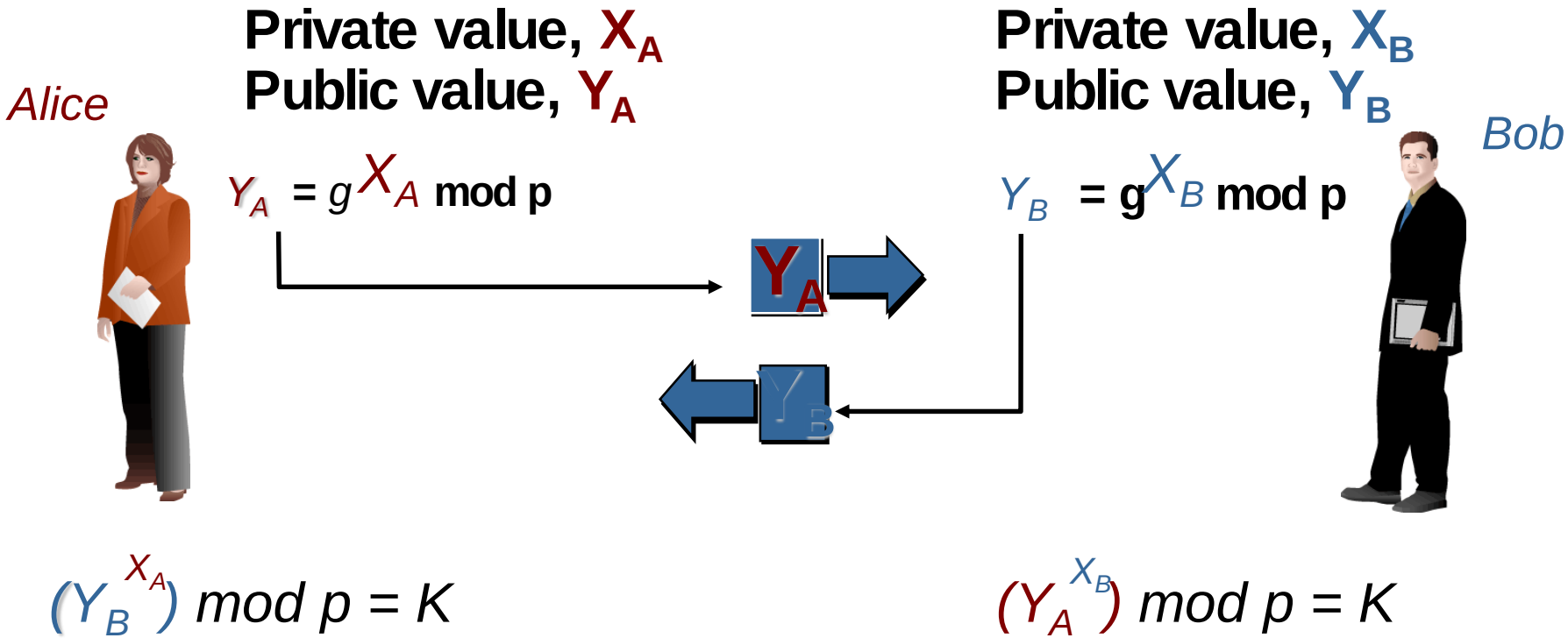



Negotiates matching IKE policies to protect IKE exchange

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# IKE Phase 1 – Second Exchange

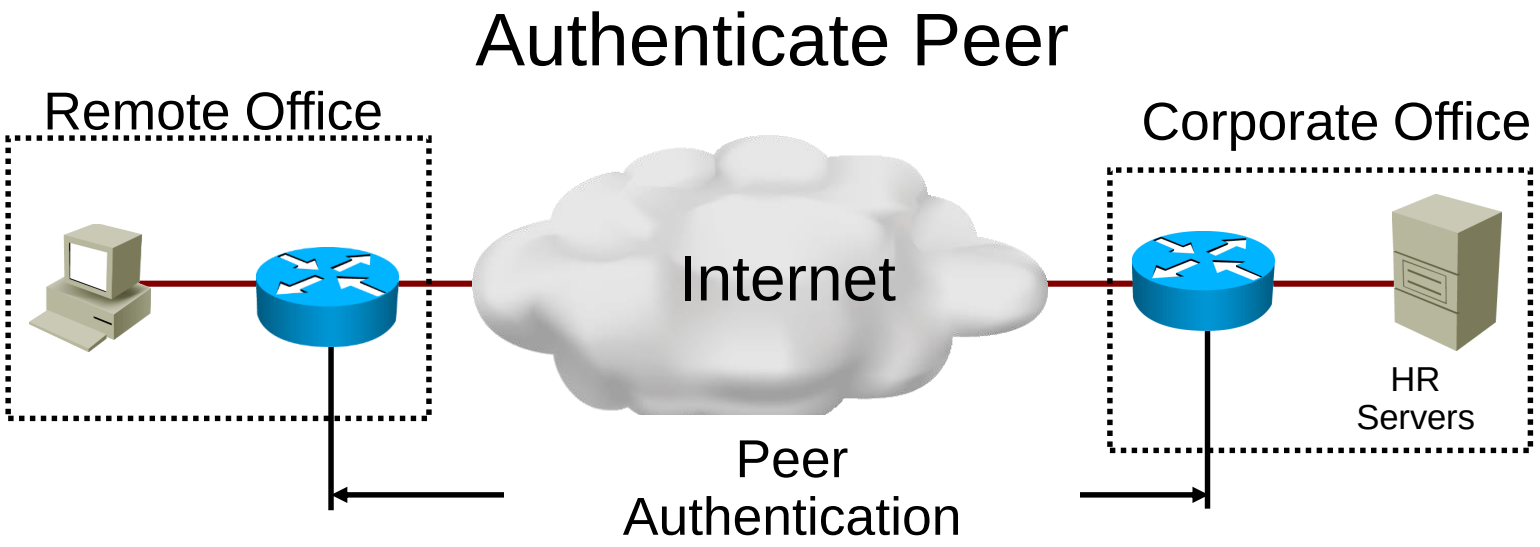
## Establish DH Key




A DH exchange is performed to establish keying material.

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# IKE Phase 1 – Third Exchange



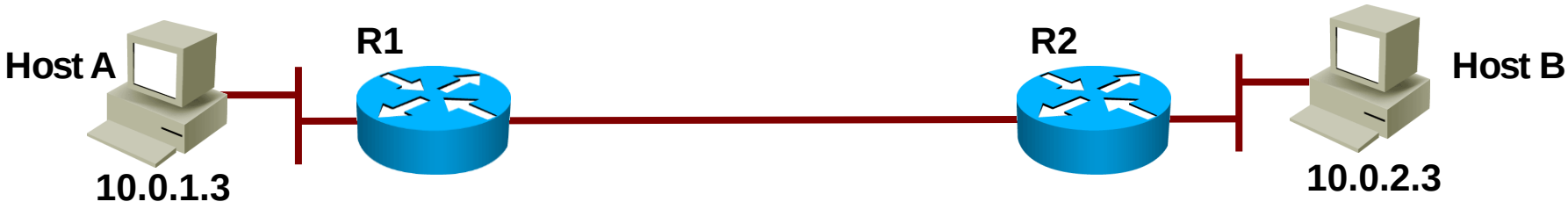

## Peer authentication methods

- PSKs
- RSA signatures
- RSA encrypted nonces

A bidirectional IKE SA is now established.

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# IKE Phase 1 – Aggressive Mode




## IKE Phase 1 Aggressive Mode Exchange

1.Send IKE policy set and R1's DH key

Policy 10
DES
MD5
pre-share
DH1
lifetime

Policy **15**
DES
MD5
pre-share
DH1
lifetime

2. Confirm IKE policy set, calculate shared secret and send R2's DH key

3.Calculate shared secret, verify peer identify, and confirm with peer

4. Authenticate peer and begin Phase 2.

## IKE Phase 2 Exchange

Negotiate IPsec policy ⇄ Negotiate IPsec policy

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# IKE Phase 2



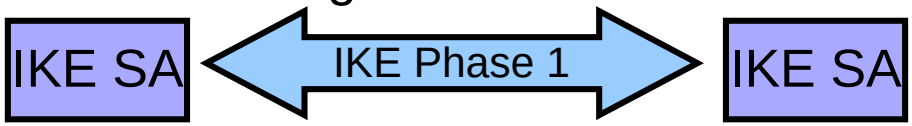

- IKE negotiates matching IPsec policies.
- Upon completion, unidirectional IPsec Security Associations(SA) are established for each protocol and algorithm combination.

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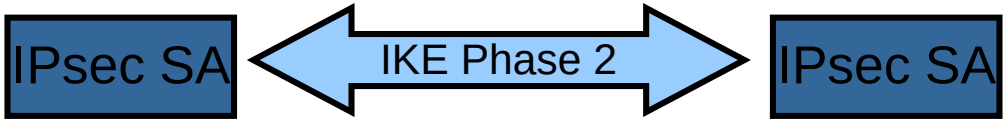
# IPSec VPN Negotiation




1. Host A sends interesting traffic to Host B.
2. R1 and R2 negotiate an IKE Phase 1 session.




3. R1 and R2 negotiate an IKE Phase 2 session.




4. Information is exchanged via IPsec tunnel.




5. The IPsec tunnel is terminated.

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# Configuring IPsec

## Tasks to Configure IPsec:

Task 1: Ensure that ACLs are compatible with IPsec.

Task 2: Create ISAKMP (IKE) policy.

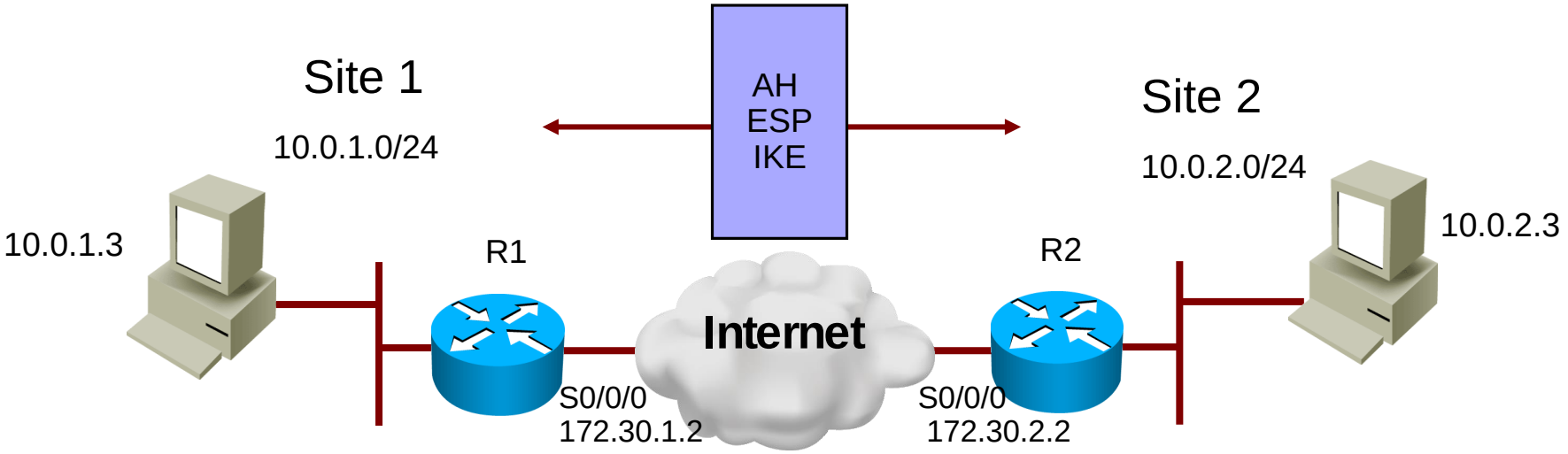
Task 3: Configure IPsec transform set.

Task 4: Create a crypto ACL.

Task 5: Create and apply the crypto map.

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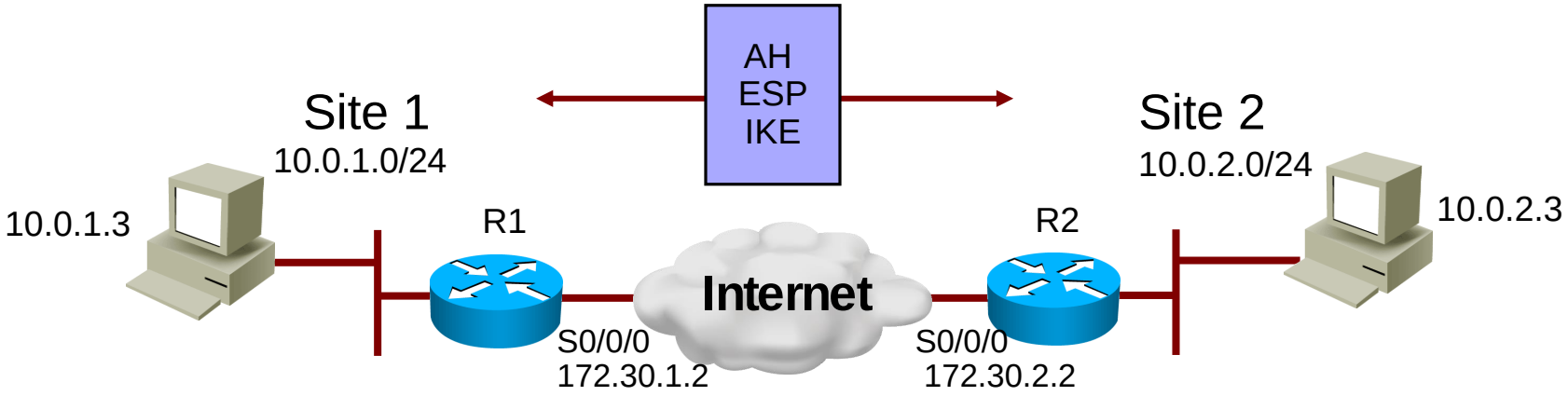
# Task 1
# Configure Compatible ACLs




- Ensure that protocols 50 (ESP), 51 (AH) and UDP port 500 (ISAKMP) traffic are not blocked by incoming ACLs on interfaces used by IPsec.

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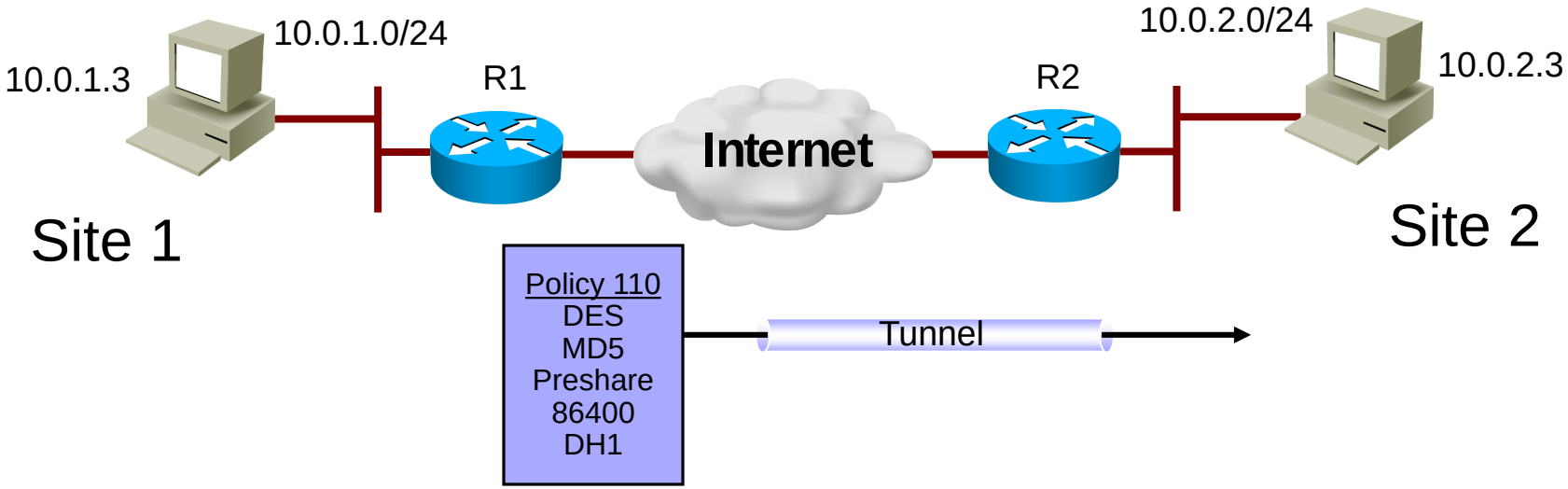
# Permitting Traffic




```
R1(config)# access-list 102 permit ahp host 172.30.2.2 host 172.30.1.2
R1(config)# access-list 102 permit esp host 172.30.2.2 host 172.30.1.2
R1(config)# access-list 102 permit udp host 172.30.2.2 host 172.30.1.2 eq isakmp
R1(config)#
R1(config)# interface Serial0/0/0
R1(config-if)# ip address 172.30.1.2 255.255.255.0
R1(config-if)# ip access-group 102 in
!
R1(config)# exit
R1#
R1# show access-lists
    access-list 102 permit ahp host 172.30.2.2 host 172.30.1.2
    access-list 102 permit esp host 172.30.2.2 host 172.30.1.2
    access-list 102 permit udp host 172.30.2.2 host 172.30.1.2 eq isakmp
R1#
```

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# Task 2
# Configure IKE




**router(config)#**

**crypto isakmp policy *priority***

Defines the parameters within the IKE policy

```
R1(config)# crypto isakmp policy 110
R1(config-isakmp)# authentication pre-share
R1(config-isakmp)# encryption des
R1(config-isakmp)# group 1
R1(config-isakmp)# hash md5
R1(config-isakmp)# lifetime 86400
```

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# ISAKMP Parameters

| Parameter | Keyword | Accepted Values | Default Value | Description |
|---|---|---|---|---|
| **encryption** | **des**<br>**3des**<br>**aes**<br>**aes 192**<br>**aes 256** | 56-bit Data Encryption Standard<br>Triple DES<br>128-bit AES<br>192-bit AES<br>256-bit AES | **des** | Message encryption algorithm |
| **hash** | **sha**<br>**md5** | SHA-1 (HMAC variant)<br>MD5 (HMAC variant) | **sha** | Message integrity (Hash) algorithm |
| **authentication** | **pre-share**<br>**rsa-encr**<br>**rsa-sig** | preshared keys<br>RSA encrypted nonces<br>RSA signatures | **rsa-sig** | Peer authentication method |
| **group** | **1**<br>**2**<br>**5** | 768-bit Diffie-Hellman (DH)<br>1024-bit DH<br>1536-bit DH | **1** | Key exchange parameters (DH group identifier) |
| **lifetime** | *seconds* | Can specify any number of seconds | 86,400 sec (one day) | ISAKMP-established SA lifetime |

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# Multiple Policies




**R1(config)#**

```
crypto isakmp policy 100
  hash md5
  authentication pre-share
!
crypto isakmp policy 200
  hash sha
  authentication rsa-sig
!
crypto isakmp policy 300
  hash md5
  authentication pre-share
```

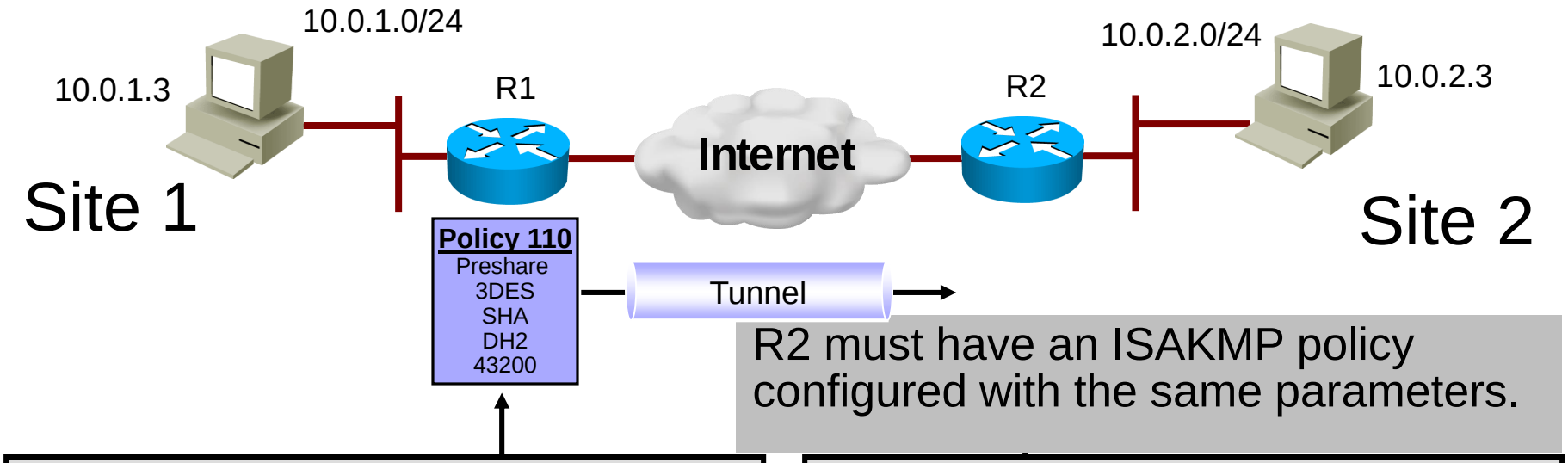
**R2(config)#**

```
crypto isakmp policy 100
  hash md5
  authentication pre-share
!
crypto isakmp policy 200
  hash sha
  authentication rsa-sig
!
crypto isakmp policy 300
  hash md5
  authentication rsa-sig
```

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# Policy Negotiations

R1 attempts to establish a VPN tunnel with R2 and sends its IKE policy parameters




R2 must have an ISAKMP policy configured with the same parameters.

```
R1(config)# crypto isakmp policy 110
R1(config-isakmp)# authentication pre-share
R1(config-isakmp)# encryption 3des
R1(config-isakmp)# group 2
R1(config-isakmp)# hash sha
R1(config-isakmp)# lifetime 43200
```

```
R2(config)# crypto isakmp policy 100
R2(config-isakmp)# authentication pre-share
R2(config-isakmp)# encryption 3des
R2(config-isakmp)# group 2
R2(config-isakmp)# hash sha
R2(config-isakmp)# lifetime 43200
```

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# Crypto ISAKMP Key

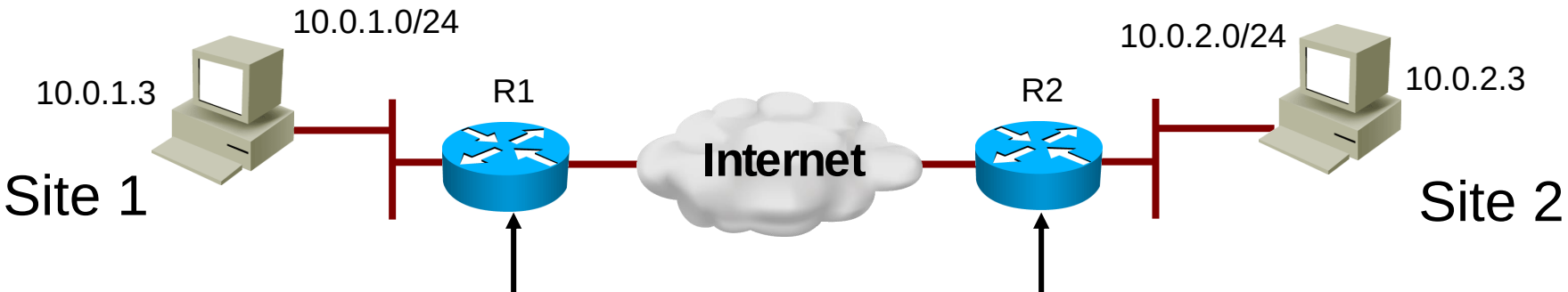
```
router(config)#
crypto isakmp key keystring address peer-address
router(config)#
crypto isakmp key keystring hostname hostname
```

| Parameter | Description |
|---|---|
| *keystring* | This parameter specifies the PSK. Use any combination of alphanumeric characters up to 128 bytes. This PSK must be identical on both peers. |
| *peer-address* | This parameter specifies the IP address of the remote peer. |
| *hostname* | This parameter specifies the hostname of the remote peer. This is the peer hostname concatenated with its domain name (for example, myhost.domain.com). |

- The *peer-address* or *peer-hostname* can be used, but must be used consistently between peers.
- If the *peer-hostname* is used, then the **crypto isakmp identity hostname** command must also be configured.

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# Sample Configuration




```
R1(config)# crypto isakmp policy 110
R1(config-isakmp)# authentication pre-share
R1(config-isakmp)# encryption 3des
R1(config-isakmp)# group 2
R1(config-isakmp)# hash sha
R1(config-isakmp)# lifetime 43200
R1(config-isakmp)# exit
R1(config)# crypto isakmp key cisco123 address 172.30.2.2
R1(config)#
```

**Note:**

- The keystring **cisco1234** matches.
- The address identity method is specified.
- The ISAKMP policies are compatible.
- Default values do not have to be configured.

```
R2(config)# crypto isakmp policy 110
R2(config-isakmp)# authentication pre-share
R2(config-isakmp)# encryption 3des
R2(config-isakmp)# group 2
R2(config-isakmp)# hash sha
R2(config-isakmp)# lifetime 43200
R2(config-isakmp)# exit
R2(config)# crypto isakmp key cisco123 address 172.30.1.2
R2(config)#
```

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# Task 3
# Configure the Transform Set

```
router(config)#
crypto ipsec transform-set transform-set-name
   transform1 [transform2] [transform3]]
```

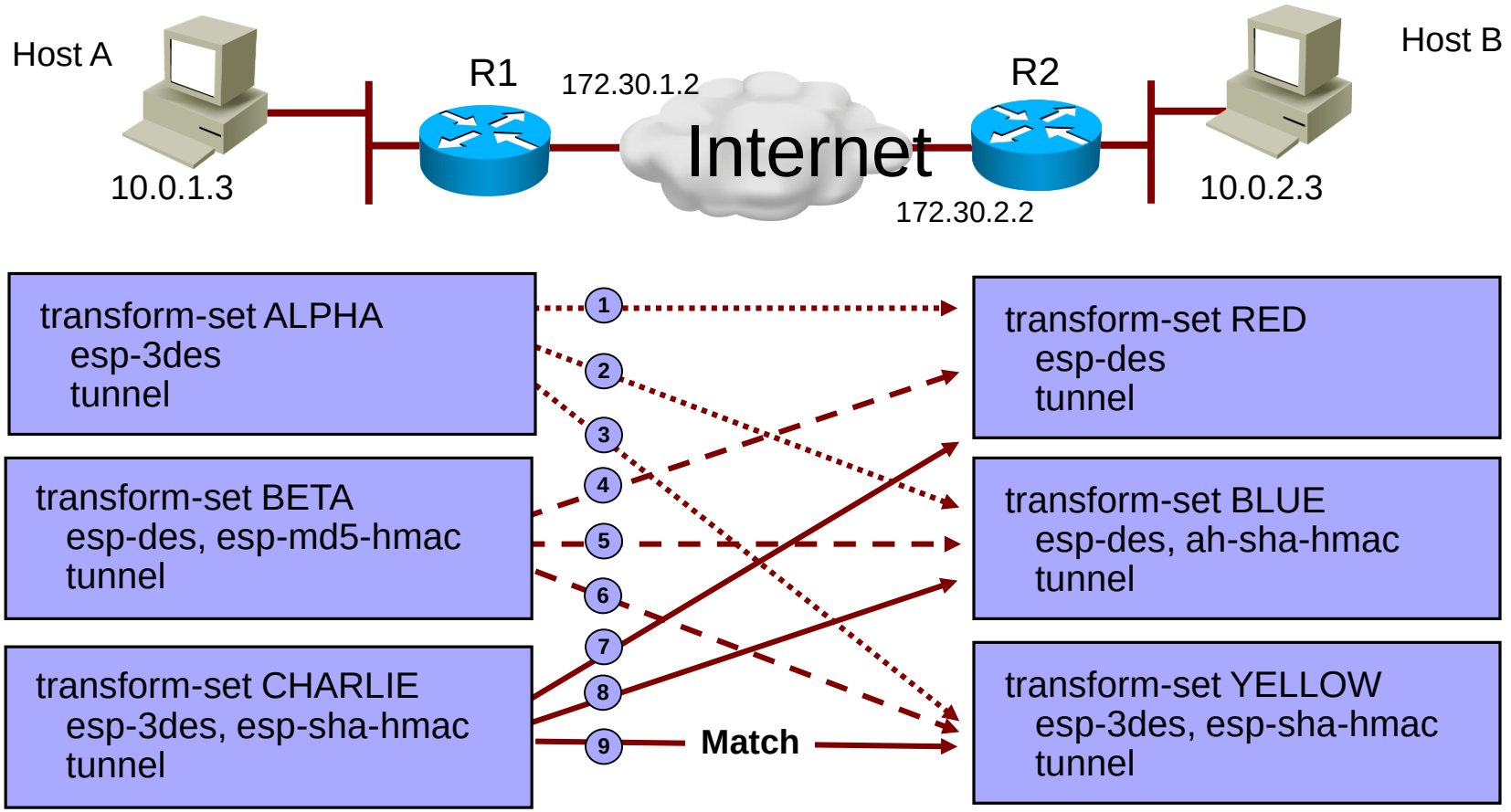
**crypto ipsec transform-set Parameters**

| Command | Description |
|---|---|
| *transform-set-name* | This parameter specifies the name of the transform set to create (or modify). |
| *transform1*, *transform2*, *transform3* | Type of transform set. You may specify up to four "transforms": one Authentication Header (AH), one Encapsulating Security Payload (ESP) encryption, one ESP authentication. These transforms define the IP Security (IPSec) security protocols and algorithms. |

A transform set is a combination of IPsec transforms that enact a security policy for traffic.

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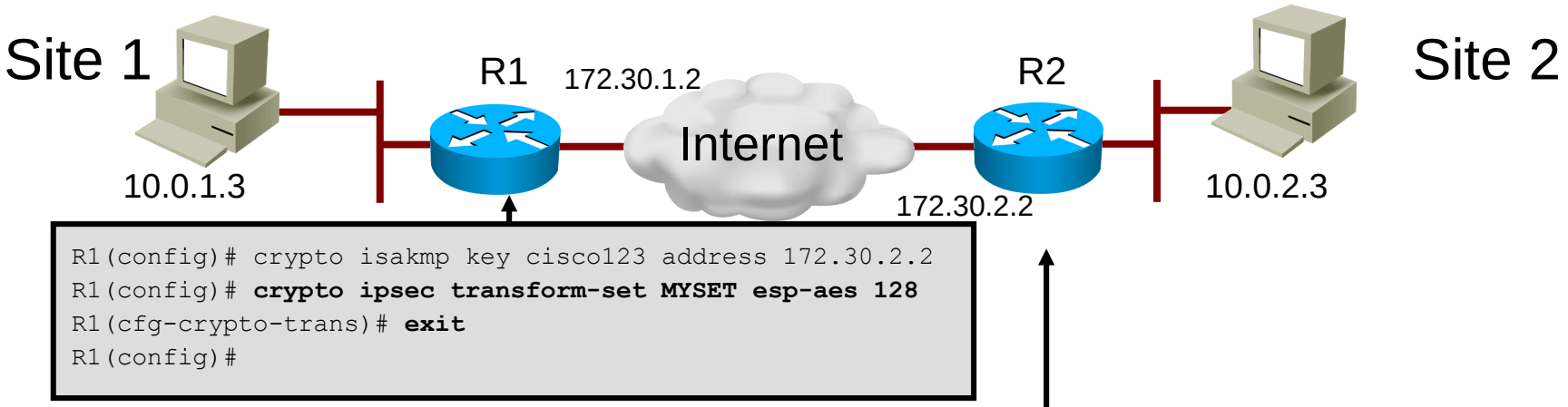
# Transform Sets




- Transform sets are negotiated during IKE Phase 2.
- The 9th attempt found matching transform sets (CHARLIE - YELLOW).

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# Sample Configuration

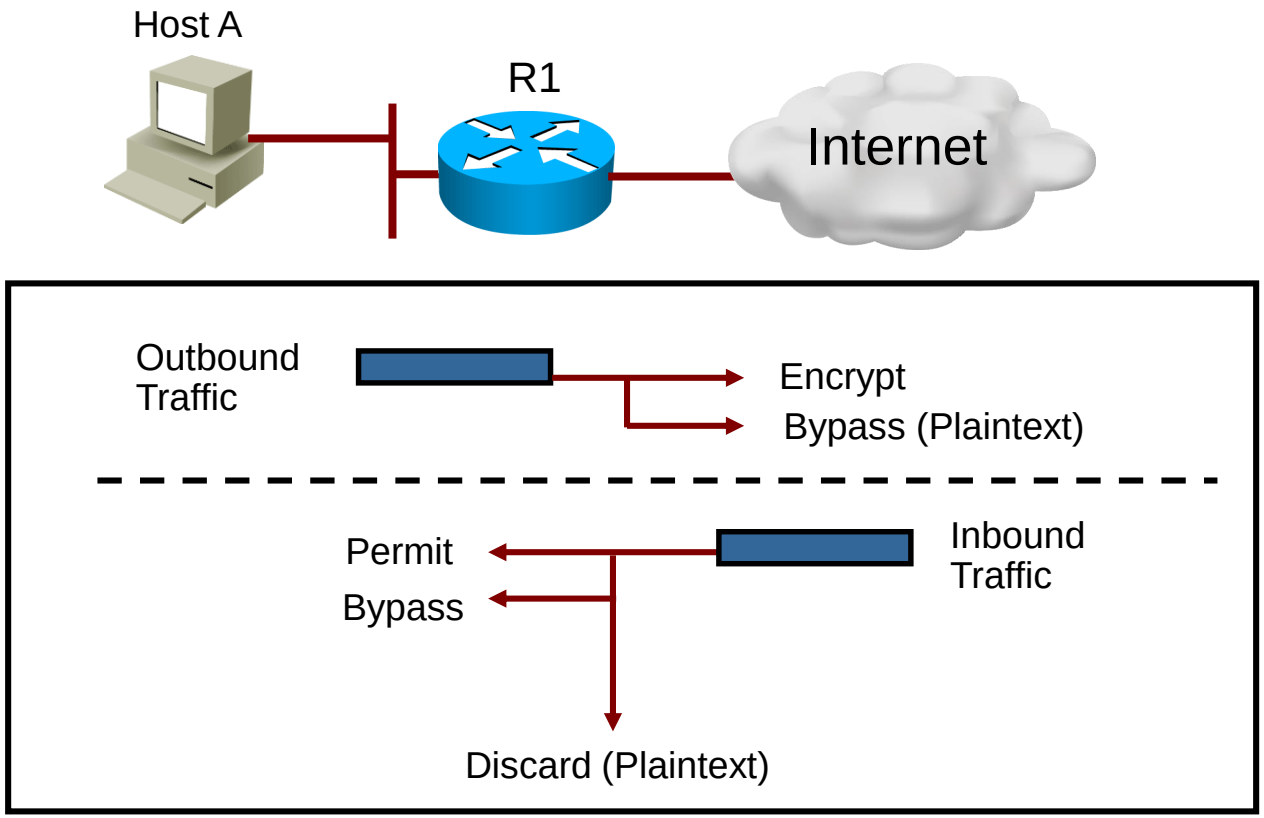



**Note:**

- Peers must share the same transform set settings.
- Names are only locally significant.

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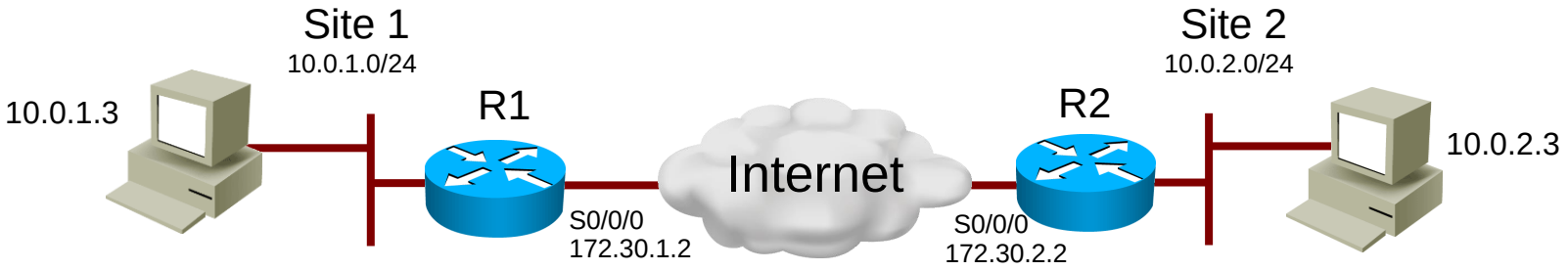
# Task 4
# Configure the Crypto ACLs




- Outbound indicates the data flow to be protected by IPsec.
- Inbound filters and discards traffic that should have been protected by IPsec.

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# Command Syntax




**router(config)#**

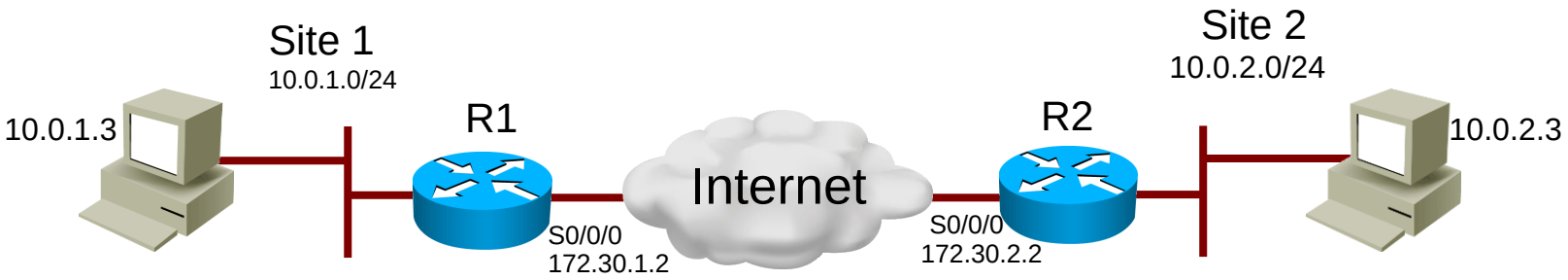
```
access-list access-list-number [dynamic dynamic-name [timeout minutes]]{deny |
permit} protocol source source-wildcard destination destination-wildcard
[precedence precedence] [tos tos] [log]
```

**access-list access-list-number Parameters**

| **access-list** *access-list-number* **Command** | **Description** |
|---|---|
| **permit** | This option causes all IP traffic that matches the specified conditions to be protected by cryptography, using the policy described by the corresponding crypto map entry. |
| **deny** | This option instructs the router to route traffic in plaintext. |
| *protocol* | This option specifies which traffic to protect by cryptography based on the protocol, such as TCP, UDP, or ICMP. If the protocol is IP, then all traffic IP traffic that matches that permit statement is encrypted. |
| *source* **and** *destination* | If the ACL statement is a permit statement, these are the networks, subnets, or hosts between which traffic should be protected. If the ACL statement is a deny statement, then the traffic between the specified source and destination is sent in plaintext. |

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# Symmetric Crypto ACLs




```
Applied to R1 S0/0/0 outbound traffic:
R1(config)# access-list 110 permit tcp 10.0.1.0 0.0.0.255 10.0.2.0 0.0.0.255

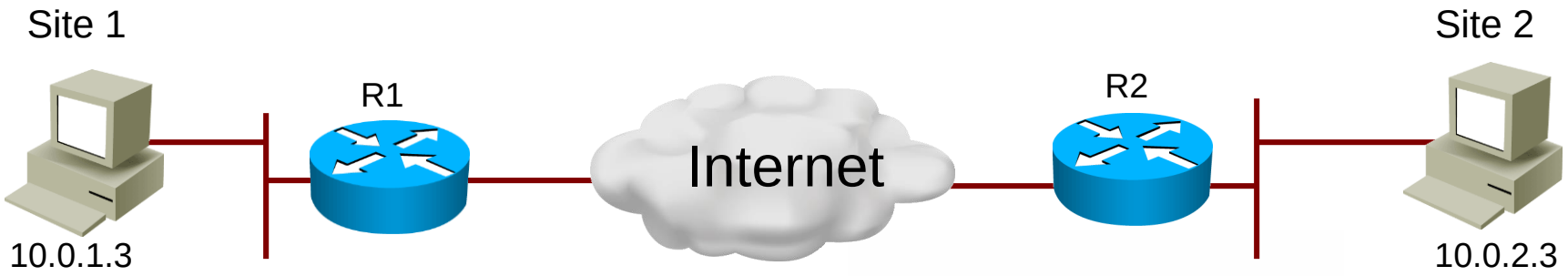
(when evaluating inbound traffic- source: 10.0.2.0, destination: 10.0.1.0)
```

```
Applied to R2 S0/0/0 outbound traffic:
R2(config)# access-list 101 permit tcp 10.0.2.0 0.0.0.255 10.0.1.0 0.0.0.255

(when evaluating inbound traffic- source: 10.0.1.0, destination: 10.0.2.0)
```

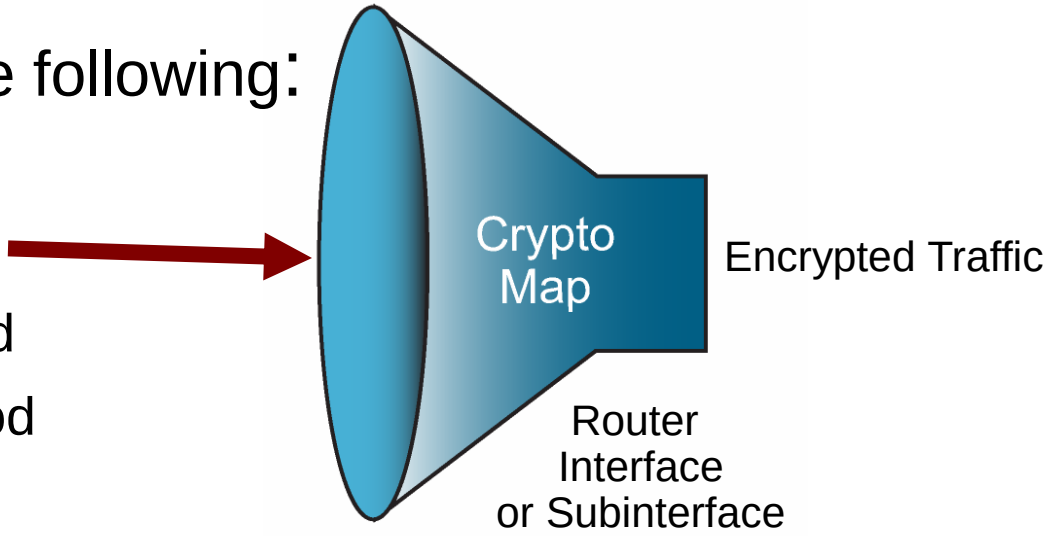
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# Task 5
# Apply the Crypto Map




Crypto maps define the following:

- ACL to be used
- Remote VPN peers
- Transform set to be used
- Key management method
- SA lifetimes




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# Crypto Map Command

**router(config)#**

**crypto map** *map-name seq-num* **ipsec-manual**

**crypto map** *map-name seq-num* **ipsec-isakmp [dynamic** *dynamic-map-name***]**

### crypto map Parameters

| Command Parameters | Description |
|---|---|
| *map-name* | Defines the name assigned to the crypto map set or indicates the name of the crypto map to edit. |
| *seq-num* | The number assigned to the crypto map entry. |
| **ipsec-manual** | Indicates that ISAKMP will not be used to establish the IPsec SAs. |
| **ipsec-isakmp** | Indicates that ISAKMP will be used to establish the IPsec SAs. |
| **cisco** | (Default value) Indicates that CET will be used instead of IPsec for protecting the traffic. |
| **dynamic** | (Optional) Specifies that this crypto map entry references a preexisting static crypto map. If this keyword is used, none of the crypto map configuration commands are available. |
| *dynamic-map-name* | (Optional) Specifies the name of the dynamic crypto map set that should be used as the policy template. |

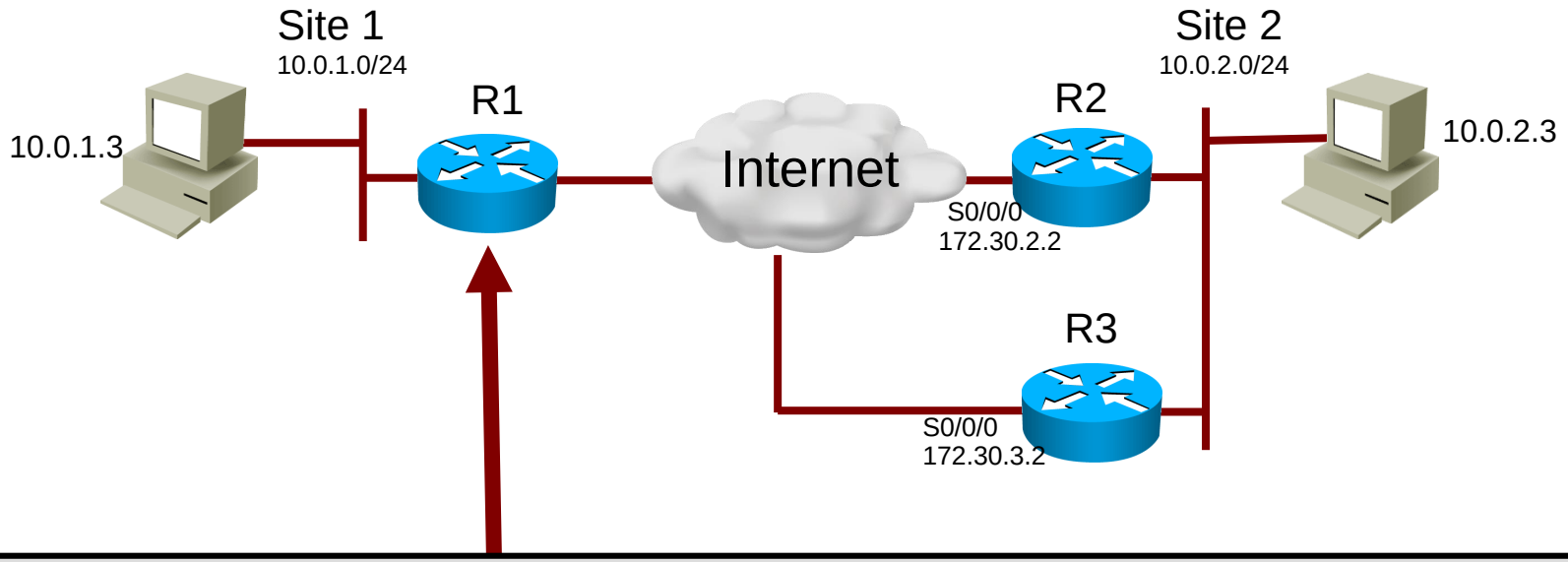
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# Crypto Map Configuration Mode Commands

| Command | Description |
|---|---|
| **set** | Used with the **peer, pfs, transform-set,** and **security-association** commands**.** |
| **peer** [*hostname* \| *ip-address*] | Specifies the allowed IPsec peer by IP address or hostname. |
| **pfs** [**group1** \| **group2**] | Specifies DH Group 1 or Group 2. |
| **transform-set** [*set_name(s)*] | Specify list of transform sets in priority order. When the **ipsec-manual** parameter is used with the **crypto map** command, then only one transform set can be defined. When the **ipsec-isakmp** parameter or the **dynamic** parameter is used with the **crypto map** command, up to six transform sets can be specified. |
| **security-association lifetime** | Sets SA lifetime parameters in seconds or kilobytes. |
| **match address** [*access-list-id* \| *name*] | Identifies the extended ACL by its name or number. The value should match the access-list-number or name argument of a previously defined IP-extended ACL being matched. |
| **no** | Used to delete commands entered with the **set command.** |
| **exit** | Exits crypto map configuration mode. |

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# Sample Configuration

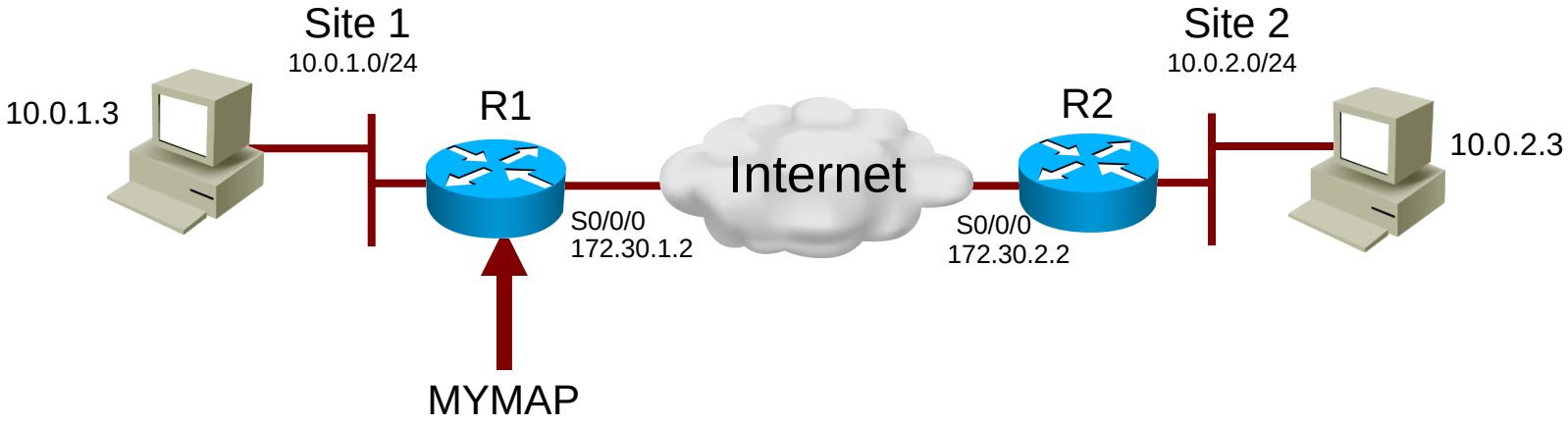



```
R1(config)# crypto map MYMAP 10 ipsec-isakmp
R1(config-crypto-map)# match address 110
R1(config-crypto-map)# set peer 172.30.2.2 default
R1(config-crypto-map)# set peer 172.30.3.2
R1(config-crypto-map)# set pfs group1
R1(config-crypto-map)# set transform-set mine
R1(config-crypto-map)# set security-association lifetime seconds 86400
```

Multiple peers can be specified for redundancy.

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# Assign the Crypto Map Set




**router(config-if)#**

```
crypto map map-name
```

```
R1(config)# interface serial0/0/0
R1(config-if)# crypto map MYMAP
```

- Applies the crypto map to outgoing interface
- Activates the IPsec policy

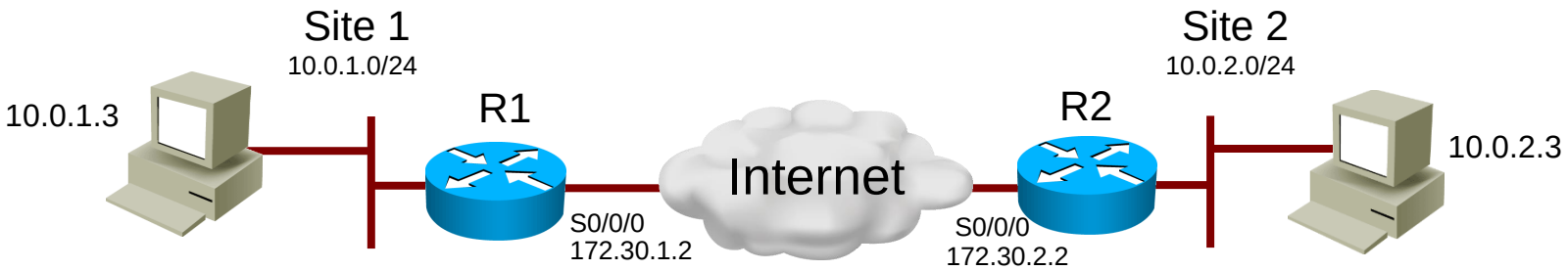
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# CLI Commands

| Show Command | Description |
|---|---|
| `show crypto map` | Displays configured crypto maps |
| `show crypto isakmp policy` | Displays configured IKE policies |
| `show crypto ipsec sa` | Displays established IPsec tunnels |
| `show crypto ipsec transform-set` | Displays configured IPsec transform sets |
| `debug crypto isakmp` | Debugs IKE events |
| `debug crypto ipsec` | Debugs IPsec events |

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# show crypto map



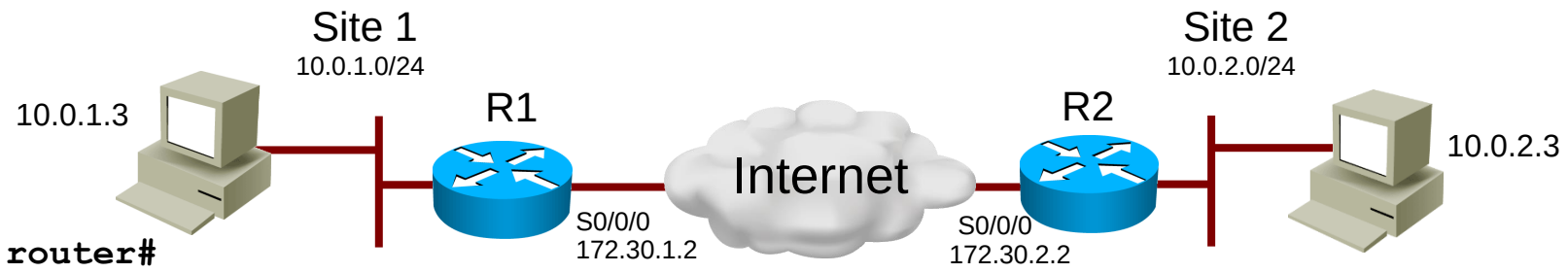

```
show crypto map
```

Displays the currently configured crypto maps

```
R1# show crypto map
Crypto Map "MYMAP" 10 ipsec-isakmp
            Peer = 172.30.2.2
            Extended IP access list 110
                 access-list 102 permit ip host 10.0.1.3 host 10.0.2.3
            Current peer: 172.30.2.2
            Security association lifetime: 4608000 kilobytes/3600 seconds
            PFS (Y/N): N
            Transform sets={ MYSET, }
```

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# show crypto isakmp policy

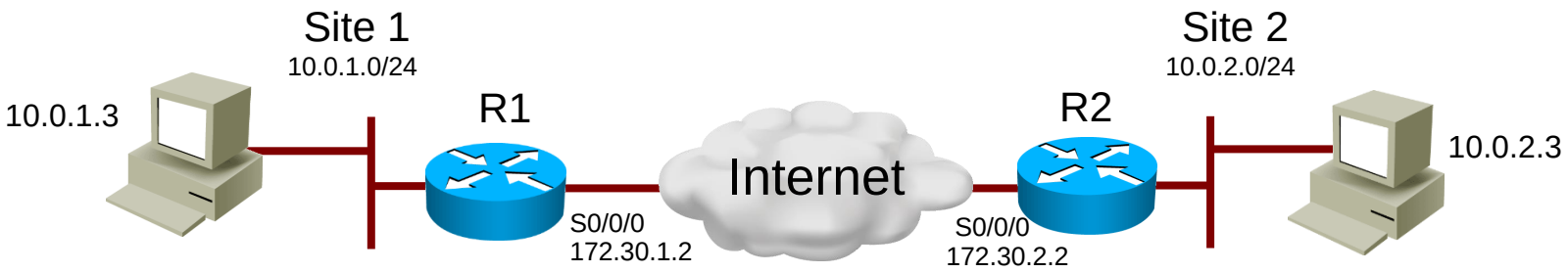



```
show crypto isakmp policy
```

```
R1# show crypto isakmp policy
Protection suite of priority 110
      encryption algorithm:   3DES - Data Encryption Standard (168 bit keys).
      hash algorithm:         Secure Hash Standard
      authentication method:  preshared
      Diffie-Hellman group:   #2 (1024 bit)
      lifetime:               86400 seconds, no volume limit
Default protection suite
      encryption algorithm:   DES - Data Encryption Standard (56 bit keys).
      hash algorithm:         Secure Hash Standard
      authentication method:  Rivest-Shamir-Adleman Signature
      Diffie-Hellman group:   #1 (768 bit)
      lifetime:               86400 seconds, no volume limit
```

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# show crypto ipsec transform-set



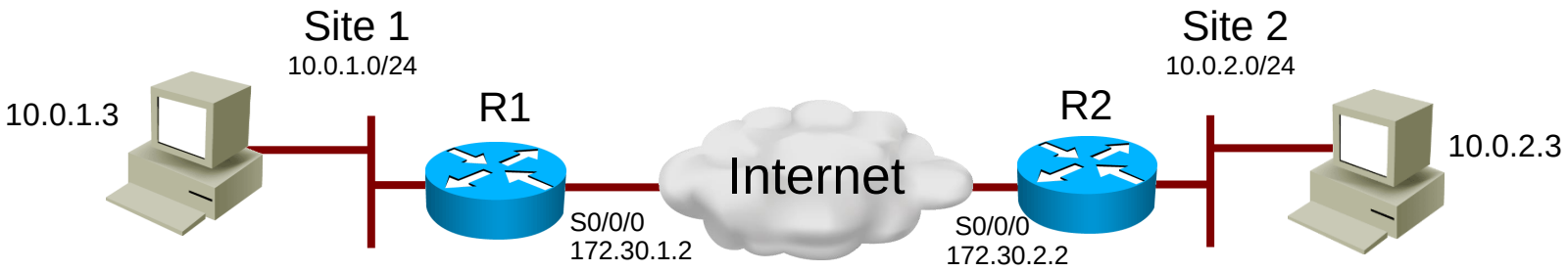

**show crypto ipsec transform-set**

Displays the currently defined transform sets

```
R1# show crypto ipsec transform-set
Transform set AES_SHA: { esp-128-aes esp-sha-hmac }
      will negotiate = { Tunnel,  },
```

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# show crypto ipsec sa




```
R1# show crypto ipsec sa
Interface: Serial0/0/0
        Crypto map tag: MYMAP, local addr. 172.30.1.2
        local ident (addr/mask/prot/port): (172.30.1.2/255.255.255.255/0/0)
       remote ident (addr/mask/prot/port): (172.30.2.2/255.255.255.255/0/0)
       current_peer: 172.30.2.2
        PERMIT, flacs={origin_is_acl,}
       #pkts encaps: 21, #pkts encrypt: 21, #pkts digest 0
       #pkts decaps: 21, #pkts decrypt: 21, #pkts verify 0
       #send errors 0, #recv errors 0
        local crypto endpt.: 172.30.1.2, remote crypto endpt.: 172.30.2.2
        path mtu 1500, media mtu 1500
        current outbound spi: 8AE1C9C
```

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# debug crypto isakmp

**router#**

```
debug crypto isakmp
```

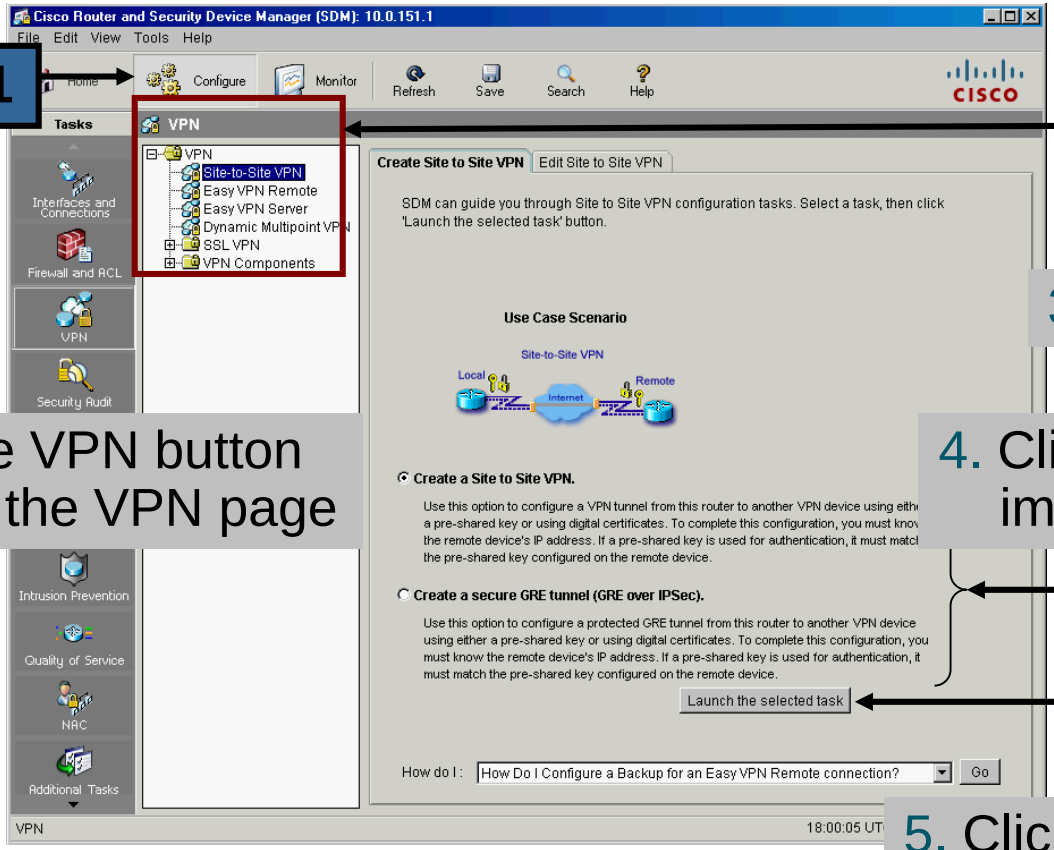
```
1d00h: ISAKMP (0:1): atts are not acceptable. Next payload is 0 1d00h: ISAKMP (0:1); no
offers accepted!
1d00h: ISAKMP (0:1): SA not acceptable!
1d00h: %CRYPTO-6-IKMP_MODE_FAILURE: Processing of Main Mode failed with peer at 172.30.2.2
```

- This is an example of the Main Mode error message.
- The failure of Main Mode suggests that the Phase I policy does not match on both sides.
- Verify that the Phase I policy is on both peers and ensure that all the attributes match.

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# Starting a VPN Wizard

1. Click Configure in main toolbar




2. Click the VPN button to open the VPN page

3. Choose a wizard

Wizards for IPsec Solutions, includes type of VPNs and Individual IPsec components

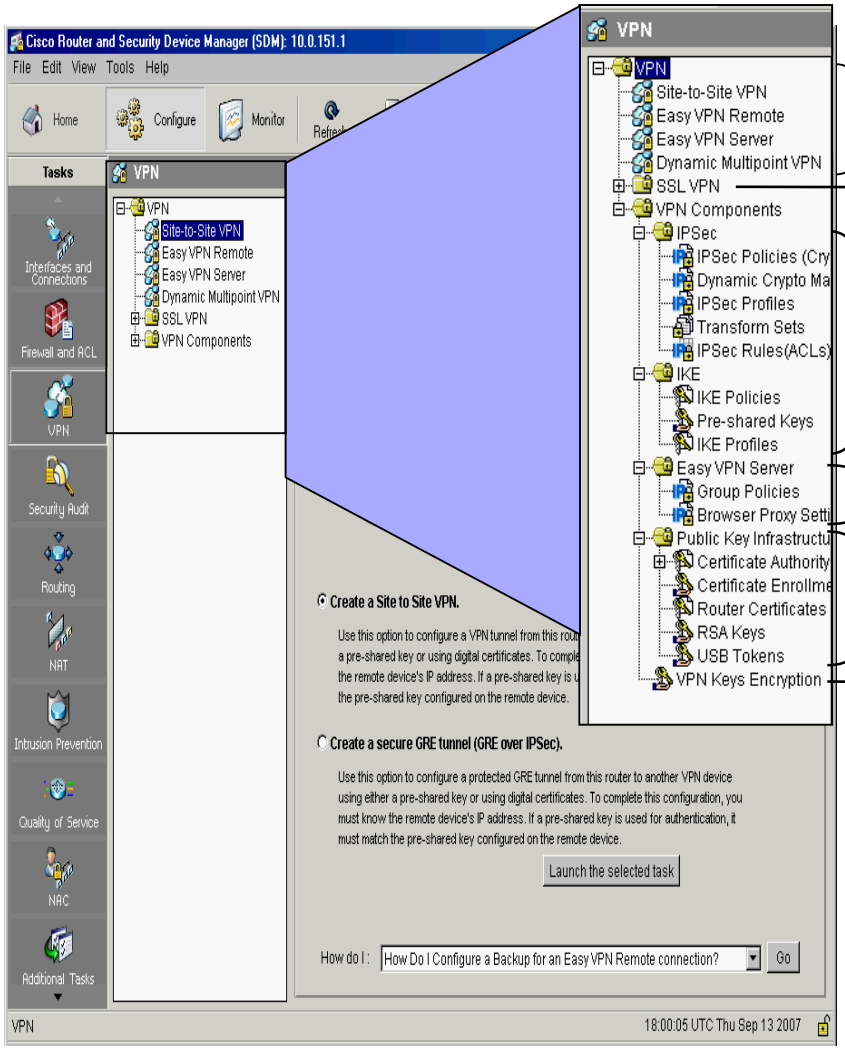
4. Click the VPN implementation subtype

VPN implementation Subtypes. Vary based On VPN wizard chosen.

5. Click the Launch the Selected Task button

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# VPN Components




**VPN Wizards**

**SSL VPN parameters**

Individual IPsec components used to build VPNs

Easy VPN server parameters

Public key certificate parameters

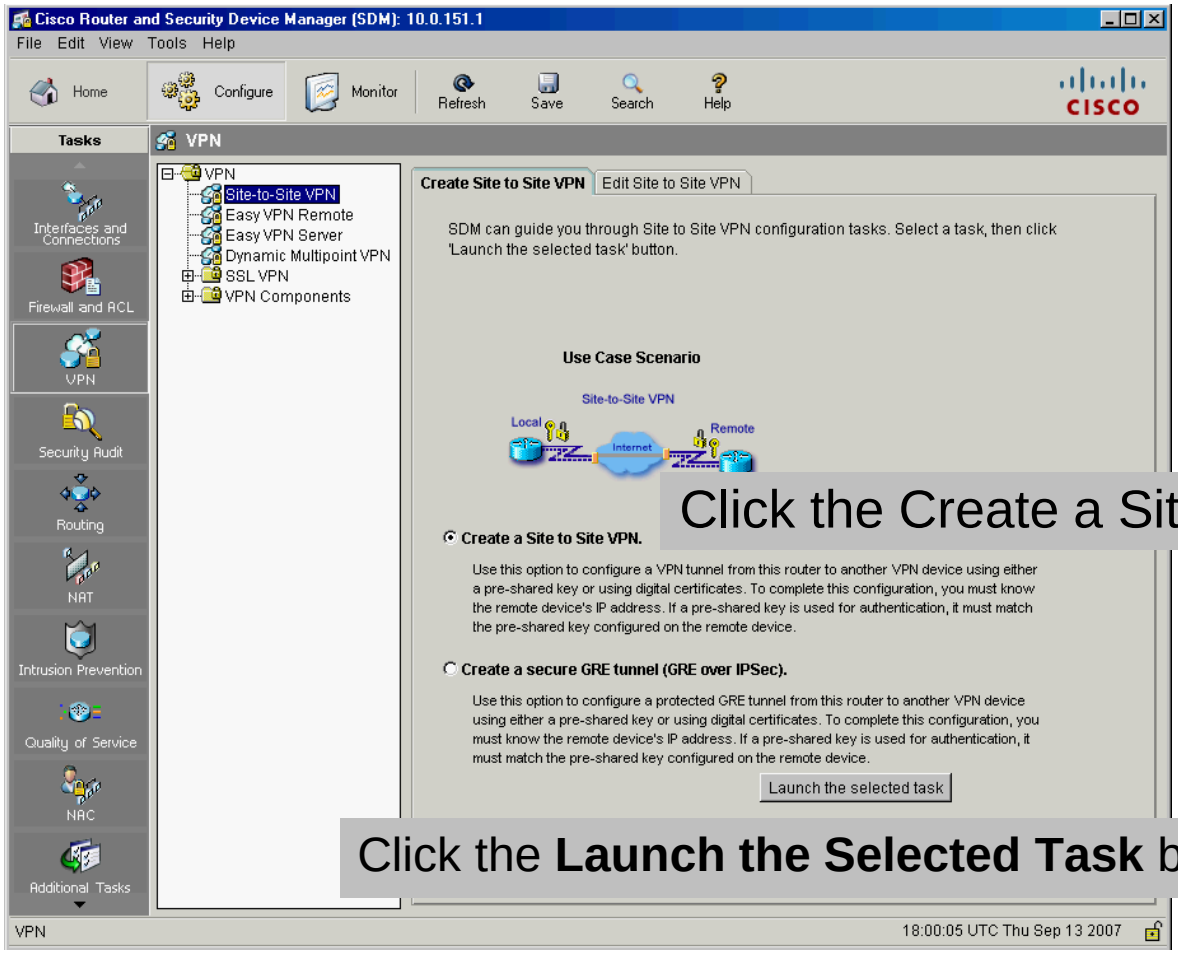
Encrypt VPN passwords

**VPN Components**

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# Configuring a Site-to-Site VPN

Choose Configure > VPN > Site-to-Site VPN

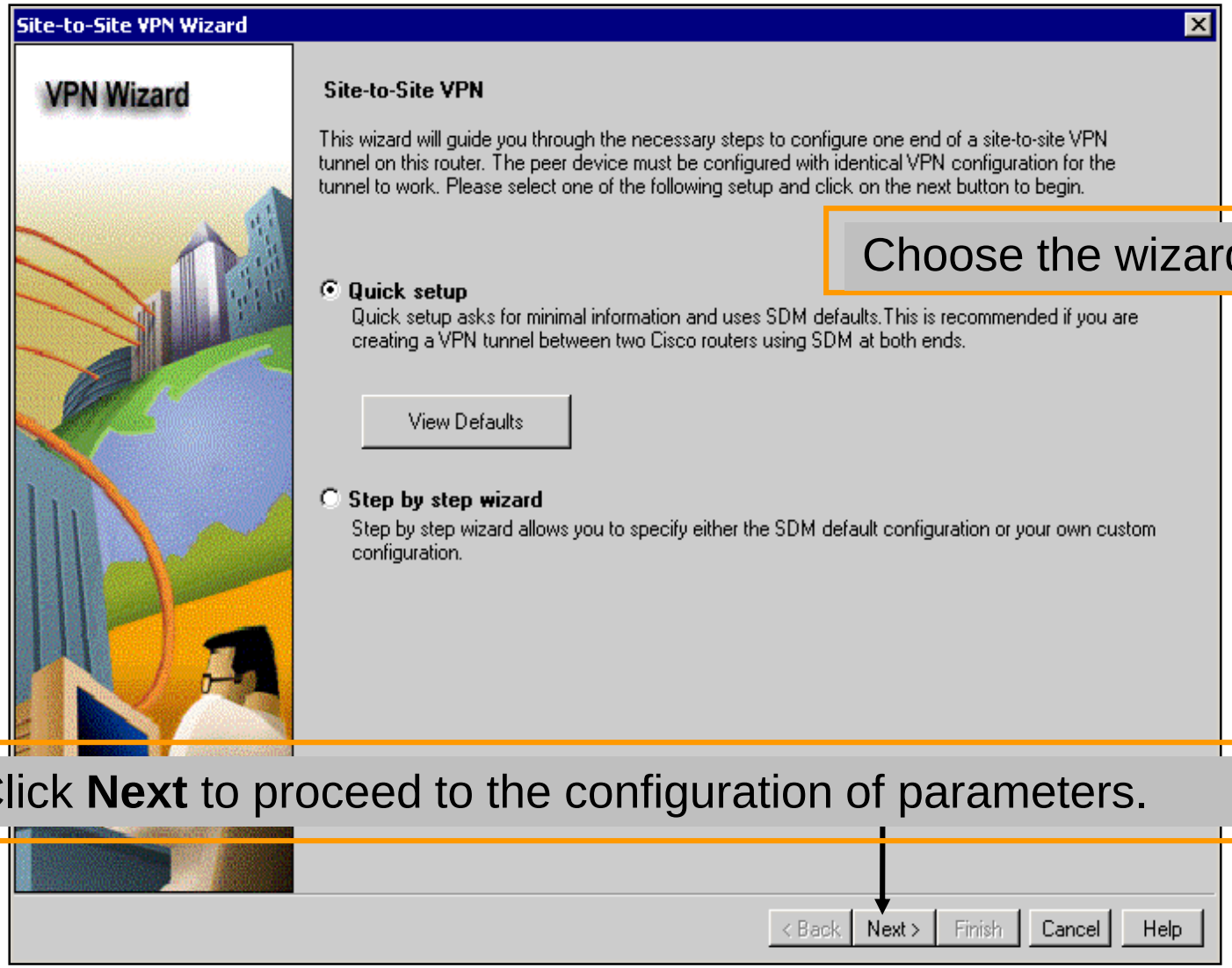



Click the Create a Site-to-Site VPN

Click the **Launch the Selected Task** button

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# Site-to-Site VPN Wizard

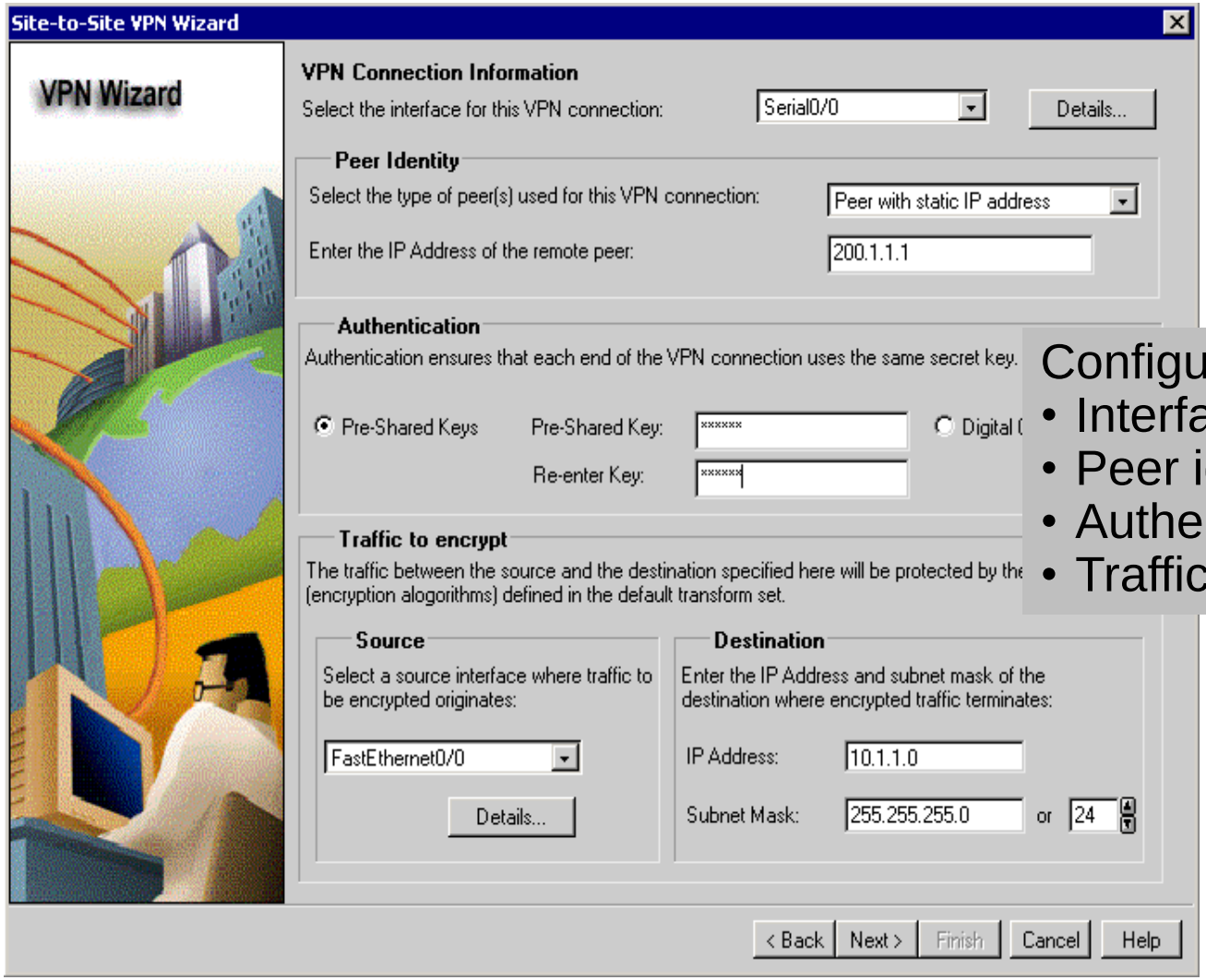



Choose the wizard mode

Click **Next** to proceed to the configuration of parameters.

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# Quick Setup

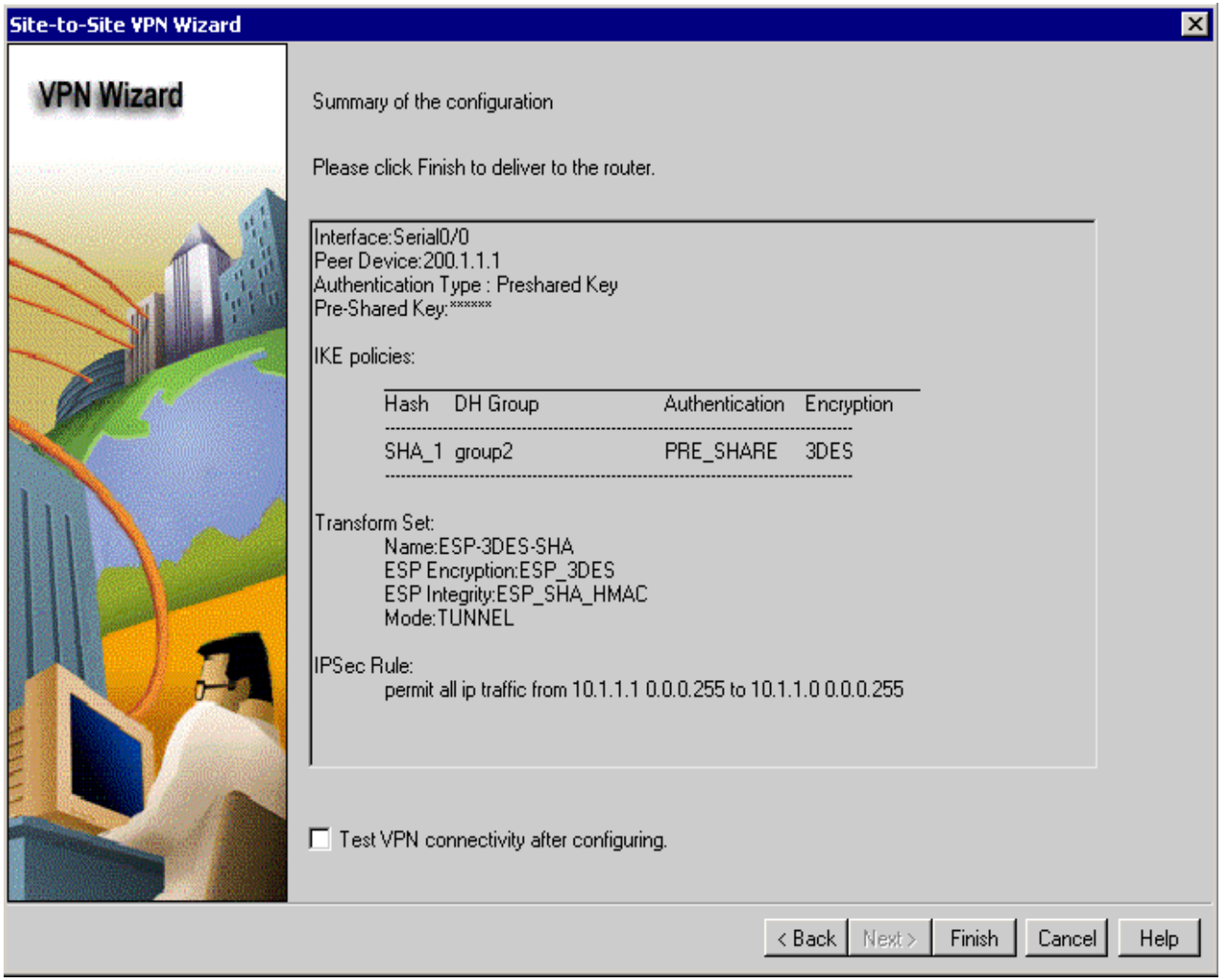



Configure the parameters

- Interface to use
- Peer identity information
- Authentication method
- Traffic to encrypt

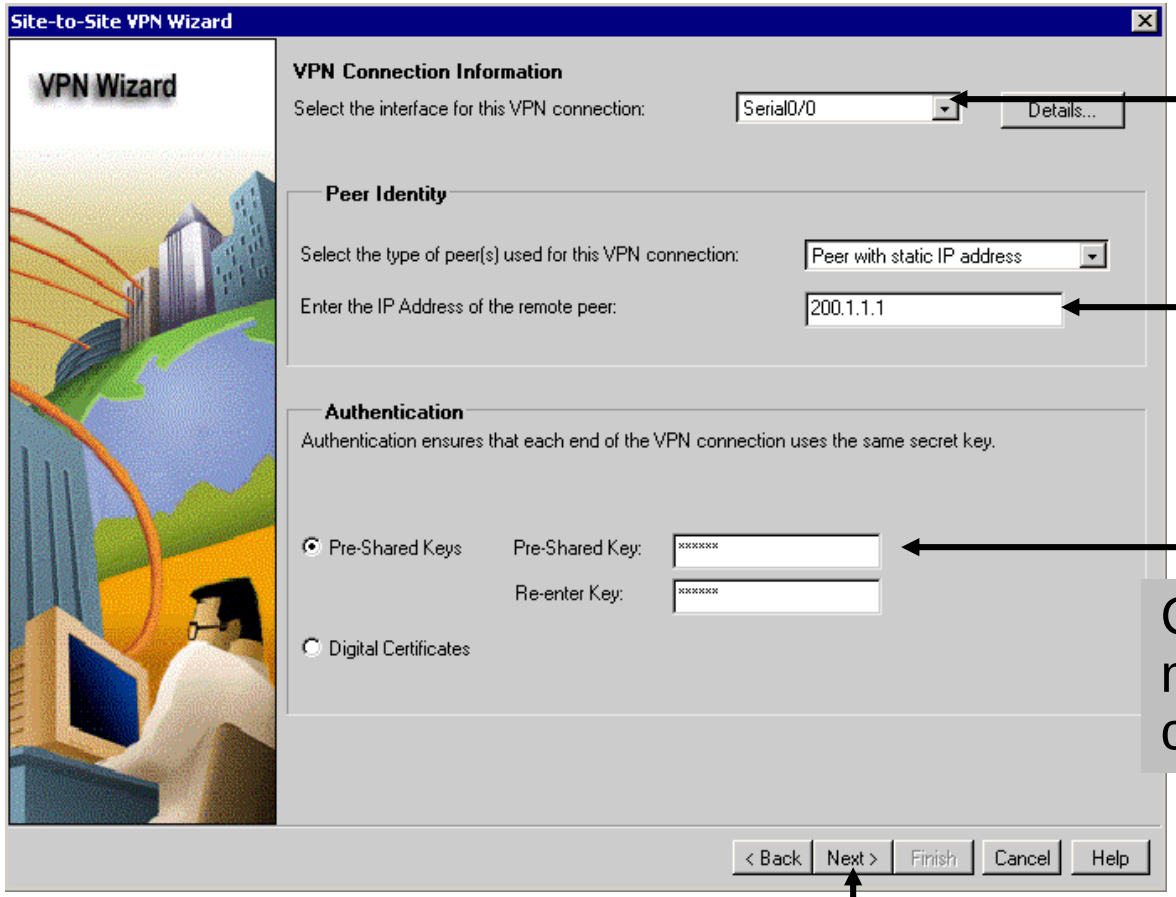
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# Verify Parameters




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# Step-by-Step Wizard




1 Choose the outside interface that is used to connect to the IPSec peer

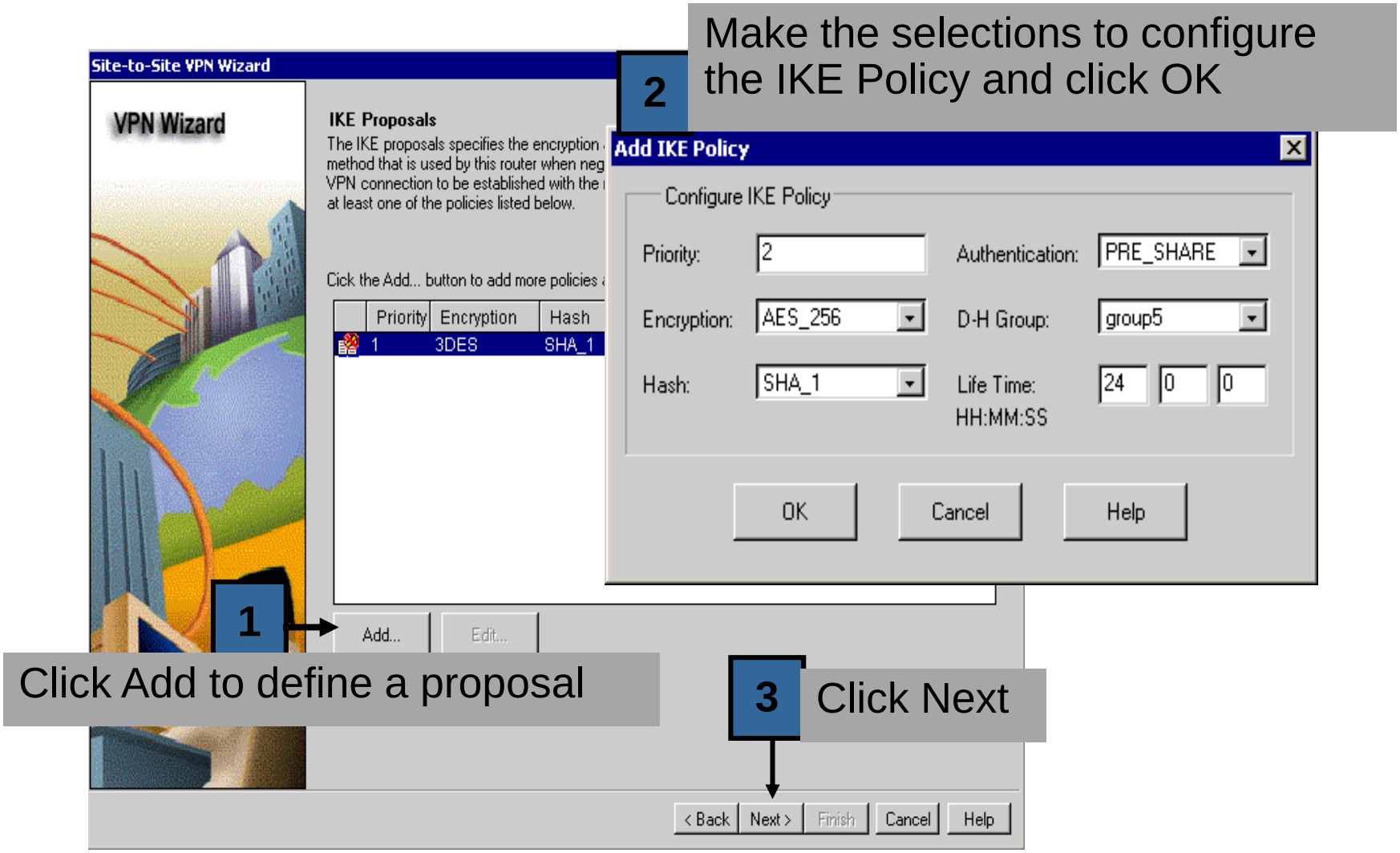
2 Specify the IP address of the peer

3 Choose the authentication method and specify the credentials

4 Click Next

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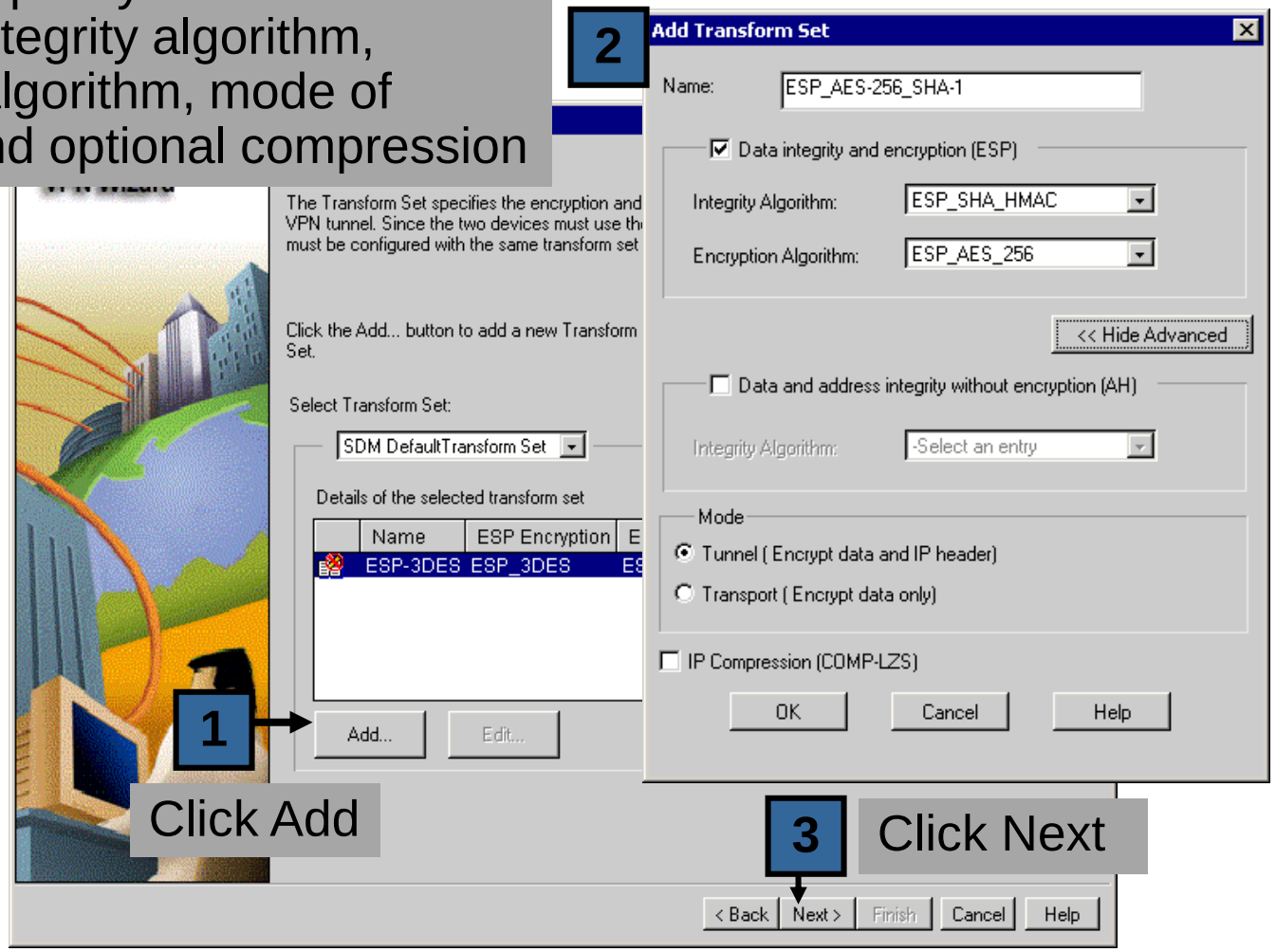
# Creating a Custom IKE Proposal




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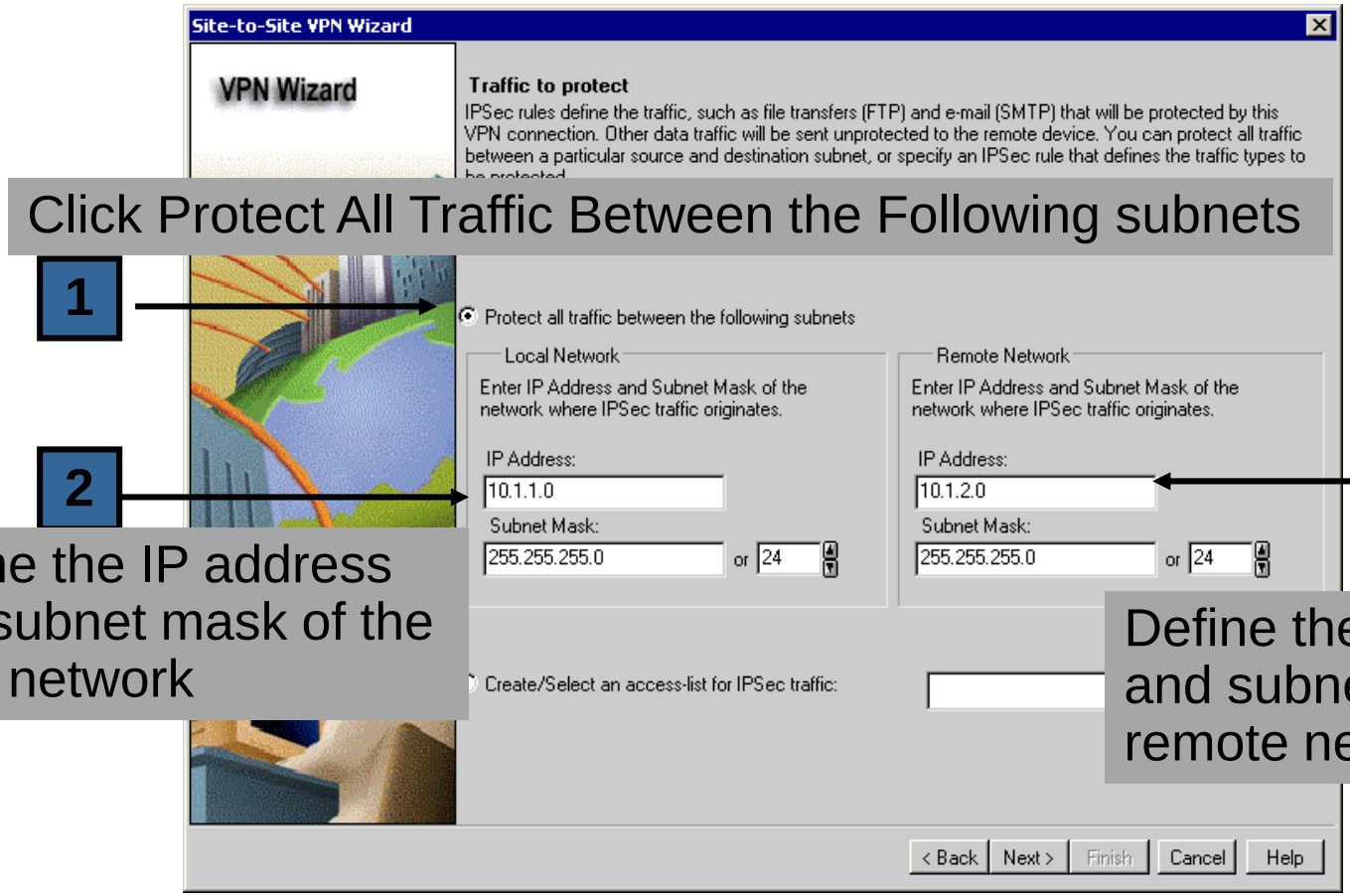
# Creating a Custom IPSec Transform Set

Define and specify the transform set name, integrity algorithm, encryption algorithm, mode of operation and optional compression



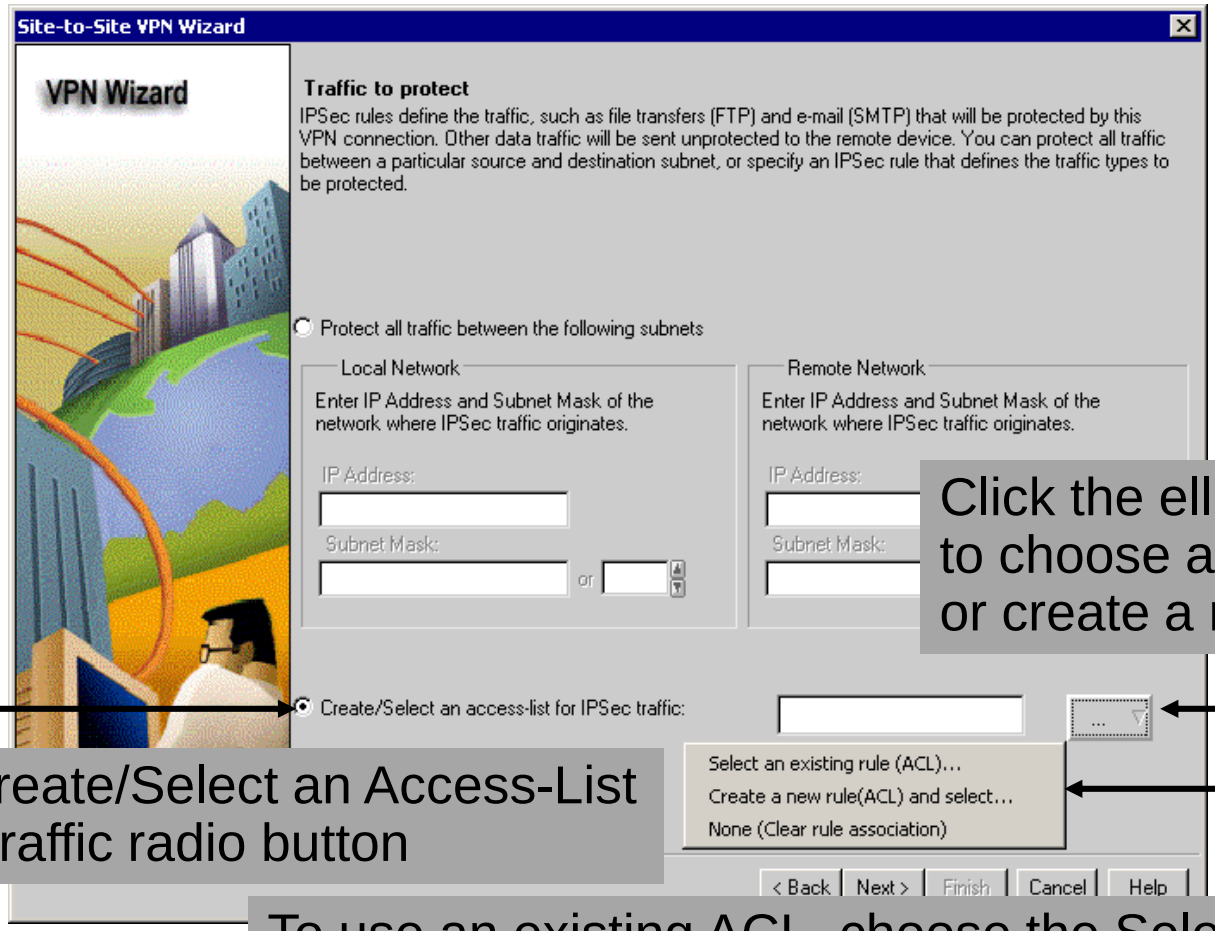

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# Protecting Traffic Subnet to Subnet




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# Protecting Traffic
# Custom ACL




1

Click the Create/Select an Access-List for IPSec Traffic radio button

Click the ellipses button to choose an existing ACL or create a new one

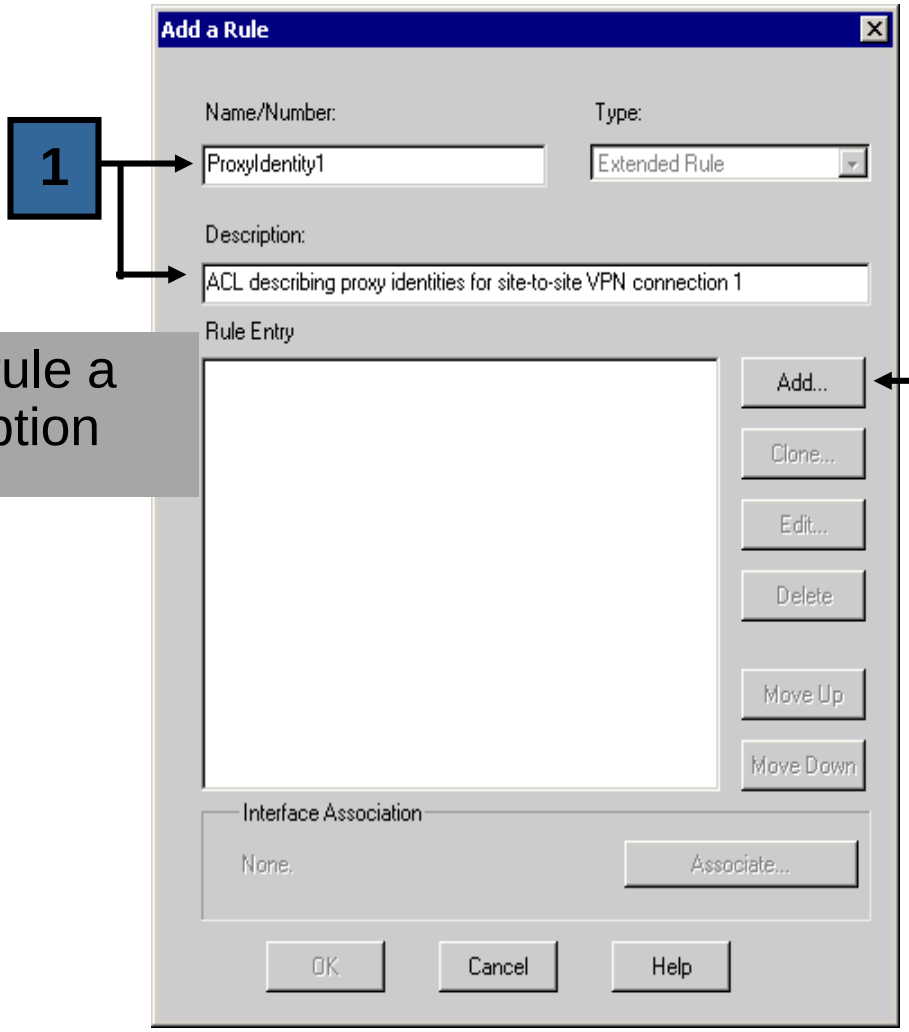
2

3

To use an existing ACL, choose the Select an Existing Rule (ACL) option. To create a new ACL, choose the Create a New Rule (ACL) and Select option

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# Add a Rule



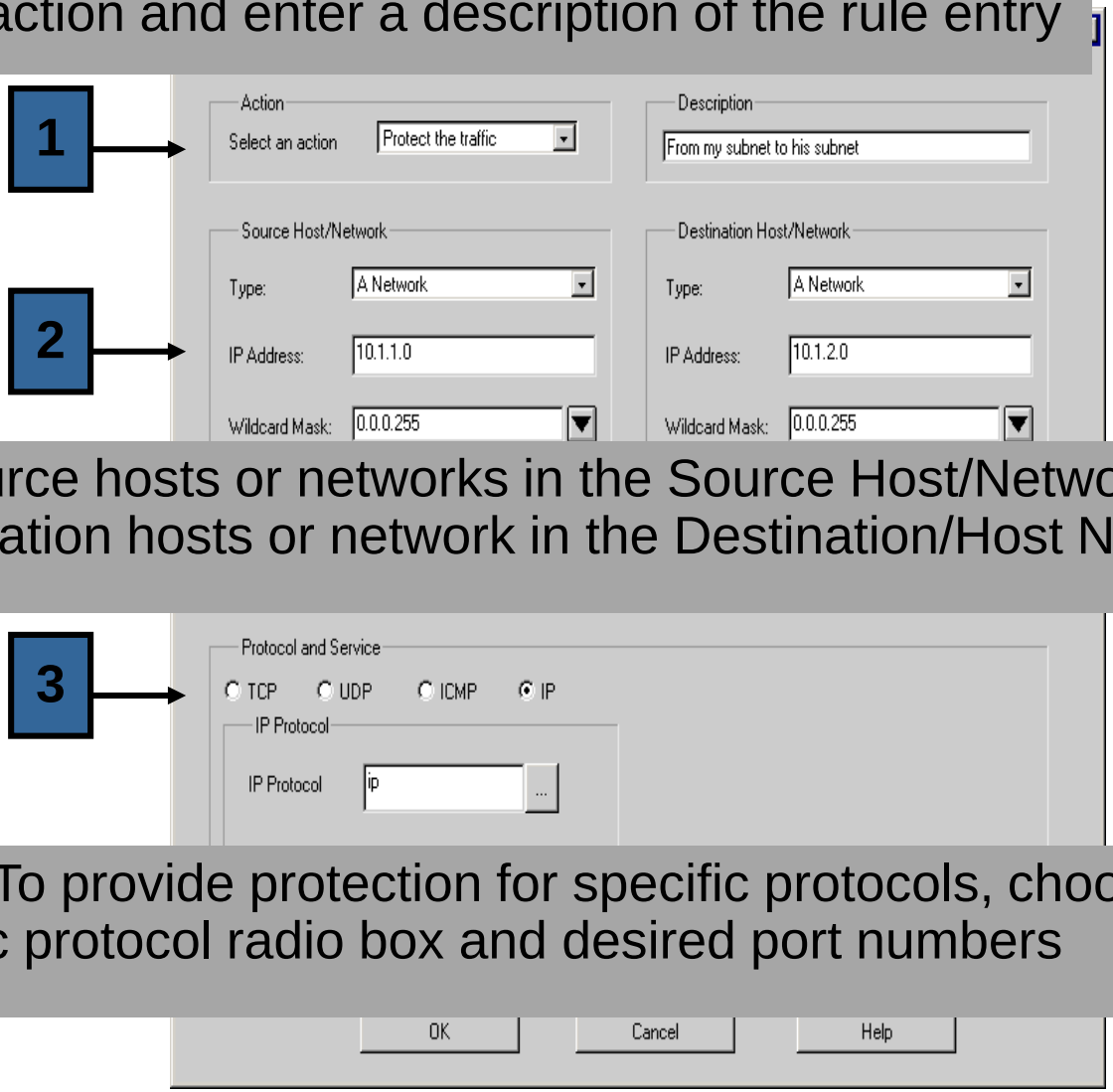

Give the access rule a name and description

Click Add

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# Configuring a New Rule Entry

Choose an action and enter a description of the rule entry

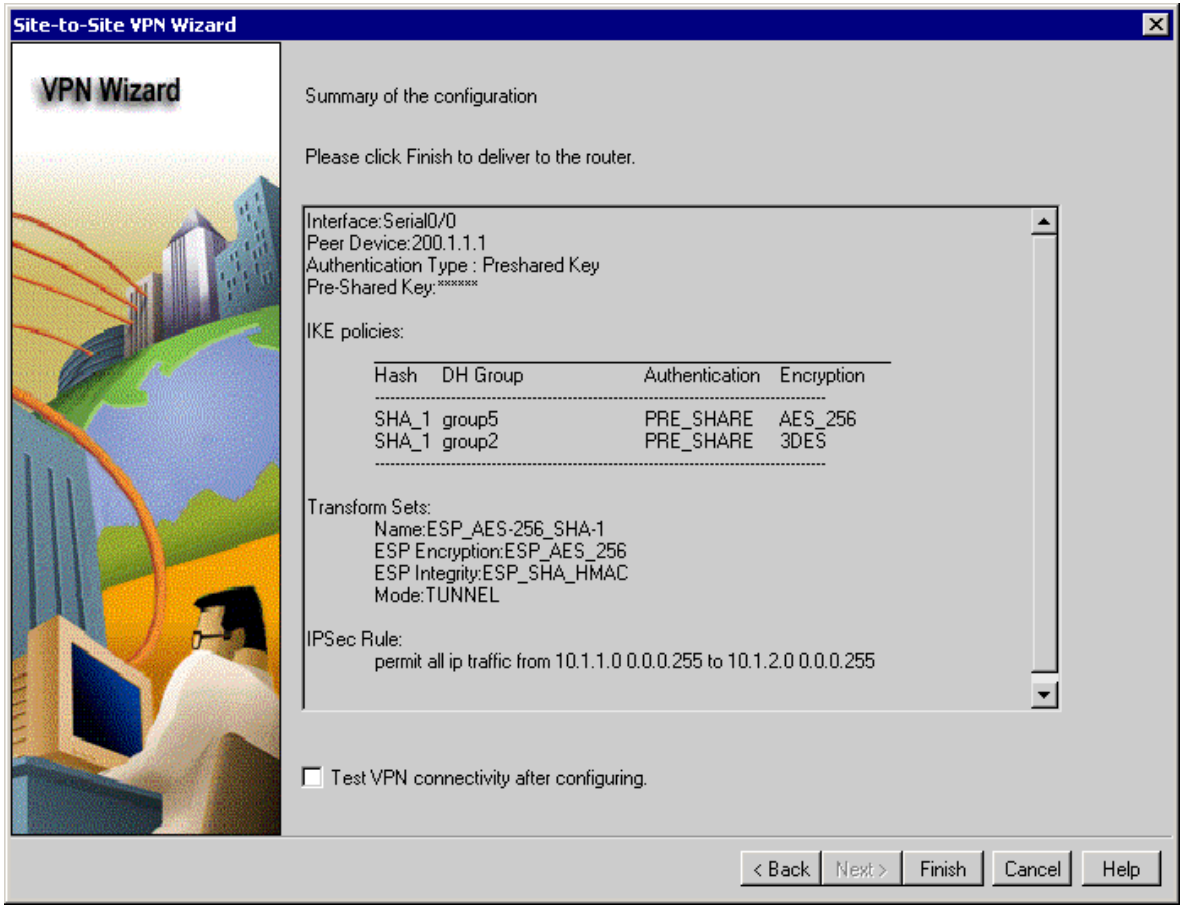



Define the source hosts or networks in the Source Host/Network pane and the destination hosts or network in the Destination/Host Network pane

(Optional) To provide protection for specific protocols, choose the specific protocol radio box and desired port numbers

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# Configuration Summary

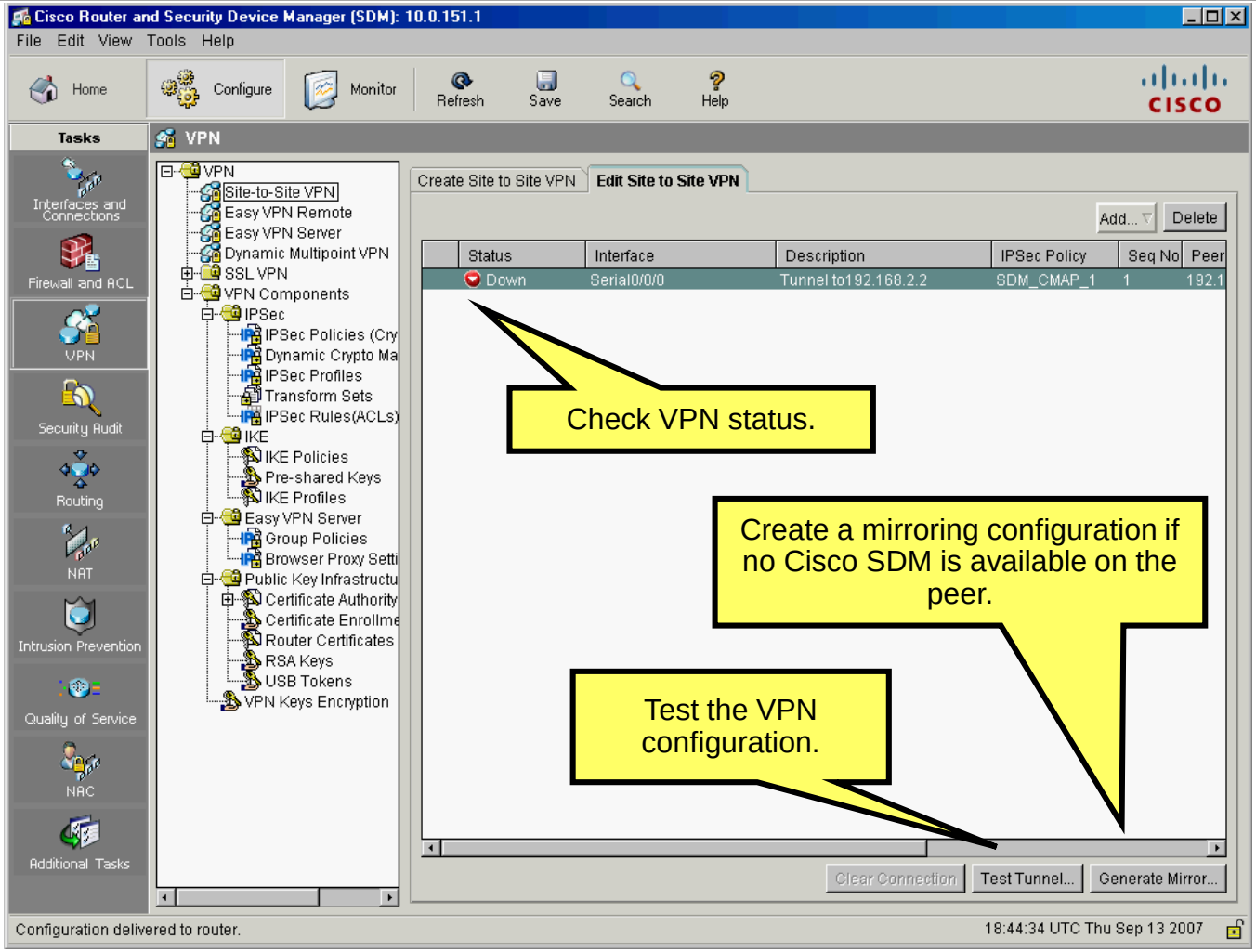



- Click **Back** to modify the configuration.
- Click **Finish** to complete the configuration.

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# Verify VPN Configuration

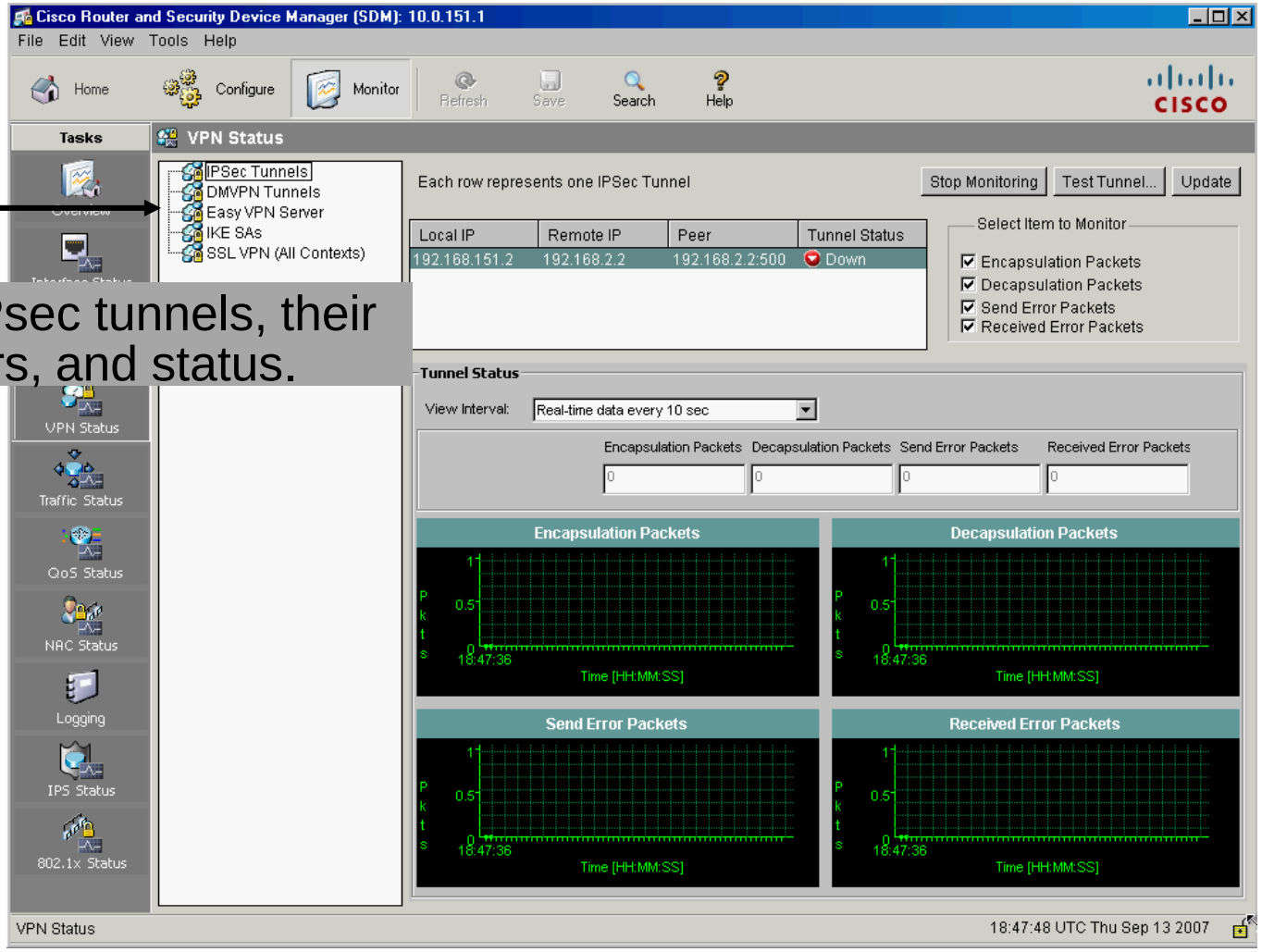
Choose Configure > VPN > Site-to-Site VPN > Edit Site-to-Site VPN




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# Monitor

## Choose Monitor > VPN Status > IPSec Tunnels




1

Lists all IPsec tunnels, their parameters, and status.

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# Telecommuting

- Flexibility in working location and working hours
- Employers save on real-estate, utility and other overhead costs
- Succeeds if program is voluntary, subject to management discretion, and operationally feasible





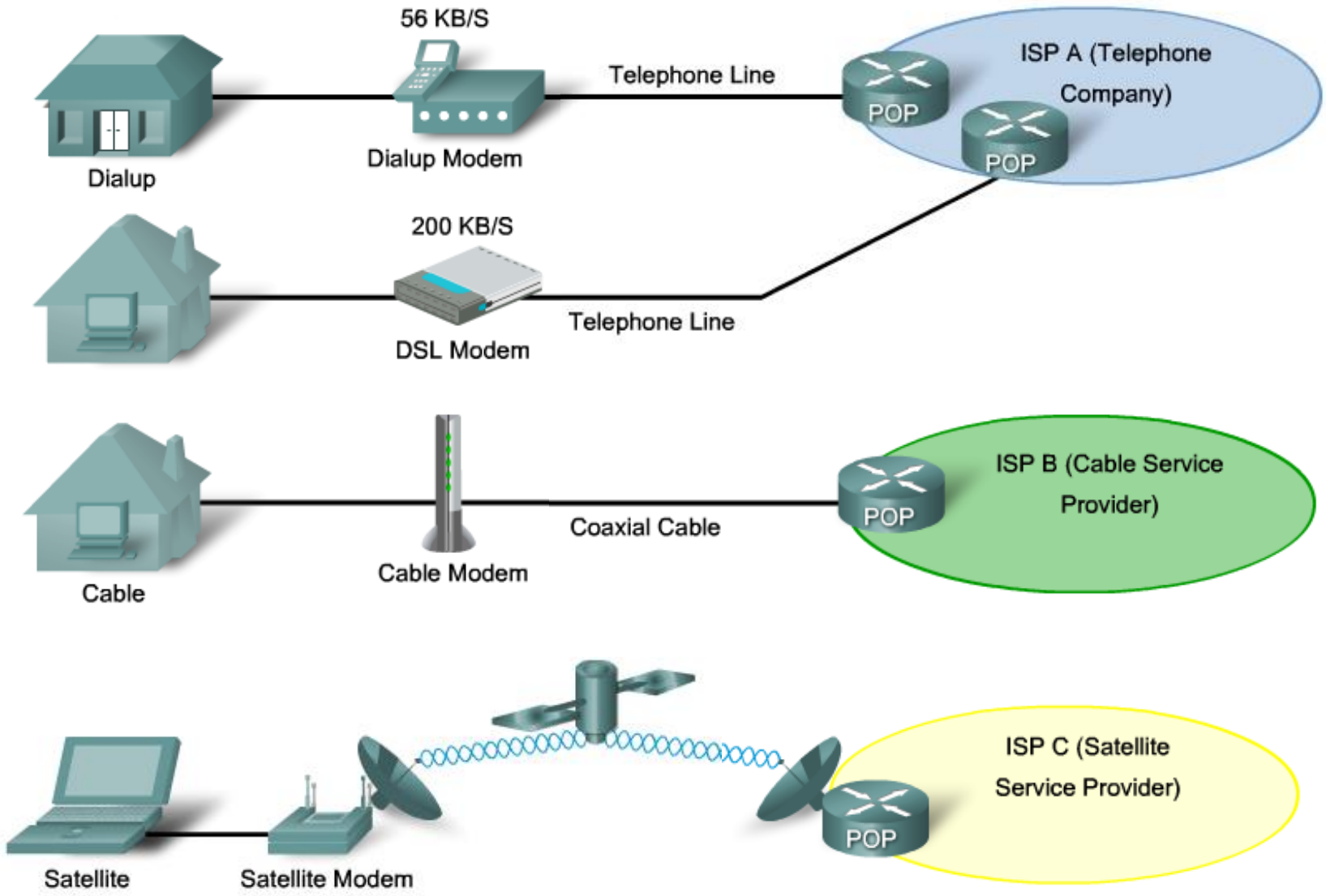
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# Telecommuting Benefits

- Organizational benefits:
  - Continuity of operations
  - Increased responsiveness
  - Secure, reliable, and manageable access to information
  - Cost-effective integration of data, voice, video, and applications
  - Increased employee productivity, satisfaction, and retention
- Social benefits:
  - Increased employment opportunities for marginalized groups
  - Less travel and commuter related stress
- Environmental benefits:
  - Reduced carbon footprints, both for individual workers and organizations

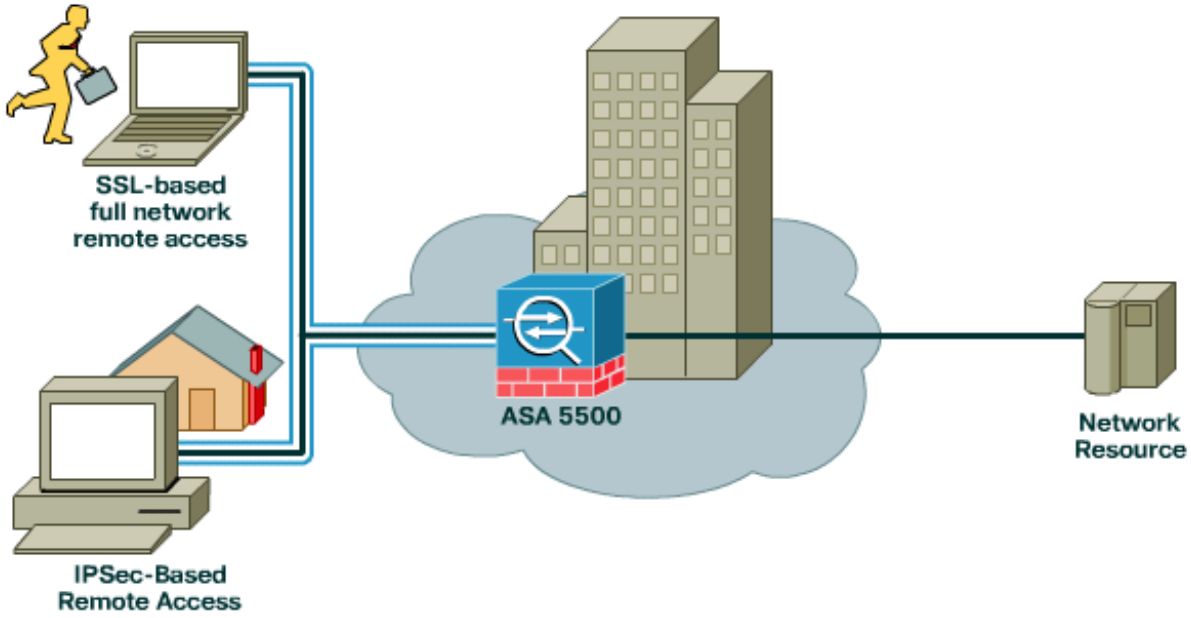
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# Implementing Remote Access




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# Methods for Deploying Remote Access




**IPsec Remote Access VPN**

**Any Application** ⟷ **Anywhere Access**

**SSL-Based VPN**

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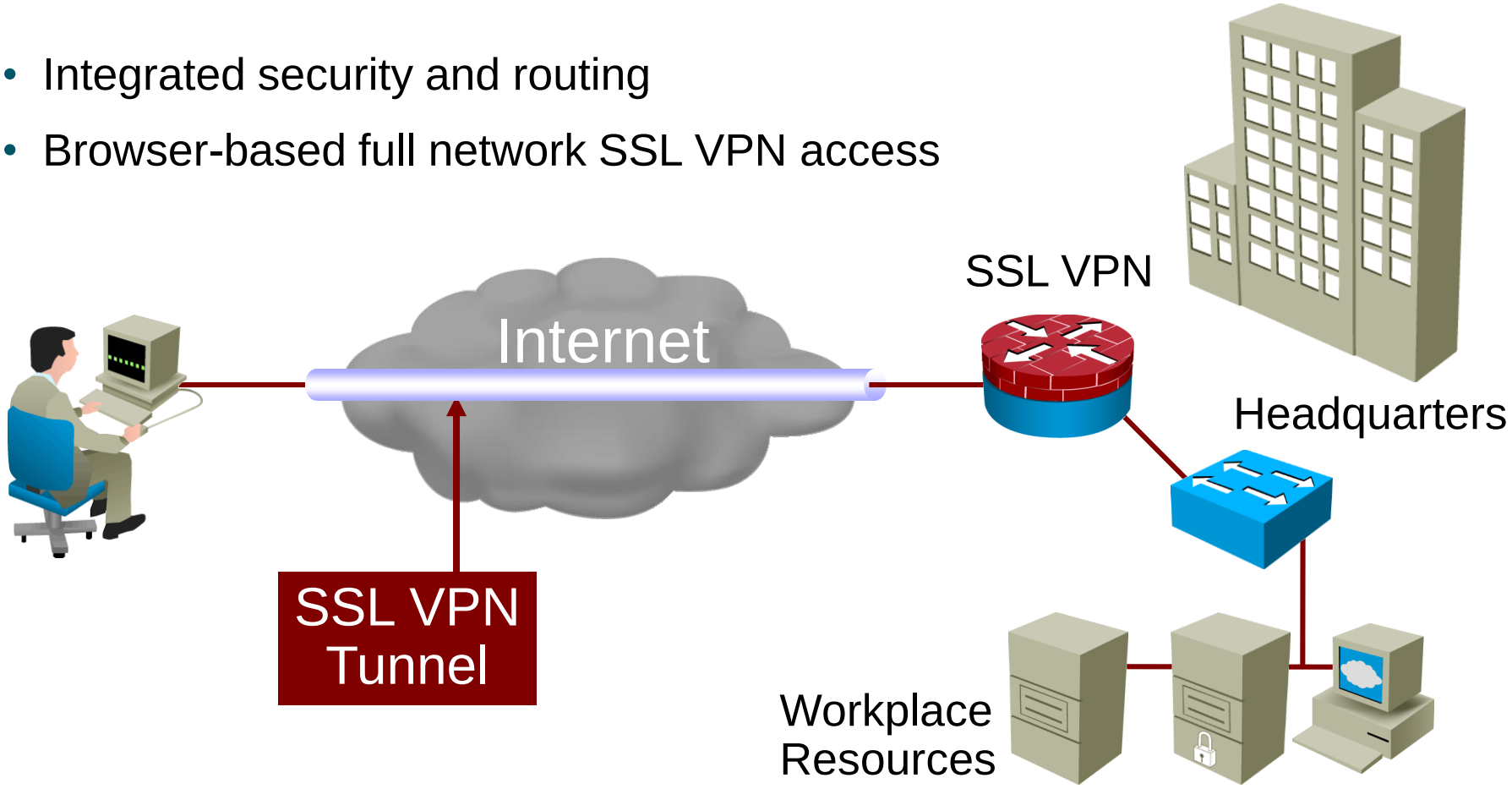
# Comparison of SSL and IPSec

| | SSL | IPsec |
|---|---|---|
| **Applications** | Web-enabled applications, file sharing, e-mail | All IP-based applications |
| **Encryption** | **Moderate**<br>Key lengths from 40 bits to 128 bits | **Stronger**<br>Key lengths from 56 bits to 256 bits |
| **Authentication** | **Moderate**<br>One-way or two-way authentication | **Strong**<br>Two-way authentication using shared secrets or digital certificates |
| **Ease of Use** | **Very high** | **Moderate**<br>Can be challenging to nontechnical users |
| **Overall Security** | **Moderate**<br>Any device can connect | **Strong**<br>Only specific devices with specific configurations can connect |

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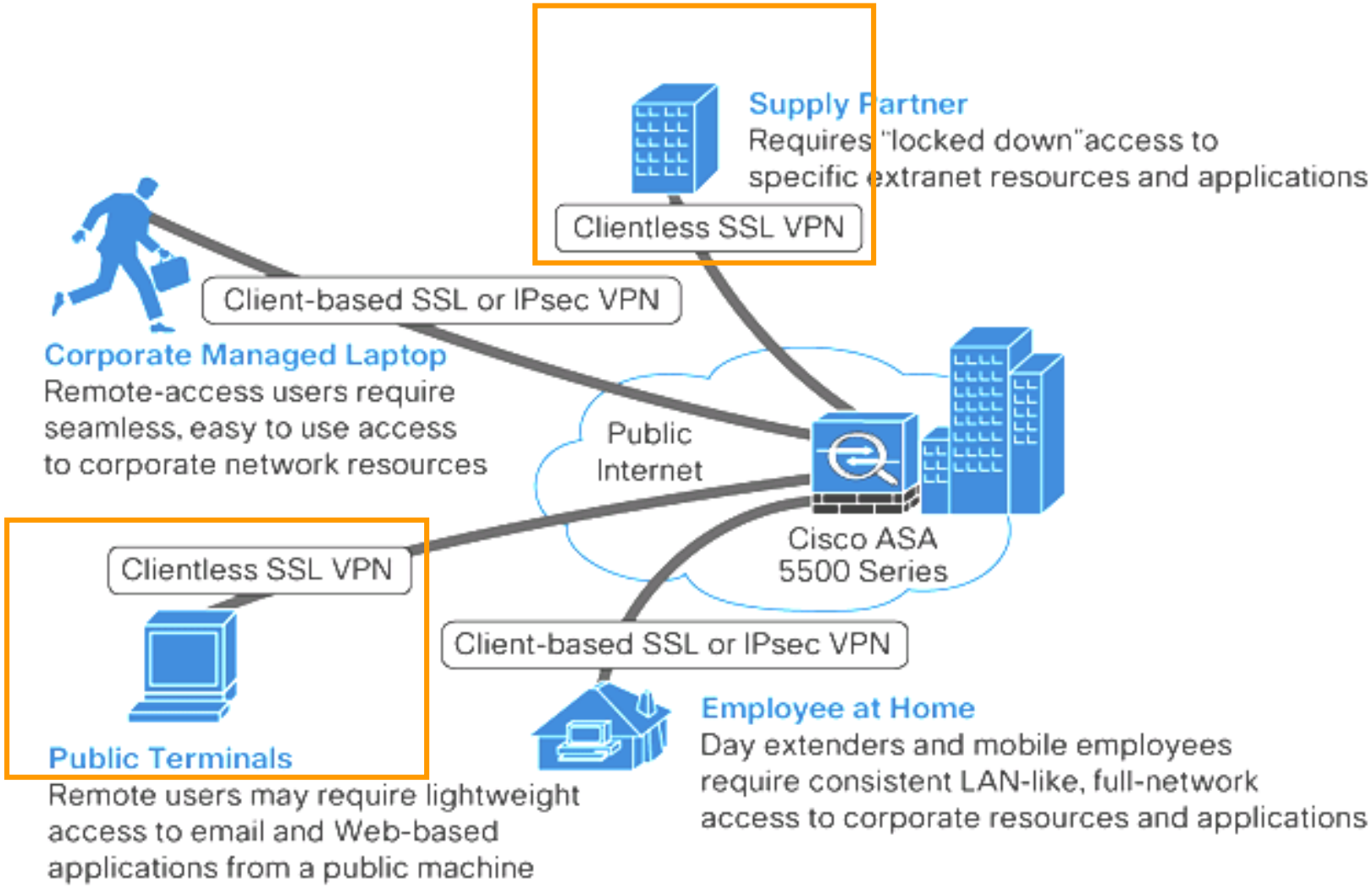
# SSL VPNs

- Integrated security and routing
- Browser-based full network SSL VPN access



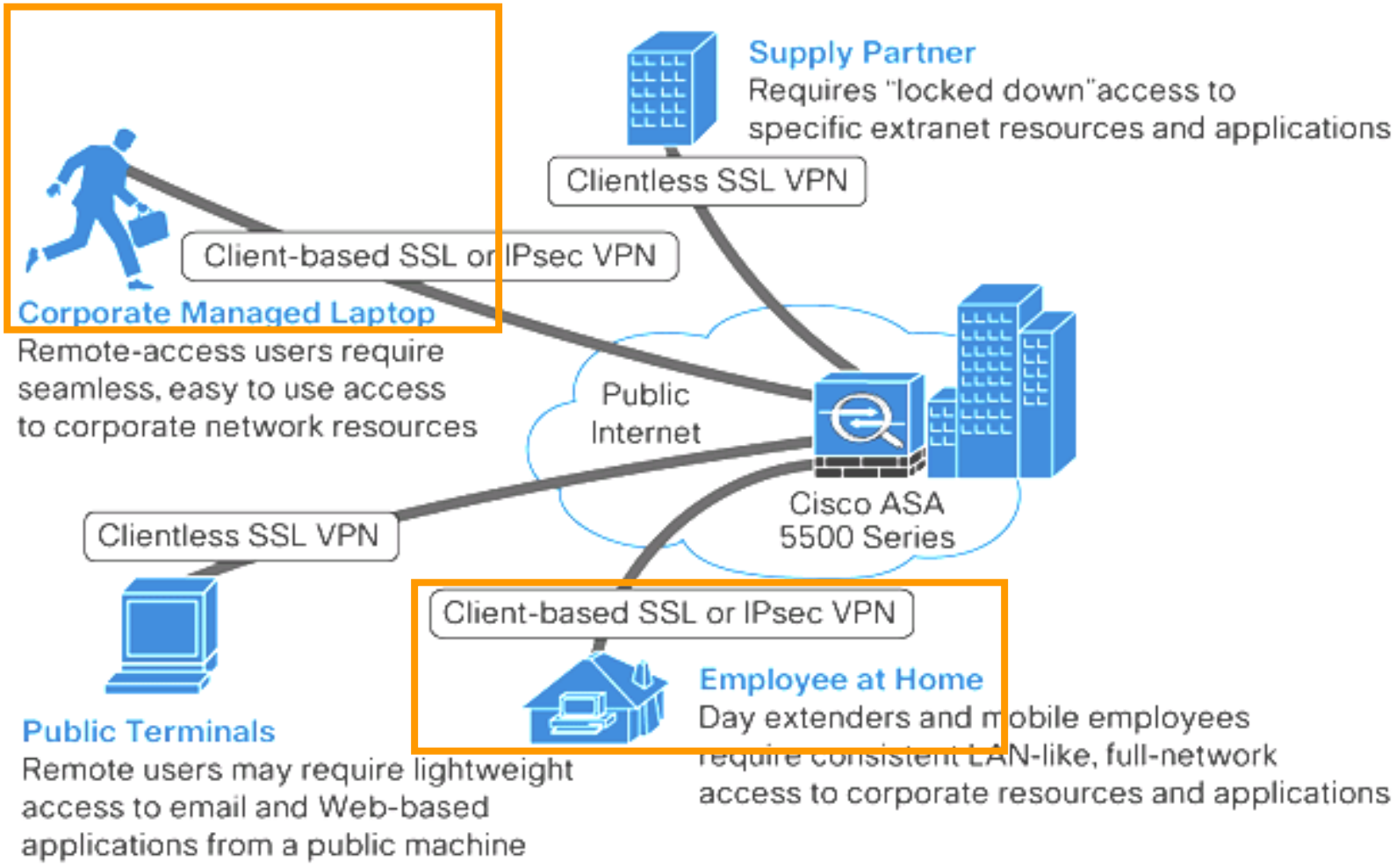

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# Types of Access



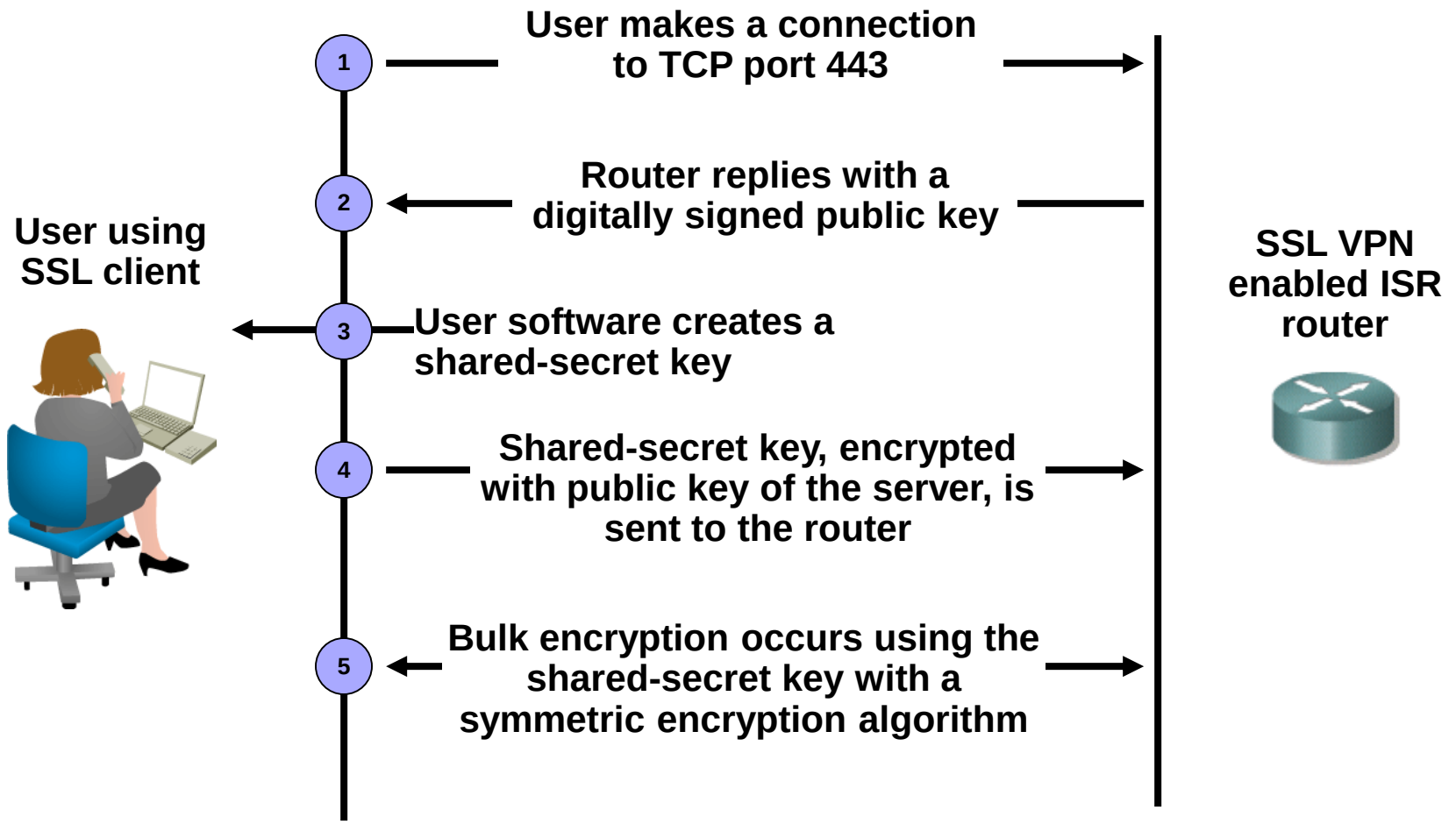

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# Full Tunnel Client Access Mode




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# Establishing an SSL Session




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# SSL VPN Design Considerations

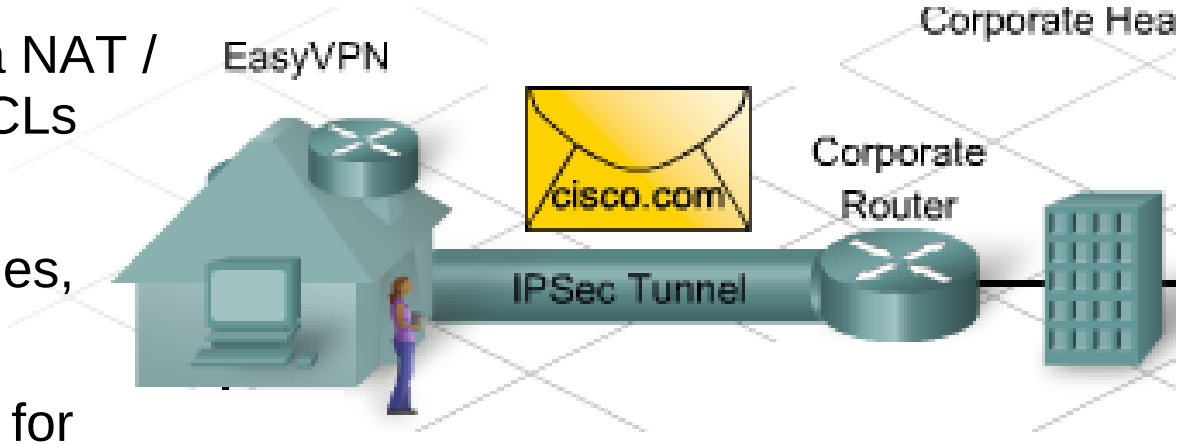
- User connectivity
- Router feature
- Infrastructure planning
- Implementation scope



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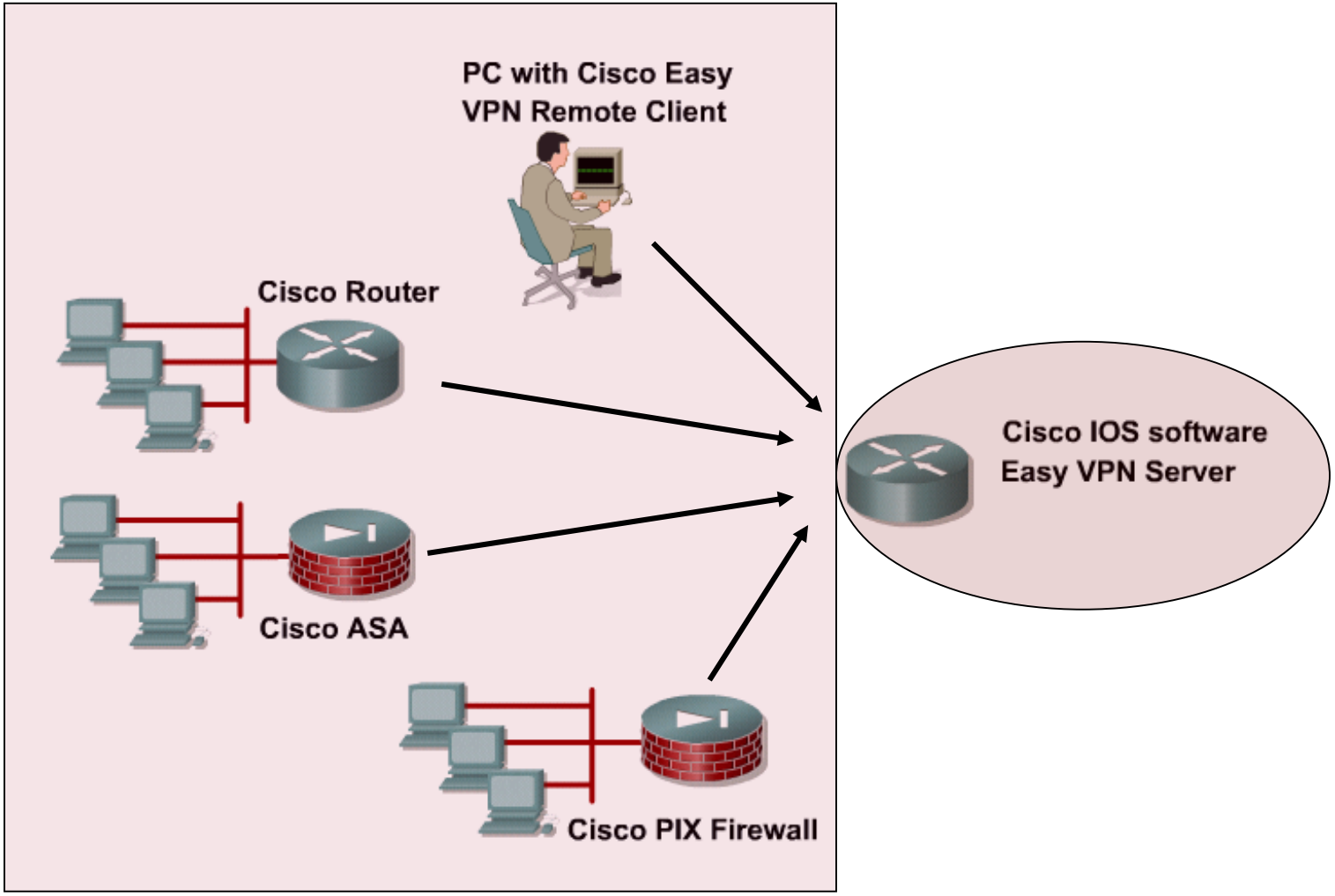
# Cisco Easy VPN

- Negotiates tunnel parameters
- Establishes tunnels according to set parameters
- Automatically creates a NAT / PAT and associated ACLs
- Authenticates users by usernames, group names, and passwords
- Manages security keys for encryption and decryption
- Authenticates, encrypts, and decrypts data through the tunnel



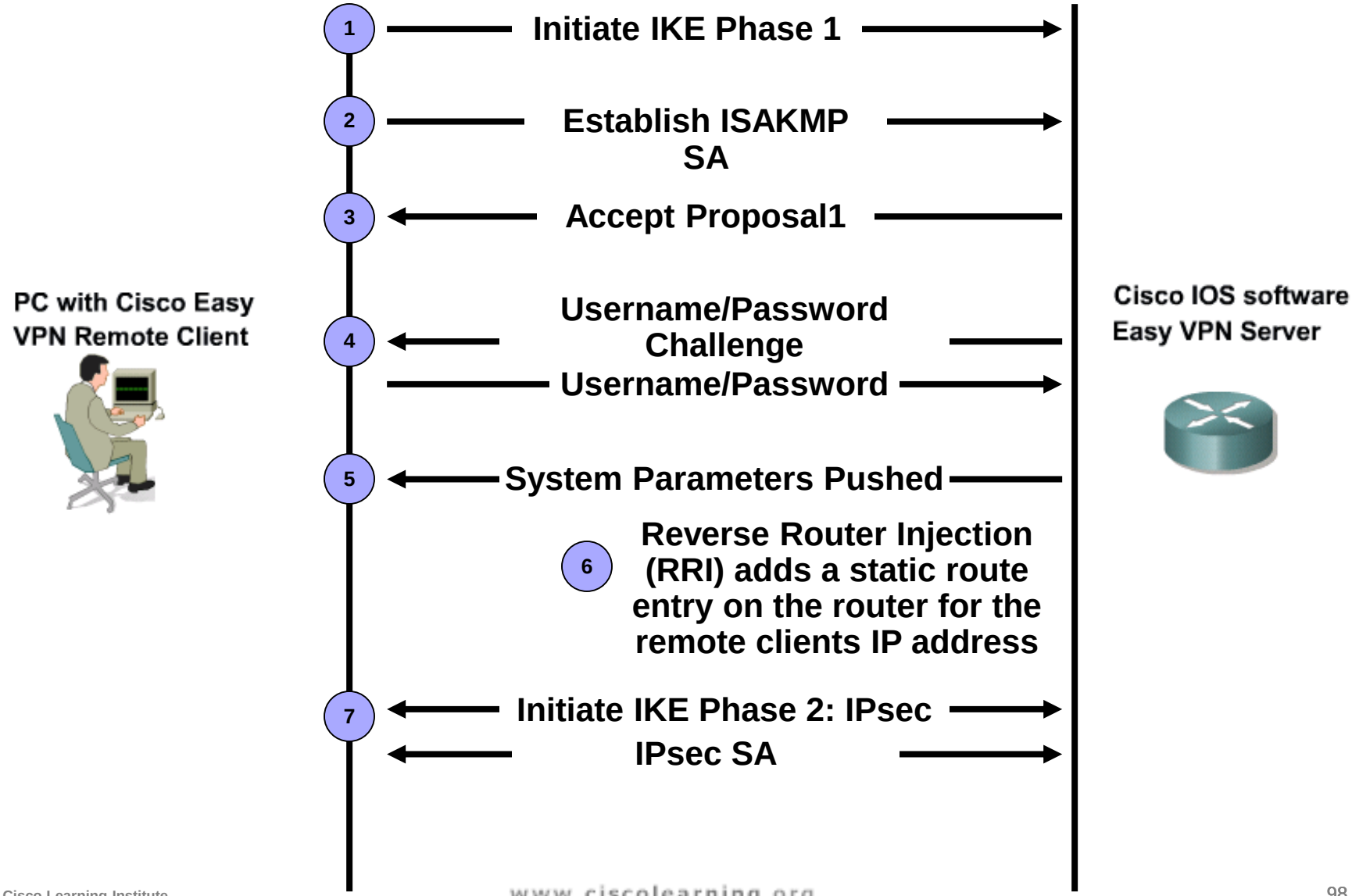

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# Cisco Easy VPN



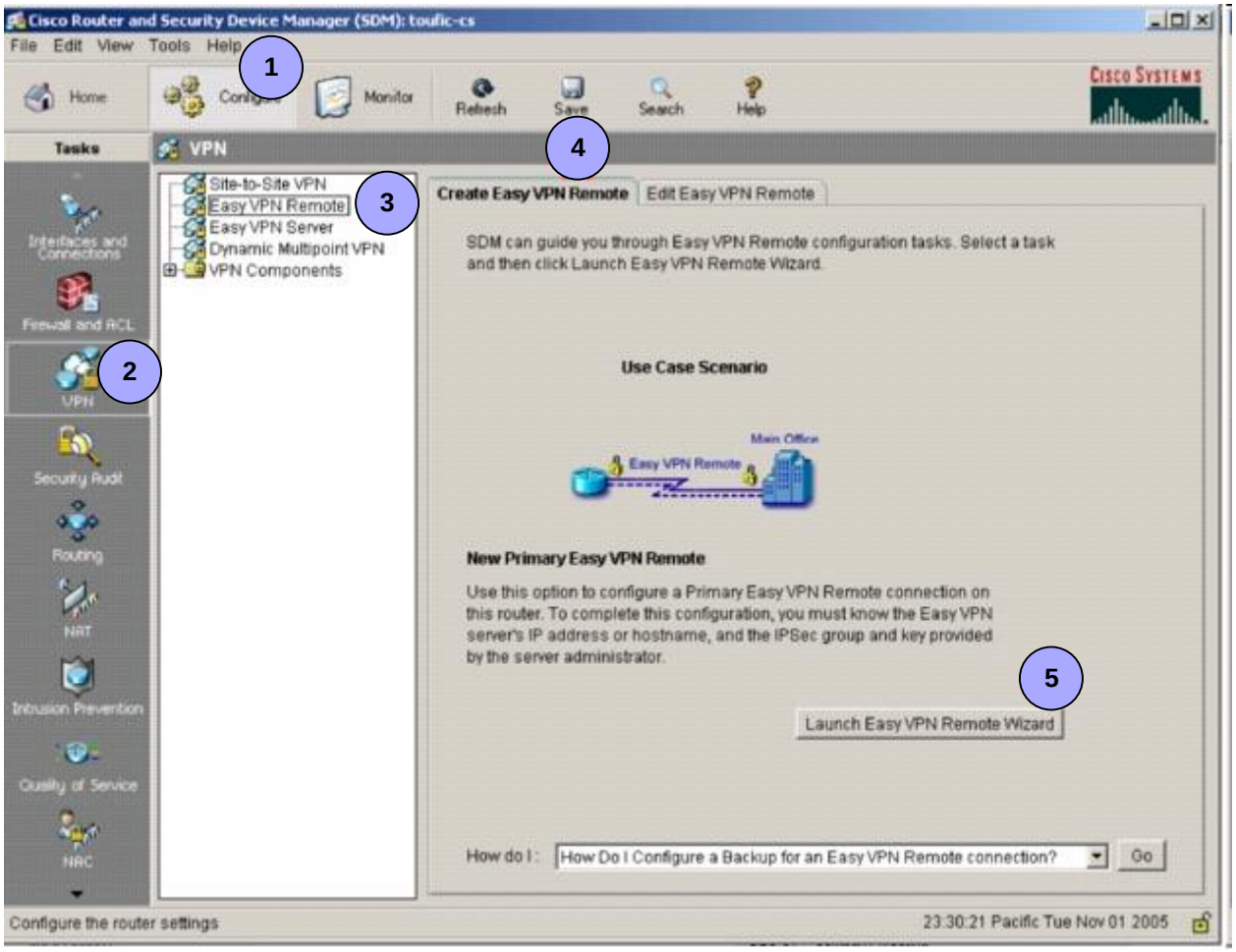

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# Securing the VPN



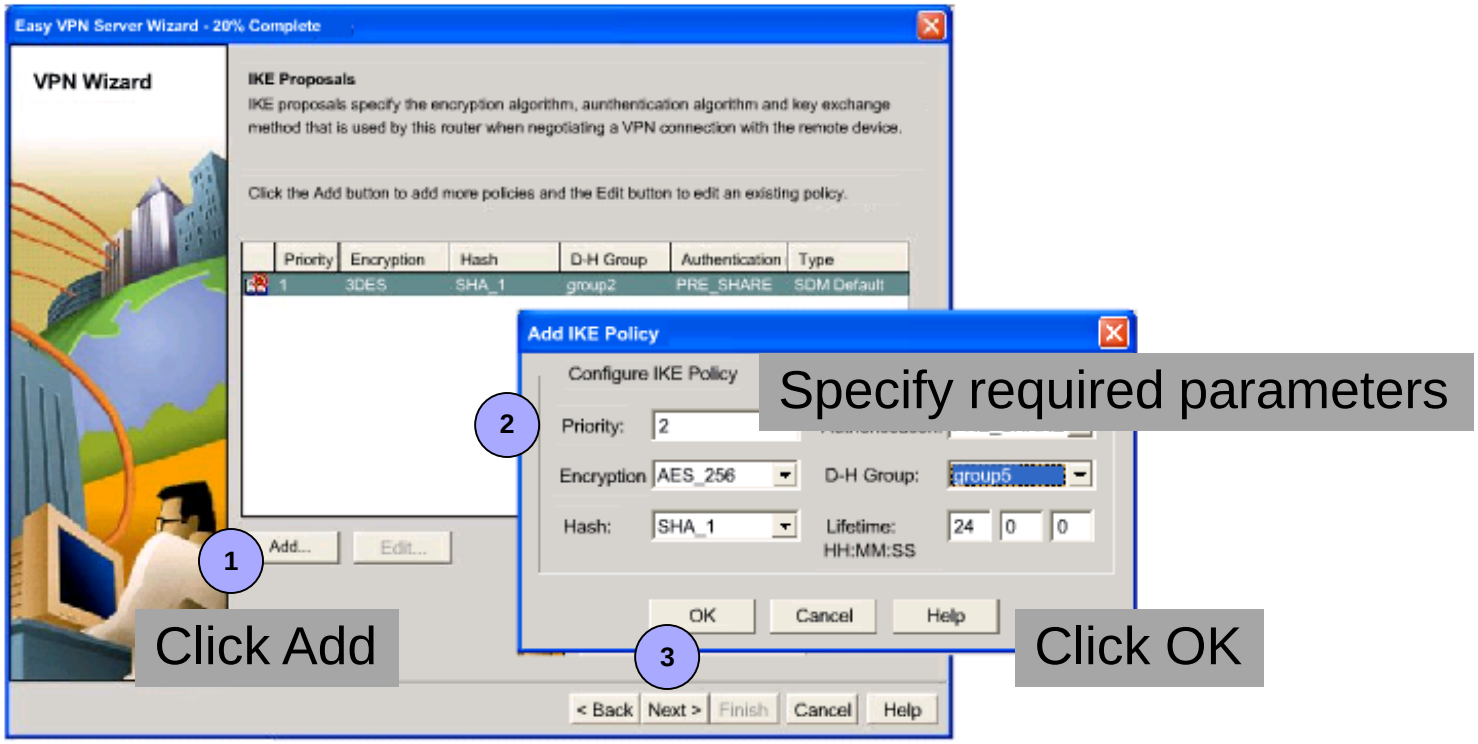

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# Configuring Cisco Easy VPN Server



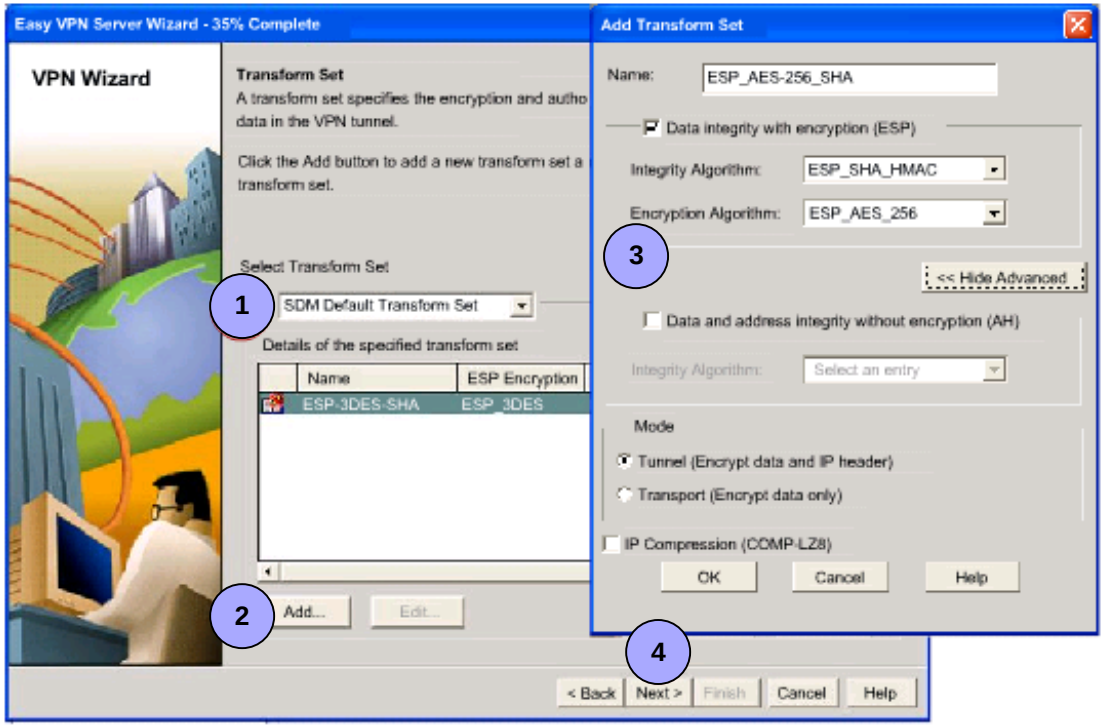

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# Configuring IKE Proposals



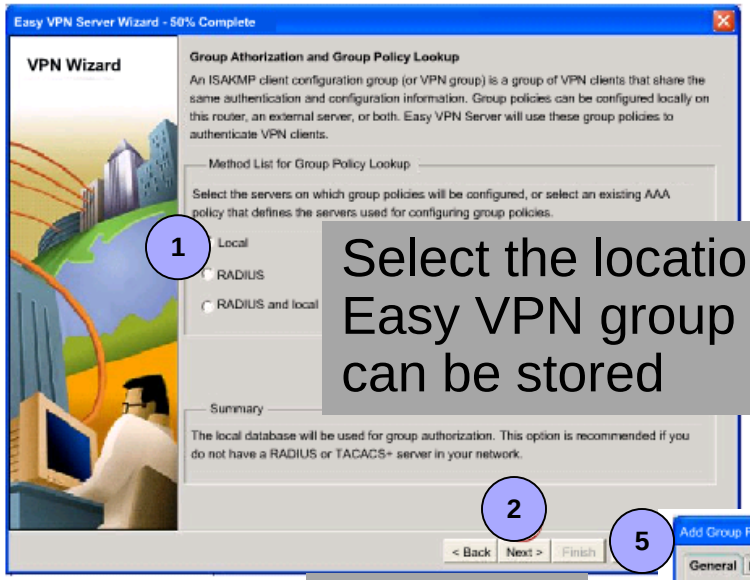

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# Creating an IPSec Transform Set




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# Group Authorization and Group Policy Lookup

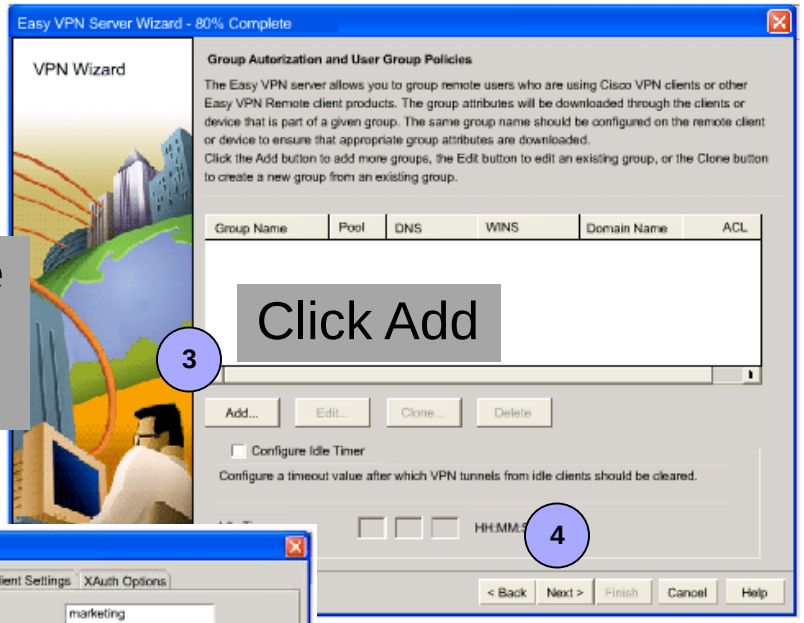



1

Select the location where Easy VPN group policies can be stored

2

Click Next



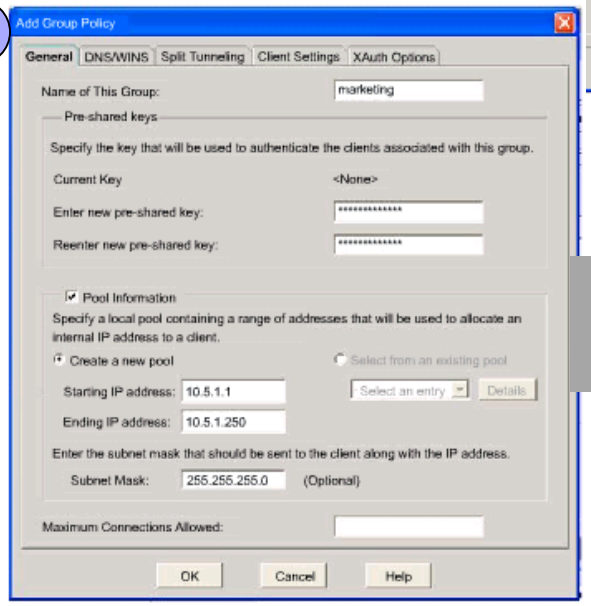

3

Click Add

4

Click Next

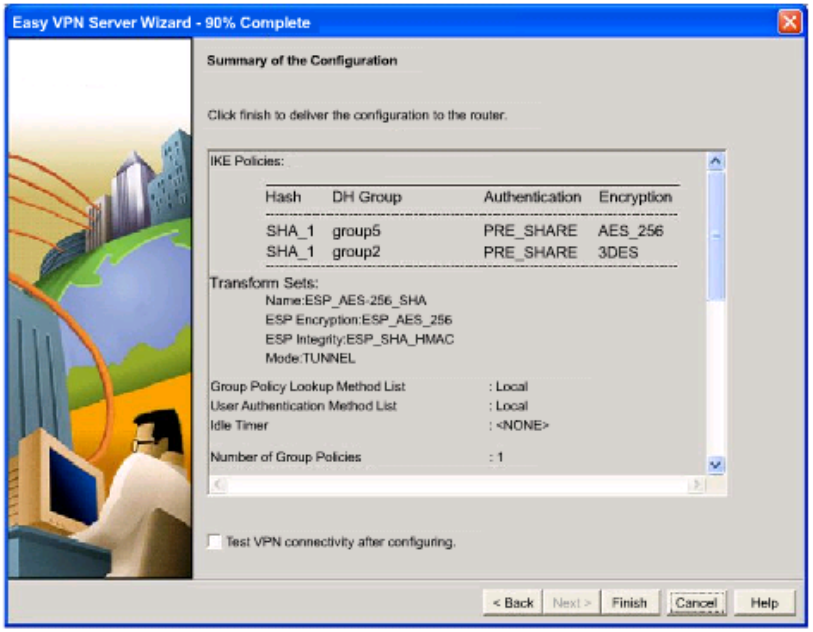
5

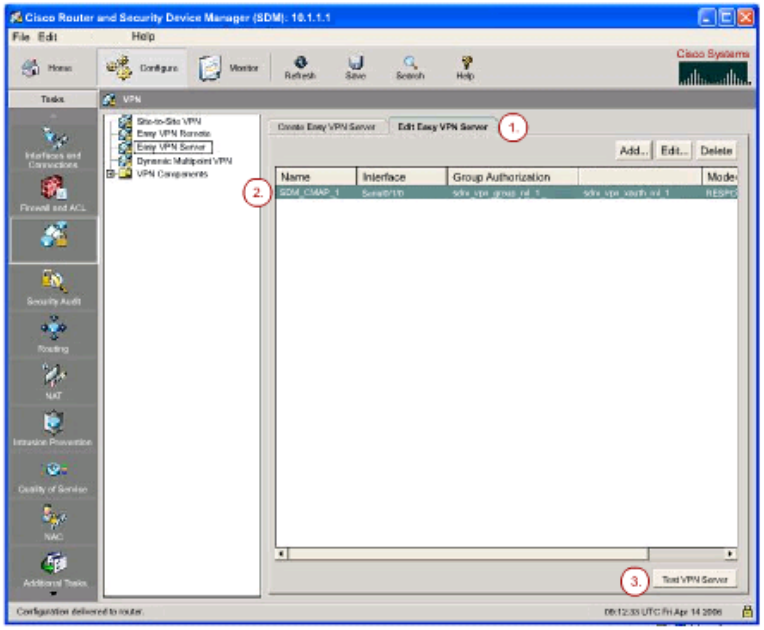


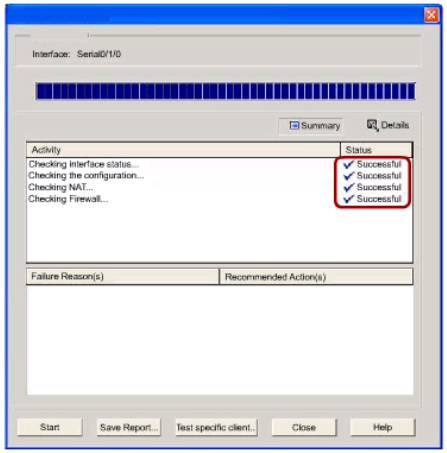

Configure the local group policies

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# Summary of Configuration Parameters

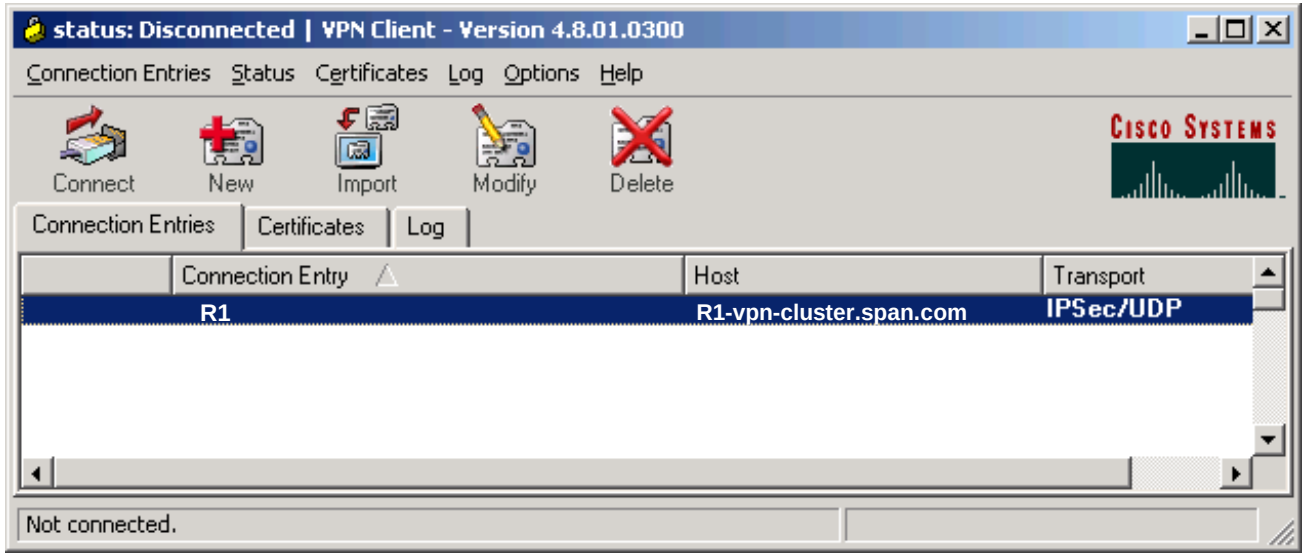



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# VPN Client Overview

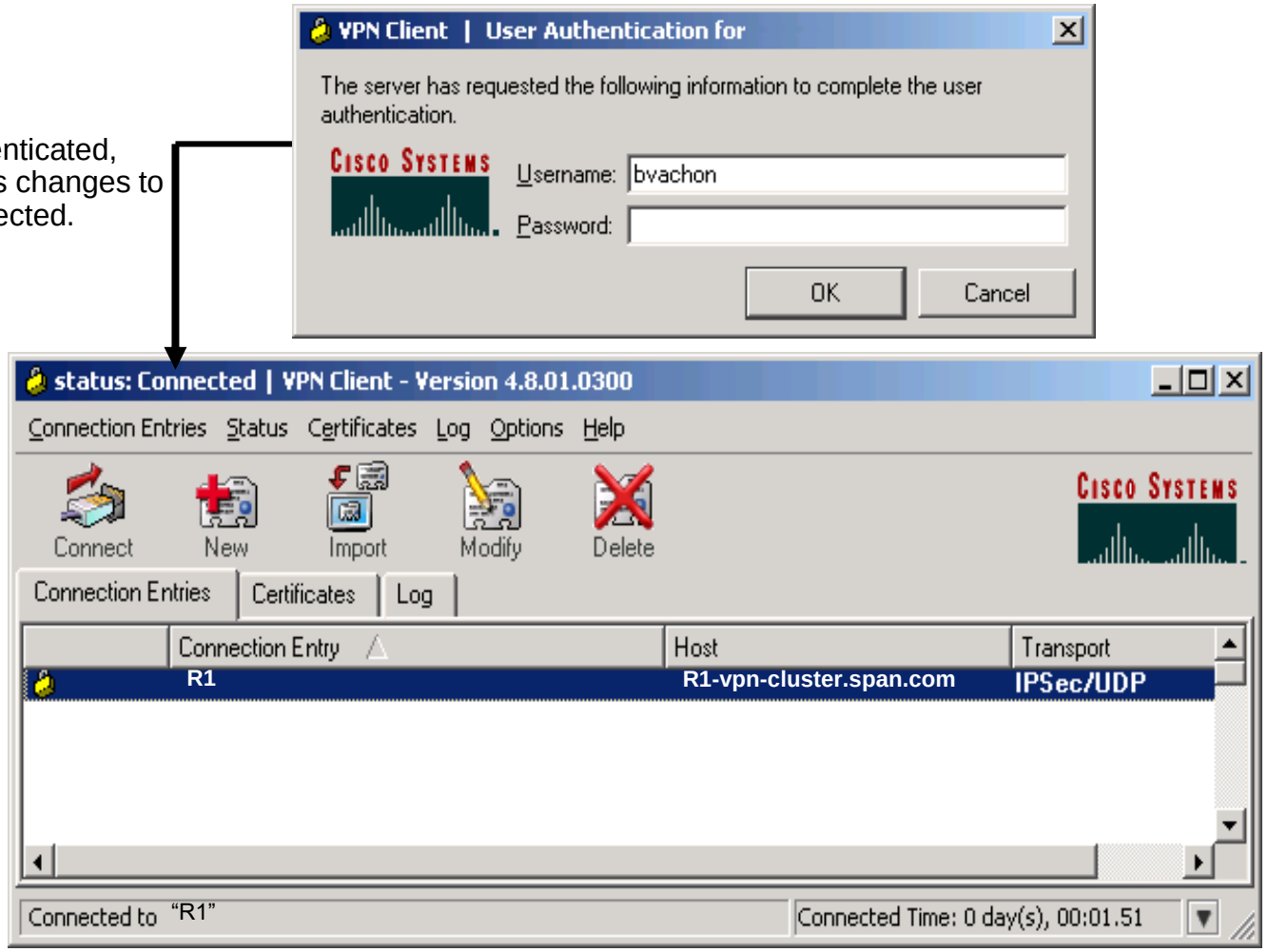



- Establishes end-to-end, encrypted VPN tunnels for secure connectivity
- Compatible with all Cisco VPN products
- Supports the innovative Cisco Easy VPN capabilities

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# Establishing a Connection

Once authenticated, status changes to connected.




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